```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

```
    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    #  开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
#  ==== app/__init__.py ====
#  -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    #  模板里展示时间统一走 localdt：库里存 UTC，展示成本地时间
    from app.utils.timefmt import to_local
    app.jinja_env.filters["localdt"] = to_local
    #  注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    #  全局登录保护：除登录页与静态资源外，均需登录
```

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from flask import request, redirect, url_for,                    101
send_from_directory                                              102
from flask_login import current_user, login_user                 103
PUBLIC_ENDPOINTS = {"auth.login", "auth.register",               104
"static", "favicon"}                                             105
#  浏览器默认会请求根路径下的 /favicon.ico。                      106
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会  107
#  回退到这个路径，补一条路由免得日志里一直刷 404。               108
@app.route("/favicon.ico")                                       109
def favicon():                                                   110
    return send_from_directory(app.static_folder,                111
"favicon.ico",                                                   112
                                       mimetype="image/vnd.micro 113
soft.icon")                                                      114
@app.before_request                                              115
def require_login():                                             116
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录    117
    if app.config.get("SCREENSHOT_MODE"):                        118
        if (request.endpoint not in ("auth.login",               119
"auth.register", "static")                                       120
                and not current_user.is_authenticated):          121
            admin = User.query.filter_by(role="admin").f         122
irst()                                                           123
            if admin:                                            124
                login_user(admin)                                125
        return None                                              126
    if request.endpoint in PUBLIC_ENDPOINTS or                   127
request.endpoint is None:                                        128
        return None                                              129
    if not current_user.is_authenticated:                        130
        return redirect(url_for("auth.login"))                   131
    return None                                                  132
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。  133
#  不使用模块编号，避免出现与实际流程不一致的代号。              134
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）  135
#  与可选的待办角标（该步骤还没做完的图斑数）。                  136
FLOW_LINKS = [                                                   137
    ("项目与矿山档案", "/project/", "folder"),                   138
    ("政策与标准知识库", "/standard/", "book"),                  139
    ("限制性指标判定", "/constraint/", "shield"),                140
    ("微地质单元", "/unit/", "layers"),                          141
    ("指标体系与权重", "/ahp/", "sliders"),                      142
    ("指标数据采集", "/indicator/", "database"),                 143
    ("适宜性定量评价", "/evaluate/", "target"),                  144
    ("报告与结果输出", "/report/", "file"),                      145
]                                                                146
#  「用户与操作日志」已并入「个人设置」，不再是独立菜单，        147
#  因此系统管理组只留一个入口。                                  148
ADMIN_LINKS = [                                                  149
    ("个人设置", "/profile", "user"),                            150
```

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]
@app.context_processor
def inject_nav():
    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
        return path.startswith(prefix.rstrip("/"))
    badges = {}
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        badges = dash.nav_badges()
    def build(links):
        return [{"label": label, "href": href, "icon": icon,
                 "active": hit(href), "badge": badges.get(href, 0)}
                for label, href, icon in links]
    groups = [
        {"title": "评价流程", "links": build(FLOW_LINKS)},
        {"title": "系统管理", "links": build(ADMIN_LINKS)},
    ]
    # 面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
    if path == "/":
        page_title = "工作台"
    elif path.startswith("/profile"):
        page_title = "个人设置"
    elif path.startswith(("/users", "/logs")):
        page_title = "账号与日志"
    elif path.startswith("/search"):
        page_title = "搜索结果"
    else:
        page_title = "工作台"
        for label, href, _icon in FLOW_LINKS:
            if hit(href):
                page_title = label
                break
    # 层级引导用：项目是归档单位，图斑才是评价对象。
    # 下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
```

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                                                    if p.patches.count()
    == 0],
        }
    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
```

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    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024          # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」里自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 昵称
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    gender = db.Column(db.String(8), nullable=True)
    # 男 / 女 / 保密
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def is_admin(self):
        return self.role == "admin"
    @property
    def shown_name(self):
        """界面上显示的名字：优先用昵称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字占位。"""
        return (self.shown_name or "?")[0].upper()
```

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    def to_dict(self):                                                    301
        return {"id": self.id, "username": self.username,                 302
                "display_name": self.display_name,                        303
    "gender": self.gender,                                                304
                "role": self.role, "active": self.active}                 305
class OperationLog(db.Model):                                             306
    """操作日志。"""                                                       307
    __tablename__ = "operation_logs"                                      308
    id = db.Column(db.Integer, primary_key=True)                          309
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"              310
    ), nullable=True)                                                     311
    action = db.Column(db.String(128), nullable=False)                    312
    # 操作类型                                                            313
    target_type = db.Column(db.String(64), nullable=True)                 314
    # 对象类型                                                            315
    target_id = db.Column(db.Integer, nullable=True)                      316
    # 对象主键                                                            317
    detail = db.Column(db.Text, nullable=True)                            318
    # 操作详情                                                            319
    created_at = db.Column(db.DateTime,                                   320
    default=datetime.utcnow, index=True)                                  321
class Project(db.Model):                                                  322
    """评价项目（项目与矿山档案）。"""                                     323
    __tablename__ = "projects"                                            324
    id = db.Column(db.Integer, primary_key=True)                          325
    name = db.Column(db.String(128), nullable=False)                      326
    code = db.Column(db.String(64), nullable=True)                        327
    # 项目编号                                                            328
    description = db.Column(db.Text, nullable=True)                       329
    status = db.Column(db.String(20), default="进行中")                   330
    # 进行中 / 已完成 / 已归档                                            331
    created_by = db.Column(db.Integer,                                    332
    db.ForeignKey("users.id"), nullable=True)                             333
    created_at = db.Column(db.DateTime,                                   334
    default=datetime.utcnow)                                              335
    patches = db.relationship("MinePatch",                                336
    backref="project", lazy="dynamic",                                    337
                                  cascade="all, delete-orphan")           338
class MinePatch(db.Model):                                                339
    """矿山图斑。"""                                                       340
    __tablename__ = "mine_patches"                                        341
    id = db.Column(db.Integer, primary_key=True)                          342
    project_id = db.Column(db.Integer,                                    343
    db.ForeignKey("projects.id"), nullable=False)                         344
    name = db.Column(db.String(128), nullable=False)                      345
    admin_region = db.Column(db.String(128), nullable=True)               346
    # 所在行政区                                                          347
    center_x = db.Column(db.Float, nullable=True)                         348
    # 中心经度 / 横坐标                                                   349
    center_y = db.Column(db.Float, nullable=True)                         350
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    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)    # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
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    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：> >= < <= ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")  # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    policy_pass = db.Column(db.Boolean, nullable=True)
    # 政策约束
    location_pass = db.Column(db.Boolean, nullable=True)
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
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    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
```

```
    created_at = db.Column(db.DateTime,                                    501
    default=datetime.utcnow)                                               502
class WeightItem(db.Model):                                                503
    """权重明细：方案内的各项权重。"""                                        504
    __tablename__ = "weight_items"                                         505
    id = db.Column(db.Integer, primary_key=True)                           506
    scheme_id = db.Column(db.Integer,                                      507
    db.ForeignKey("weight_schemes.id"), nullable=False)                    508
    indicator_code = db.Column(db.String(32),                              509
    nullable=False)       # 指标编码                                       510
    weight = db.Column(db.Float, nullable=False)                           511
    # 权重值                                                               512
class ExpertMatrix(db.Model):                                              513
    """专家判断矩阵：多专家打分的原始矩阵。"""                                 514
    __tablename__ = "expert_matrices"                                      515
    id = db.Column(db.Integer, primary_key=True)                           516
    scheme_id = db.Column(db.Integer,                                      517
    db.ForeignKey("weight_schemes.id"), nullable=True)                     518
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵                    519
    unit_type = db.Column(db.String(32), nullable=True)                    520
    # 微地质单元类型                                                       521
    l1_name = db.Column(db.String(64), nullable=True)                      522
    # 所属一级指标（二级矩阵用）                                           523
    expert_name = db.Column(db.String(64), nullable=True)                  524
    # 专家姓名                                                             525
    level = db.Column(db.Integer, nullable=False,                          526
    default=1)          # 矩阵对应指标层级                                 527
    matrix_json = db.Column(db.Text, nullable=False)                       528
    # 判断矩阵（JSON 二维数组）                                            529
    cr = db.Column(db.Float, nullable=True)                                530
    # 该专家矩阵的一致性比例                                               531
    created_at = db.Column(db.DateTime,                                    532
    default=datetime.utcnow)                                               533
class IndicatorData(db.Model):                                             534
    """指标实测数据：按微地质单元组织。"""                                    535
    __tablename__ = "indicator_data"                                       536
    id = db.Column(db.Integer, primary_key=True)                           537
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni               538
    ts.id"), nullable=False, unique=True)                                  539
    soil_ratio = db.Column(db.Float, nullable=True)                        540
    # 土石比                                                               541
    thickness = db.Column(db.Float, nullable=True)                         542
    # 土层厚度 cm                                                          543
    ph = db.Column(db.Float, nullable=True)                                544
    organic_matter = db.Column(db.Float, nullable=True)                    545
    # 有机质 g/kg                                                          546
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null      547
    heavy_metals = db.Column(db.Text, nullable=True)                       548
    coverage = db.Column(db.Float, nullable=True)                          549
    # 植被覆盖度 %                                                         550
```

```
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡 / 阴阳坡 / 阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
```

```
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
#  邮箱格式：够用的轻量校验。
#  WTForms 自带的 Email() 需要额外安装 email_validator,
#  为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
                                       Length(max=64,

    message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
```

```
                                    validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称
    "),
                                       Length(min=2, max=64,
                                              message="账号名称
    2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码
    "),
                                         Length(min=6,
    max=64,

    message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",

    message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
```

```
    """修改密码：需要先验证当前密码。"""                                    701
    old_password = PasswordField("当前密码",                                702
    validators=[DataRequired()])                                            703
    new_password = PasswordField("新密码",                                  704
                                        validators=[DataRequired(),         705
    Length(min=6, max=64,                                                   706
                                                                            707
    message="新密码至少 6 位")])                                            708
    confirm = PasswordField("确认新密码",                                   709
                                    validators=[DataRequired(),             710
                                                                            711
    EqualTo("new_password",                                                 712
                                                                            713
    message="两次输入的新密码不一致")])                                     714
    submit_pwd = SubmitField("修改密码")                                    715
#  ==== app/utils/constants.py ====                                         716
#  -*- coding: utf-8 -*-                                                    717
"""评价指标常量表。                                                        718
全部数值取自论文，作为系统计算的唯一基准：                                719
- 论文 4.2.2 节：指标标准化理想极值与取值依据                              720
- 论文 4.2.3 节 / 表 4：适宜性等级划分                                     721
- 论文表 3：按微地质单元分别确定的一级、二级指标权重                       722
- 论文 4.2.1 节：平均随机一致性指标 RI                                     723
"""                                                                        724
#  ------------------------------------------------------------------------725
    ------------------                                                      726
#  一、微地质单元类型                                                      727
#  ------------------------------------------------------------------------728
    ------------------                                                      729
#  论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡      730
#  三种地质微单元类型"                                                     731
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")                              732
#  每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）  733
#  供数据录入界面与评价引擎判断"该单元是否需要某类指标"                   734
UNIT_ENABLED_INDICATORS = {                                                735
    "岩质边坡": ("植被指标", "地形指标"),                  #  不设土壤指标  736
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),                      737
    "平台": ("土壤指标", "植被指标"),                      #  不设地形指标  738
}                                                                          739
#  ------------------------------------------------------------------------740
    ------------------                                                      741
#  二、指标标准化理想极值（论文 4.2.2 节）                                742
#  ------------------------------------------------------------------------743
    ------------------                                                      744
IDEAL_SOIL_RATIO = 0.8          #  土石比理想值                           745
IDEAL_THICKNESS = 20.0          #  土层厚度理想值 （cm）                  746
IDEAL_PH_RANGE = (5.0, 9.0)     #  pH 理想区间（适中型指标）              747
IDEAL_ORGANIC = 30.0            #  有机质理想值                           748
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值                 749
IDEAL_COVERAGE = 75.0           #  植被覆盖度参照生态系统 （%）           750
```

```
IDEAL_SHANNON = 0.7                  # Shannon-Wiener 多样性指数参照生态系统
# 重金属极值 （mg/kg），《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 Ⅳ 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、５３０、４００、４５、２２０、６００、８００
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 ４．２．１ 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 ４．２．２ 节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２ 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 ４．２．２ 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
# ------------------------------------------------------------------
    ------------------
# 三、适宜性等级划分（论文表 4）
# ------------------------------------------------------------------
    ------------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ------------------------------------------------------------------
    ------------------
```

```
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ----------------------------------------------------------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6: 1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ----------------------------------------------------------------
# 五、默认权重方案（论文表 3）
# ----------------------------------------------------------------
# 论文 4.2.2 节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
# 分别计算评价指标权重"（论文摘要、图 2 流程与表 5 均作"碎石边坡"，
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。
# 下表为论文表 3 所载的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS = {
    "岩质边坡": {
        "l1": {"植被指标": 0.45, "地形指标": 0.55},
        "l2": {
            "植被指标": {"植被覆盖度": 0.67, "生物多样性": 0.33},
            "地形指标": {"坡度": 0.64, "坡向": 0.36},
        },
    },
    "碎石边坡": {
        "l1": {"土壤指标": 0.44, "植被指标": 0.31, "地形指标": 0.25},
        "l2": {
            "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
                         "有机质": 0.17, "污染指标": 0.17},
            "植被指标": {"植被覆盖度": 0.64, "生物多样性": 0.36},
            "地形指标": {"坡度": 0.67, "坡向": 0.33},
        },
    },
    "平台": {
        "l1": {"土壤指标": 0.54, "植被指标": 0.46},
        "l2": {
            "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
                         "有机质": 0.17, "污染指标": 0.17},
            "植被指标": {"植被覆盖度": 0.61, "生物多样性": 0.39},
        },
    },
}
# ----------------------------------------------------------------
# 六、限制性指标（论文表 2 + 4.1 节流程）
# ----------------------------------------------------------------
# 论文摘要与 4.1 节：四项限制性指标；流程段另明确"是否为耕地"为约束条件
CONSTRAINT_ITEMS = ("政策", "区位约束", "土壤污染", "地质灾害", "耕地属性")
```

```
# ---------------------------------------------------------------------- 851
    ----------------- 852
# 七、账号职务 853
# ---------------------------------------------------------------------- 854
    ----------------- 855
# 只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。 856
ROLE_LABELS = {"admin": "管理员", "operator": "操作员"} 857
ROLE_CODES = ("admin", "operator") 858
DEFAULT_ROLE = "operator"  # 新注册账号的默认职务 859
ROLE_ORDER = ("admin", "operator")  # 列表排序用：管理员在前 860
# 性别（注册与个人资料中选填） 861
GENDERS = ("男", "女", "保密") 862
# ==== app/utils/schema.py ==== 863
# -*- coding: utf-8 -*- 864
"""轻量结构迁移。 865
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。 866
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上， 867
使旧数据库无需重建即可用上新字段。 868
""" 869
from sqlalchemy import text 870
# （表名， 列名， 列定义） 871
REQUIRED_COLUMNS = [ 872
    # 遥感影像挂接 873
    ("mine_patches", "image_path", "VARCHAR(256)"), 874
    ("mine_patches", "image_note", "VARCHAR(256)"), 875
    ("mine_patches", "image_uploaded_at", "DATETIME"), 876
    # 指标维护 877
    ("indicators", "enabled_units", "VARCHAR(128)"), 878
    ("indicators", "description", "TEXT"), 879
    # 多专家矩阵 880
    ("expert_matrices", "unit_type", "VARCHAR(32)"), 881
    ("expert_matrices", "l1_name", "VARCHAR(64)"), 882
    ("expert_matrices", "cr", "FLOAT"), 883
    # 用户个人资料（账户页） 884
    ("users", "display_name", "VARCHAR(64)"), 885
    ("users", "email", "VARCHAR(128)"), 886
    ("users", "phone", "VARCHAR(32)"), 887
    ("users", "gender", "VARCHAR(8)"), 888
    ("users", "avatar", "VARCHAR(256)"), 889
] 890
# 职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种， 891
# 旧数据里的后两种统一归为操作员。 892
ROLE_MIGRATION = {"evaluator": "operator", "viewer": 893
    "operator"} 894
def _add_columns(db): 895
    added = [] 896
    with db.engine.connect() as conn: 897
        for table, column, ddl in REQUIRED_COLUMNS: 898
            rows = conn.execute(text("PRAGMA 899
    table_info(%s)" % table)).fetchall() 900
```

```
                if not rows:                                                    901
                    continue                                  #                 902
        表还不存在，create_all 会建                                               903
                existing = {r[1] for r in rows}                                 904
                if column in existing:                                          905
                    continue                                                    906
                conn.execute(text(                                              907
                    "ALTER TABLE %s ADD COLUMN %s %s" % (table,                 908
        column, ddl)))                                                          909
                conn.commit()                                                   910
                added.append("%s.%s" % (table, column))                         911
        return added                                                            912
def _migrate_roles(db):                                                         913
        """把旧角色名统一为新的两种，返回改动的行数。"""                                    914
        changed = 0                                                             915
        with db.engine.connect() as conn:                                       916
            rows = conn.execute(text("PRAGMA                                    917
        table_info(users)")).fetchall()                                         918
            if not rows:                                                        919
                return 0                                                        920
            for old, new in ROLE_MIGRATION.items():                             921
                r = conn.execute(text("UPDATE users SET role =                  922
        :new "                                                                  923
                                      "WHERE role = :old"),                     924
                                     {"new": new, "old": old})                  925
                changed += r.rowcount or 0                                      926
            conn.commit()                                                       927
        return changed                                                          928
def ensure_schema(db):                                                          929
        """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""                                 930
        try:                                                                    931
            changed = _add_columns(db)                                          932
            n = _migrate_roles(db)                                              933
            if n:                                                               934
                changed.append("用户角色归一为管理员/操作员 %d 条" % n)                      935
            return changed                                                      936
        except Exception as e:                                  #               937
        迁移失败不应阻断启动                                                          938
            print("[schema] 结构检查跳过：%s" % e)                                  939
            return []                                                           940
#  ==== app/utils/uploads.py ====                                               941
#  -*- coding: utf-8 -*-                                                        942
"""上传文件处理（遥感影像、批量导入）。"""                                                 943
import os                                                                       944
import re                                                                       945
import uuid                                                                     946
from datetime import datetime                                                   947
from werkzeug.utils import secure_filename                                      948
def _safe_ext(filename):                                                        949
        return os.path.splitext(filename)[1].lower().lstrip(".")                950
```

```
def ext_allowed(filename, allowed):                                    951
    return _safe_ext(filename) in allowed                              952
def save_upload(file_storage, subdir, allowed_ext,                     953
    upload_root):                                                      954
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。    955
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。                    956
    """                                                                957
    if file_storage is None or not file_storage.filename:              958
        return None, "未选择文件"                                       959
    ext = _safe_ext(file_storage.filename)                             960
    if ext not in allowed_ext:                                         961
        return None, "不支持的文件类型：.%s" % ext                      962
    target_dir = os.path.join(upload_root, subdir)                     963
    os.makedirs(target_dir, exist_ok=True)                             964
    # 仅保留安全字符，并加时间戳与随机串防重名                                 965
    base = secure_filename(os.path.splitext(file_storage.fil           966
    ename)[0]) or "file"                                               967
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or                968
    "file"                                                             969
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M           970
    %S"),                                                              971
                                uuid.uuid4().hex[:6], ext)             972
    abs_path = os.path.join(target_dir, name)                          973
    file_storage.save(abs_path)                                        974
    rel = os.path.join("uploads", subdir,                              975
    name).replace("\\", "/")                                           976
    return rel, None                                                   977
def remove_upload(rel_path, upload_root):                              978
    """删除已保存的上传文件（相对 static 的路径）。"""                          979
    if not rel_path:                                                   980
        return                                                         981
    abs_path = os.path.join(upload_root,                               982
    rel_path.replace("uploads/", "", 1))                               983
    try:                                                               984
        if os.path.isfile(abs_path):                                   985
            os.remove(abs_path)                                        986
    except OSError:                                                    987
        pass                                                           988
# ==== app/utils/charts.py ====                                        989
# -*- coding: utf-8 -*-                                                990
"""轻量图表。                                                          991
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                                  992
- 服务端渲染，页面与打印（PDF）中都能稳定显示                               993
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）                    994
- 单色方案，符合界面「一个视图一个强调色」的约定                            995
"""                                                                    996
from html import escape                                                997
ACCENT = "#1f3a5f"                                                     998
ACCENT_SOFT = "#7f93ad"                                                999
INK = "#23262b"                                                       1000
```

```
MUTED = "#6b7280"
LINE = "#dfe3e8"
#  灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
        return str(v)
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,
    value_w=76):
    """水平条形图。items: [(标签, 数值, 显示文本)]"""
    if not items:
        return '<p class="muted">暂无数据</p>'
    n = len(items)
    height = n * (bar_h + gap) + 10
    max_v = max((abs(float(v)) for _, v, _ in items),
    default=1) or 1
    body_w = width - label_w - value_w
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"
    height="%d" '
             'role="img" aria-label="条形图" '
             'style="max-width:%dpx;font-family:inherit;">'
    % (width, height, height, width)]
    for i, (label, value, disp) in enumerate(items):
        y = i * (bar_h + gap) + 5
        frac = (abs(float(value)) / max_v) if max_v else 0
        w = max(2.0, body_w * frac)
        fill = GRAYS[i % len(GRAYS)]
        parts.append(
            '<text x="%d" y="%d" text-anchor="end"
    font-size="12" fill="%s">%s</text>'
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
    rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
            'style="font-variant-numeric:tabular-nums;">%s</
    text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
    escape(str(disp))))
    parts.append('</svg>')
    return "".join(parts)
def donut(items, size=210, thickness=34):
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。
```

```
    """                                                                 1051
    vals = [(l, float(v), d) for l, v, d in items if                    1052
float(v) > 0]                                                           1053
    if not vals:                                                        1054
        return '<p class="muted">暂无数据</p>'                           1055
    total = sum(v for _, v, _ in vals) or 1                             1056
    r = (size - thickness) / 2                                          1057
    cx = cy = size / 2                                                  1058
    import math                                                         1059
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                       1060
height="%d" role="img" '                                                1061
             'aria-label="权重分布环形图">' % (size, size, size,         1062
size)]                                                                  1063
    angle = -math.pi / 2                                                1064
    for i, (label, v, disp) in enumerate(vals):                         1065
        sweep = 2 * math.pi * (v / total)                               1066
        a0, a1 = angle, angle + sweep                                   1067
        x0, y0 = cx + r * math.cos(a0), cy + r *                        1068
math.sin(a0)                                                            1069
        x1, y1 = cx + r * math.cos(a1), cy + r *                        1070
math.sin(a1)                                                            1071
        large = 1 if sweep > math.pi else 0                             1072
        parts.append(                                                   1073
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f               1074
%.2f" '                                                                 1075
            'fill="none" stroke="%s" stroke-width="%d"/>'               1076
            % (x0, y0, r, r, large, x1, y1, GRAYS[i %                   1077
len(GRAYS)], thickness))                                                1078
        angle = a1                                                      1079
    parts.append('<text x="%d" y="%d" text-anchor="middle"              1080
font-size="12" '                                                        1081
                 'fill="%s">合计</text>' % (cx, cy - 2,                 1082
MUTED))                                                                 1083
    parts.append('<text x="%d" y="%d" text-anchor="middle"              1084
font-size="15" '                                                        1085
                 'fill="%s" style="font-variant-numeric:tabu            1086
lar-nums;">%s</text>'                                                   1087
                 % (cx, cy + 18, INK, _fmt(total, 4)))                  1088
    parts.append('</svg>')                                              1089
    legend = ['<table style="width:auto;margin-top:10px;"><t            1090
head><tr>'                                                              1091
              '<th scope="col">项目</th><th                             1092
scope="col">数值</th>'                                                  1093
              '<th scope="col">占比</th></tr></thead><tbody>']          1094
    for i, (label, v, disp) in enumerate(vals):                         1095
        legend.append(                                                  1096
            '<tr><td style="text-align:left;">'                         1097
            '<span style="display:inline-block;width:10px;he            1098
ight:10px;'                                                             1099
            'border-radius:2px;background:%s;margin-right:6p            1100
```

```
        x;"><span>%s</td>'                                                  1101
                '<td class="mono">%s</td><td                                1102
        class="mono">%.1f%%</td></tr>'                                      1103
                % (GRAYS[i % len(GRAYS)], escape(str(label)),               1104
        escape(str(disp)),                                                  1105
                    100.0 * v / total))                                     1106
        legend.append('</tbody></table>')                                   1107
        return ('<div style="display:flex;flex-wrap:wrap;gap:20p            1108
        x;align-items:center;">'                                            1109
                + "".join(parts) + "".join(legend) + '</div>')              1110
# ==== app/utils/browser.py ====                                            1111
# -*- coding: utf-8 -*-                                                     1112
"""HTML 转 PDF（报告导出）。                                                1113
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、             1114
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，          1115
避免中文路径在 file:// URL 下出问题。                                       1116
"""                                                                         1117
import os                                                                   1118
import shutil                                                               1119
import subprocess                                                           1120
import tempfile                                                             1121
import uuid                                                                 1122
CANDIDATES = [                                                              1123
        r"C:\Program Files\Google\Chrome\Application\chrome.exe"            1124
        ,                                                                   1125
        r"C:\Program Files (x86)\Google\Chrome\Application\chrom            1126
        e.exe",                                                             1127
        r"C:\Program Files (x86)\Microsoft\Edge\Application\msed            1128
        ge.exe",                                                            1129
        r"C:\Program Files\Microsoft\Edge\Application\msedge.exe            1130
        ",                                                                  1131
]                                                                           1132
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器                          1133
ENV_KEY = "DSH_BROWSER_PATH"                                                1134
def browser_path():                                                         1135
        """定位可用的浏览器可执行文件，找不到返回 None。"""                 1136
        override = os.environ.get(ENV_KEY)                                  1137
        if override and os.path.isfile(override):                           1138
                return override                                             1139
        for p in CANDIDATES:                                                1140
                if os.path.isfile(p):                                       1141
                        return p                                            1142
        for name in ("chrome", "msedge", "chromium"):                       1143
                found = shutil.which(name)                                  1144
                if found:                                                   1145
                        return found                                        1146
        return None                                                         1147
def html_to_pdf(html, timeout=90):                                          1148
        """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""   1149
        exe = browser_path()                                                1150
```

```
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
    text=True,
                           timeout=timeout,
    encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
```

```
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
```

```
    ----                                                                1251
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)          1252
    及对应取值。                                                        1253
    """                                                                 1254
    A = np.asarray(A, dtype=float)                                      1255
    r = weights_and_consistency(A)                                      1256
    w = r["weights"]                                                    1257
    n = A.shape[0]                                                      1258
    best = None                                                         1259
    for i in range(n):                                                  1260
        for j in range(i + 1, n):                                       1261
            expected = w[i] / w[j]                                      1262
            if expected == 0:                                           1263
                continue                                                1264
            deviation = abs(A[i, j] - expected) / expected              1265
            if best is None or deviation >                              1266
    best["deviation"]:                                                  1267
                best = {"i": i, "j": j, "value": A[i, j],               1268
                        "expected": expected, "deviation":              1269
    deviation}                                                          1270
    return best                                                         1271
def aggregate_matrices(matrices):                                       1272
    """多专家判断矩阵聚合：几何平均（保持互反性）。                     1273
    参数                                                                1274
    ----                                                                1275
    matrices : list[np.ndarray]，多个同阶判断矩阵。                     1276
    """                                                                 1277
    stack = np.stack([np.asarray(m, dtype=float) for m in               1278
    matrices])                                                          1279
    return np.prod(stack, axis=0) ** (1.0 / len(stack))                 1280
def judge_from_weights(weights):                                        1281
    """由权重向量构造完全一致判断矩阵并检验。                           1282
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量     1283
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。                          1284
    """                                                                 1285
    w = np.asarray(weights, dtype=float)                                1286
    n = w.shape[0]                                                      1287
    A = np.outer(w, 1.0 / w)                                            1288
    r = weights_and_consistency(A)                                      1289
    r["matrix"] = A                                                     1290
    return r                                                            1291
#  ==== app/services/standardization.py ====                          1292
#  -*- coding: utf-8 -*-                                              1293
"""指标标准化（的一部分）。                                           1294
按论文 4.2.2 节式(8) 实现：                                           1295
    x' = (x - x_min) / (x_max - x_min)                                  1296
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。           1297
所有理想极值取自 app.utils.constants，可配置。                        1298
"""                                                                     1299
from app.utils.constants import (                                       1300
```

```
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染(=1)处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
```

```
    None else 1.0                                                    1351
    return total / len(HEAVY_METAL_LIMIT)                            1352
def standardize_unit(unit_type, data, weights=None):                 1353
    """对单个微地质单元的全部指标做标准化。                                1354
    参数                                                             1355
    ----                                                             1356
    unit_type : str，岩质边坡 / 碎石边坡 / 平台                          1357
    data : dict，实测数据，字段见接口契约：                                1358
        soil_ratio, thickness, ph, organic_matter,                   1359
    heavy_metals(dict),                                              1360
        coverage, shannon, slope, aspect                             1361
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG1362
    HTS 结构），                                                       1363
            未提供时用默认值。                                           1364
    返回                                                             1365
    ----                                                             1366
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }      1367
    标准化值可能为 None，表示该指标不适用或方法待定。                        1368
    """                                                              1369
    from app.utils.constants import DEFAULT_WEIGHTS                  1370
    if weights is None:                                              1371
        weights = DEFAULT_WEIGHTS[unit_type]                         1372
    l1_enabled = weights["l1"].keys()                                1373
    l2w = weights["l2"]                                              1374
    result = {}                                                      1375
    # ---- 土壤指标 ----                                             1376
    if "土壤指标" in l1_enabled:                                       1377
        result["土壤指标"] = {                                         1378
            "土石比": (data.get("soil_ratio"),                       1379
                       std_benefit(data.get("soil_ratio"),           1380
    IDEAL_SOIL_RATIO),                                               1381
                       l2w["土壤指标"]["土石比"]),                     1382
            "土层厚度": (data.get("thickness"),                      1383
                         std_benefit(data.get("thickness"),          1384
    IDEAL_THICKNESS),                                                1385
                         l2w["土壤指标"]["土层厚度"]),                  1386
            "pH": (data.get("ph"),                                   1387
                   std_mid_ph(data.get("ph"),                        1388
    *IDEAL_PH_RANGE),                                                1389
                   l2w["土壤指标"]["pH"]),                           1390
            "有机质": (data.get("organic_matter"),                   1391
                       std_benefit(data.get("organic_matter")1392
    , IDEAL_ORGANIC),                                                1393
                       l2w["土壤指标"]["有机质"]),                     1394
            "污染指标": (data.get("heavy_metals") or {},             1395
                         pollution_score(data.get("heavy_meta1396
    ls") or {}),                                                     1397
                         l2w["土壤指标"]["污染指标"]),                  1398
        }                                                            1399
    # ---- 植被指标 ----                                             1400
```

```
    if "植被指标" in l1_enabled:
        result["植被指标"] = {
            "植被覆盖度": (data.get("coverage"),
                           std_benefit(data.get("coverage"),
    IDEAL_COVERAGE),
                           l2w["植被指标"]["植被覆盖度"]),
            "生物多样性": (data.get("shannon"),
                           std_benefit(data.get("shannon"),
    IDEAL_SHANNON),
                           l2w["植被指标"]["生物多样性"]),
        }
    # ---- 地形指标 ----
    if "地形指标" in l1_enabled:
        result["地形指标"] = {
            "坡度": (data.get("slope"),
                     std_slope(data.get("slope")),
                     l2w["地形指标"]["坡度"]),
            "坡向": (data.get("aspect"),
                     aspect_score(data.get("aspect")),
                     l2w["地形指标"]["坡向"]),
        }
    return result
# ==== app/services/weight_service.py ====
# -*- coding: utf-8 -*-
"""权重方案的存取与生效权重解析。
系统有两套权重来源：
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），
   是**权威取值来源**，案例复算与报告结论以此为准；
2. 通过层次分析法在本系统中实际算出的权重，保存在
   weight_schemes / weight_items 两张表中，可标记为「生效」。
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：
以论文表 3 为底，用生效方案中已保存的条目覆盖。
没有任何生效方案时，即完全使用论文表 3 的权重。
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，
而不是算完即弃。
"""
import copy
from app.models import db, WeightScheme, WeightItem
from app.utils.constants import DEFAULT_WEIGHTS
from app.services.indicator_service import TREE
# 方案名称：每个微地质单元类型固定一个可生效的方案
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"
def _code_maps():
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
```

```
            elif level == 2:                                                1451
                l2[code] = (l1.get(parent, parent), name)                   1452
    return l1, l2                                                           1453
def effective_weights(unit_type):                                           1454
    """返回该单元类型当前生效的权重。                                          1455
    返回 (weights, source_label, scheme)                                    1456
        weights      嵌套权重结构，可直接交给标准化与评价引擎                  1457
        source_label 权重来源说明，用于界面与报告标注                         1458
        scheme       生效的权重方案对象，无自定义方案时为 None                 1459
    """                                                                     1460
    base = DEFAULT_WEIGHTS.get(unit_type)                                   1461
    if base is None:                                                        1462
        return None, "未定义该微地质单元类型的权重", None                      1463
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                1464
    e,                                                                      1465
                                                                            1466
    name=ACTIVE_NAME).first()                                               1467
    if scheme is None:                                                      1468
        return copy.deepcopy(base), "论文表 3 默认权重", None               1469
    merged = copy.deepcopy(base)                                            1470
    l1_map, l2_map = _code_maps()                                           1471
    applied = 0                                                             1472
    for item in WeightItem.query.filter_by(scheme_id=scheme.                1473
    id).all():                                                              1474
        code, w = item.indicator_code, item.weight                          1475
        if w is None or w <= 0:                                             1476
            continue                                                        1477
        if code in l1_map and l1_map[code] in merged["l1"]:                 1478
            merged["l1"][l1_map[code]] = w                                  1479
            applied += 1                                                    1480
        elif code in l2_map:                                                1481
            l1_name, l2_name = l2_map[code]                                 1482
            if l1_name in merged["l2"] and l2_name in                       1483
    merged["l2"][l1_name]:                                                  1484
                merged["l2"][l1_name][l2_name] = w                          1485
                applied += 1                                                1486
    if applied == 0:                                                        1487
        return copy.deepcopy(base), "论文表 3 默认权重", None               1488
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme       1489
def save_weights(unit_type, code_weight_pairs, note=None):                  1490
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。          1491
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）              1492
    分次录入而无需一次填完。                                                1493
    """                                                                     1494
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                1495
    e,                                                                      1496
                                                                            1497
    name=ACTIVE_NAME).first()                                               1498
    if scheme is None:                                                      1499
        scheme = WeightScheme(unit_type=unit_type,                          1500
```

```
    name=ACTIVE_NAME,                                                   1501
                                      level=1, is_default=True)         1502
        db.session.add(scheme)                                          1503
        db.session.flush()                                              1504
    if note:                                                            1505
        scheme.level = scheme.level or 1                                1506
    if not scheme.is_default:                                           1507
        scheme.is_default = True                                        1508
    written = 0                                                         1509
    for code, w in code_weight_pairs:                                   1510
        if w is None or w <= 0:                                         1511
            continue                                                    1512
        item = WeightItem.query.filter_by(scheme_id=scheme.i            1513
    d,                                                                  1514
                                                                        1515
    indicator_code=code).first()                                        1516
        if item is None:                                                1517
            item = WeightItem(scheme_id=scheme.id,                      1518
    indicator_code=code)                                                1519
            db.session.add(item)                                        1520
        item.weight = float(w)                                          1521
        written += 1                                                    1522
    db.session.commit()                                                 1523
    return scheme, written                                              1524
def reset_weights(unit_type):                                           1525
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""                    1526
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty            1527
    pe,                                                                 1528
                                                                        1529
    name=ACTIVE_NAME).all()                                             1530
    n = 0                                                               1531
    for s in schemes:                                                   1532
        WeightItem.query.filter_by(scheme_id=s.id).delete(              1533
            synchronize_session=False)                                  1534
        db.session.delete(s)                                            1535
        n += 1                                                          1536
    db.session.commit()                                                 1537
    return n                                                            1538
def scheme_items(unit_type):                                            1539
    """返回该单元类型已保存的自定义权重条目 {指标编码: 权重}。"""                    1540
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ            1541
    e,                                                                  1542
                                                                        1543
    name=ACTIVE_NAME).first()                                           1544
    if scheme is None:                                                  1545
        return {}, None                                                 1546
    items = {i.indicator_code: i.weight                                 1547
             for i in WeightItem.query.filter_by(scheme_id=s            1548
    cheme.id).all()}                                                    1549
    return items, scheme                                                1550
```

```
def summary():
    """各单元类型的权重来源概览，供权重首页展示。"""
    rows = []
    for unit_type in DEFAULT_WEIGHTS:
        weights, label, scheme = effective_weights(unit_type)
        rows.append({
            "unit_type": unit_type,
            "label": label,
            "custom": scheme is not None,
        })
    return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式（９） 实现：
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:
    [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
```

```
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    #  一级指标权重按实际参与计算的指标归一化。
    #  论文表 3 的权重本身已归一（和为 1），此处为恒等变换；
    #  若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in
    l1_weights)
    if lw_sum > 0:
        unit_score = sum(l1_weights[k] * l1_scores[k]
                         for k in l1_scores if k in
    l1_weights) / lw_sum
    else:
        unit_score = 0.0
    return {"l1_scores": l1_scores, "unit_score":
    unit_score, "details": details}
def score_mine(units, weights_map=None):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型: 权重结构}。未提供的单元类型
        使用论文表 3 的默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]))
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
```

```
                "l1_scores": r["l1_scores"],
                "details": r["details"],
            })
        grade_info = grade_of(total)
        return {
            "final_score": total,
            "grade": grade_info["grade"],
            "grade_info": grade_info,
            "unit_results": unit_results,
        }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
from app.utils.timefmt import to_local, now_utc
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
```

```
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
    return t
def build_report(patch, mine_result):
    """生成评价报告 Word 文档，返回 BytesIO。
    参数
    ----
    patch : MinePatch，图斑对象
    mine_result : evaluation_service.score_mine 的返回 dict
    """
    doc = Document()
    # 页边距
    for section in doc.sections:
        section.top_margin = Cm(2.5)
        section.bottom_margin = Cm(2.5)
        section.left_margin = Cm(2.5)
        section.right_margin = Cm(2.5)
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0",
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)
    _para(doc, "生成时间：%s" % to_local(now_utc(), "%Y-%m-%d
%H：%M：%S"),
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,
    space_after=12)
    # 图斑信息
    _heading(doc, "一、图斑基本信息", 1)
    info = [
        ("图斑名称", patch.name),
        ("所在行政区", patch.admin_region or "—"),
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is
    not None else "—"),
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
    ]
    _table(doc, ["项目", "内容"], info)
    # 评价结论
    _heading(doc, "二、评价结论", 1)
    gi = mine_result["grade_info"]
    concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
    ]
    _table(doc, ["项目", "结果"], concl)
```

```
    # 各单元得分
    _heading(doc, "三、各微地质单元得分", 1)
    rows = []
    for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"] * 100),
                     "%.4f" % u["unit_score"], "%.4f" % u["weighted"]))
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"], rows)
    # 完整得分明细
    _heading(doc, "四、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"], u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f" % d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else "%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std, w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重", "贡献"],
               detail_rows)
        # 一级指标得分
        l1_row = []
        for k, v in u["l1_scores"].items():
            l1_row.append("%s=%.4f" % (k, v))
        _para(doc, "一级指标得分：" + "，".join(l1_row), size=10.5, space_after=10)
    buf = io.BytesIO()
    doc.save(buf)
    buf.seek(0)
    return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
```

```
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
    IDEAL_PH_RANGE, False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
    "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
    str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
     "论文 4.2.2 节：参照生态系统取值 75%", False),
    ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
    "benefit", ALL_UNITS,
     "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
    # ---- 二级指标：地形 ----
    ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
    SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
     "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
    为陡坡重要分界", False),
    ("aspect", "坡向", 2, "terrain", "—",
    "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
     "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
```

```
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
    str(_limit), "cost",
        "碎石边坡,平台",
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
    _limit,
        False,
    ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg， As， Ni， Cu， Cd， Pb， Cr， Zn，
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
```

```
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
#  表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    #  示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
```

```
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
    None)
                        for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict], 每项 {unit_type, fields{...},
    heavy_metals{}}
      errors : list[str], 逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
    "、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
```

```
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
"、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
            continue
        fields, heavy = {}, {}
        ok = True
        for key, val in rec.items():
            if key in ("unit_type",):
                continue
            if key.startswith("hm_"):
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx,
key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
                s = str(val).strip() if val is not None
else ""
                if s == "":
                    fields["aspect"] = None
                elif s in ASPECT_TYPES:
                    fields["aspect"] = s
                else:
                    errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                  % (idx, s,
"、".join(ASPECT_TYPES)))
                    ok = False
                continue
            n, err = _num(val)
            if err:
                errors.append("第 %d 行：%s %s" % (idx, key,
err))
                ok = False
```

```
                else:                                                    2051
                    fields[key] = n                                      2052
            if ok:                                                       2053
                parsed.append({"unit_type": unit_type,                   2054
        "fields": fields,                                                2055
                                    "heavy_metals": heavy})              2056
        return parsed, errors                                            2057
# ==== app/services/workflow.py ====                                     2058
# -*- coding: utf-8 -*-                                                  2059
"""评价流程前置条件（定量评价与报告输出的流程门禁）。                    2060
论文 4.1 节确定的评价流程存在严格的先后关系：                            2061
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比    2062
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告             2063
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的      2064
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种      2065
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）    2066
在前，微地质单元划分在后，顺序不可颠倒。                                2067
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，              2068
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，                2069
避免在条件不具备时产出看似合理的评价结论。                              2070
"""                                                                      2071
from app.models import (db, MicroUnit, IndicatorData,                    2072
    ConstraintCheck,                                                     2073
                                EvaluationResult)                        2074
# 指标数据非空字段（用于判断某单元是否已录入数据）                       2075
DATA_FIELDS = ("soil_ratio", "thickness", "ph",                          2076
    "organic_matter",                                                    2077
                "coverage", "shannon", "slope", "aspect")                2078
def _has_data(ind):                                                      2079
    if ind is None:                                                      2080
        return False                                                     2081
    if any(getattr(ind, f, None) is not None for f in                    2082
    DATA_FIELDS):                                                        2083
        return True                                                      2084
    return bool(ind.heavy_metals_dict())                                 2085
def patch_progress(patch_id):                                            2086
    """检查某图斑的评价流程完成情况。                                    2087
    返回 dict：                                                          2088
        units          {done, count, ratio_sum, detail}                  2089
        constraint     {done, passed, detail}                            2090
        data           {done, ready, missing, detail}                    2091
        blocking       未满足的前置步骤列表（按流程顺序）                2092
        ready          是否可进入定量评价                                2093
    """                                                                  2094
    units = MicroUnit.query.filter_by(patch_id=patch_id).all             2095
    ()                                                                   2096
    ratio_sum = sum(u.ratio or 0.0 for u in units)                       2097
    # ---- 步骤一：微地质单元 ----                                       2098
    units_step = {                                                       2099
        "done": bool(units),                                             2100
```

```
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    # ---- 步骤二：限制性条件判定 ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)
    \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
                           "detail": "限制性条件判定未通过（%s）" %
    (cc.reason or "")}
    # ---- 步骤三：指标数据 ----
    missing = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        if not _has_data(ind):
            missing.append(u.unit_type)
    if not units:
        data_step = {"done": False, "ready": False,
    "missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
    "missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    else:
        data_step = {"done": True, "ready": True,
    "missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
    len(units)}
    ratio_ok = ratio_sum > 0.999 if units else False
    if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 １００%，请检查面积录入）"
    # ---- 汇总未满足的前置条件（按论文 ４．１ 节的流程顺序：限制性判定优先） ----
    # reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
    # 缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
    blocking = []
    if not constraint_step["done"]:
        blocking.append({
```

```
                "step": "constraint", "label": "限制性条件判定未执行",
                "reason": "五项限制性条件尚未逐项判定。",
                "fix_url": "/constraint/%d" % patch_id,
        "fix_label": "去执行判定",
            })
        elif not constraint_step["passed"]:
            blocking.append({
                "step": "constraint", "label": "限制性条件判定未通过",
                "reason": cc.reason or "",
                "fix_url": "/constraint/%d" % patch_id,
        "fix_label": "去复核判定结果",
            })
        if not units_step["done"]:
            blocking.append({
                "step": "units", "label": "未划分微地质单元",
                "reason": "该图斑下没有任何微地质单元。",
                "fix_url": "/unit/%d" % patch_id, "fix_label":
        "去划分",
            })
        elif not ratio_ok:
            blocking.append({
                "step": "units", "label": "面积占比合计不为 100%",
                "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
                "fix_url": "/unit/%d" % patch_id, "fix_label":
        "去核对面积",
            })
        if units_step["done"] and not data_step["ready"]:
            blocking.append({
                "step": "data", "label": "指标数据未录入完整",
                "reason": data_step["detail"],
                "fix_url": "/indicator/", "fix_label": "去录入",
            })
        return {
            "units": units_step,
            "constraint": constraint_step,
            "data": data_step,
            "blocking": blocking,
            "ready": not blocking,
        }
def latest_result(patch_id):
    """取该图斑最近一次的评价记录。"""
    return EvaluationResult.query.filter_by(patch_id=patch_i
    d) \
        .order_by(EvaluationResult.id.desc()).first()
def report_progress(patch_id):
    """报告输出的前置条件：评价流程齐备，且已有评价记录。
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
    导出时读取已保存的评价结果，而不是在导出环节重新计算。
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
    同一份评价出现两个数字，结论无法追溯。
```

```
    返回 dict：{ready, blocking, result, flow}
        result 为最近一次评价记录（EvaluationResult），无记录时为空。
    """
    flow = patch_progress(patch_id)
    blocking = list(flow["blocking"])
    record = latest_result(patch_id)
    if not blocking and record is None:
        blocking.append({
            "step": "evaluate", "label": "尚未执行适宜性评价",
            "reason": "该图斑还没有评价记录。",
            "fix_url": "/evaluate/%d" % patch_id,
            "fix_label": "去执行评价",
        })
    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
#  ==== app/services/dashboard.py ====
#  -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
          href="/project/",
          desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
          href="/standard/",
          desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
    dict(key="constraint", no="03", title="限制性指标判定",
    icon="shield",
          href="/constraint/",
```

```
            desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),
        dict(key="unit", no="04", title="微地质单元", icon="layers",
            href="/unit/",
            desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
        dict(key="ahp", no="05", title="指标体系与权重",
        icon="sliders",
            href="/ahp/",
            desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
        dict(key="indicator", no="06", title="指标数据采集",
        icon="database",
            href="/indicator/",
            desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
        dict(key="evaluate", no="07", title="适宜性定量评价",
        icon="target",
            href="/evaluate/",
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
        dict(key="report", no="08", title="报告与结果输出",
        icon="file",
            href="/report/",
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
    ]
def _has_data(ind):
    return wf._has_data(ind)
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）
MODULES = [
    ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),
]
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
```

```
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
        MicroUnit.patch_id.in_(patch_ids)).all() if
patch_ids else []
    unit_by_patch = {}
    for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
    unit_ids = [u.id for u in units]
    data_unit_ids = set()
    if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
        for r in EvaluationResult.query.filter(
                EvaluationResult.patch_id.in_(patch_ids)).al
l():
            evaluated.add(r.patch_id)
    # ---- 各步骤的待办数量 ----
    pending = {
        "constraint": sum(1 for p in patches if p.id not in
passed_constraint),
        "unit": sum(1 for p in patches if not
unit_by_patch.get(p.id)),
        "indicator": 0,
        "evaluate": 0,
    }
    ready_patches = 0
    for p in patches:
        us = unit_by_patch.get(p.id, [])
        if us and any(u.id not in data_unit_ids for u in
us):
            pending["indicator"] += 1
        pr = wf.patch_progress(p.id)
        if pr["ready"]:
            ready_patches += 1
            if p.id not in evaluated:
                pending["evaluate"] += 1
    # ---- 步骤可用性：前置数据不具备时置灰 ----
```

```
    avail = {
        "project": True,
        "standard": True,
        "constraint": bool(patches),
        "unit": bool(patches),
        "ahp": True,
        "indicator": bool(units),
        "evaluate": ready_patches > 0,
        "report": bool(evaluated),
    }
    notes = {
        "constraint": "需先建立矿山图斑",
        "unit": "需先建立矿山图斑",
        "indicator": "需先划分微地质单元",
        "evaluate": "需前置步骤齐备的图斑",
        "report": "需先执行适宜性评价",
    }
    steps = []
    for s in STEPS:
        steps.append(dict(
            s,
            available=avail[s["key"]],
            pending=pending.get(s["key"], 0),
            note=None if avail[s["key"]] else
notes.get(s["key"], ""),
        ))
    unit_types = len({u.unit_type for u in units})
    stats = {
        "projects": projects,
        "patches": len(patches),
        "unit_types": unit_types,
        "standards": Standard.query.count(),
    }
    return {
        "stats": stats,
        "steps": steps,
        "badges": {k: v for k, v in pending.items() if v},
        "progress": {
            "patches": len(patches),
            "units": len(units),
            "data_units": len(data_unit_ids),
            "ready": ready_patches,
            "evaluated": len(evaluated),
        },
    }
def nav_badges():
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
    snap = snapshot()
    return {
        "/constraint/": snap["badges"].get("constraint", 0),
```

```
            "/unit/": snap["badges"].get("unit", 0),
            "/indicator/": snap["badges"].get("indicator", 0),
            "/evaluate/": snap["badges"].get("evaluate", 0),
        }
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.register": "注册账号",
    "auth.logout": "退出登录",
    "auth.add_user": "新增账号",
    "auth.edit_user": "编辑账号",
    "auth.delete_user": "删除账号",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
```

```
#  无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    #  个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
        ))
        db.session.commit()
    except Exception:
        #  审计失败不应影响业务响应
        db.session.rollback()
#  ==== app/views/auth.py ====
#  -*- coding: utf-8 -*-
"""账号与权限 —— 登录、注册、个人资料、用户维护、操作日志。
页面组织：
  /login          登录页（含「点我注册」入口）
  /register       自助注册（默认「操作员」）
  /profile        个人设置，分类切换：info / security / users / logs
```

```
                后两个分类仅管理员可见，由本模块统一管控。
"""
from flask import (Blueprint, render_template, redirect,
    url_for, request,
                   flash, abort, current_app)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import (LoginForm, EmptyForm, ProfileForm,
    PasswordForm,
                       RegisterForm, email_ok)
from app.models import db, User, OperationLog
from app.utils import uploads as up
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE,
    GENDERS
bp = Blueprint("auth", __name__)
# 「个人设置」的四个分类；后两个是管理员专属
PROFILE_TABS = ("info", "security", "users", "logs")
ADMIN_TABS = ("users", "logs")
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
def _intro_data():
    """登录页下方的系统介绍所用的静态内容。"""
    from app.services import dashboard as dash
    return {"modules": dash.MODULES}
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html",
    dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "toast-error")
            else:
                login_user(user, remember=form.remember.data
    )
```

```
                    #  用  toast  类别：渲染为页面中上方的浮层，淡入停留后淡出。
                    #  登录成功只是告知性反馈，不需要用户停下来阅读。
                    flash("登录成功", "toast")
                    return redirect(url_for("auth.index"))
                else:
                    flash("账户名称或密码错误", "toast-error")
        #  字段为空之类的校验失败，错误留在表单里逐项显示，不走浮层：
        #  那类消息要对着输入框看，一闪而过反而让人以为「点了没反应」。
        return render_template("auth/login.html", form=form,
        **_intro_data())
@bp.route("/register", methods=["GET", "POST"])
def register():
        """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""
        if current_user.is_authenticated:
            return redirect(url_for("auth.index"))
        form = RegisterForm()
        if form.validate_on_submit():
            username = form.username.data.strip()
            if User.query.filter_by(username=username).first():
                form.username.errors.append("该账户名称已被使用")
            else:
                user = User(username=username,
                            role=DEFAULT_ROLE,
                            password_hash=generate_password_hash
        (form.password.data),
                            display_name=(form.display_name.data
        or "").strip() or None,
                            gender=(form.gender.data or
        "").strip() or None,
                            email=(form.email.data or
        "").strip() or None,
                            phone=(form.phone.data or
        "").strip() or None)
                rel, err = _save_avatar(request.files.get("avata
        r"))
                if err:
                    flash(err, "error")
                else:
                    user.avatar = rel
                    db.session.add(user)
                    db.session.commit()
                    login_user(user)
                    flash("注册成功，欢迎使用", "toast")
                    return redirect(url_for("auth.index"))
        return render_template("auth/register.html", form=form)
@bp.route("/logout")
@login_required
def logout():
        logout_user()
        flash("已退出登录", "ok")
```

```
    return redirect(url_for("auth.login"))
# ---------------------------- 个人设置 ----------------------------
    ----
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。
@bp.route("/profile", methods=["GET", "POST"])
@login_required
def profile():
    is_admin = current_user.role == "admin"
    tab = request.values.get("tab") or "info"
    if tab not in PROFILE_TABS or (tab in ADMIN_TABS and
    not is_admin):
        tab = "info"
    # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名
    info_form = ProfileForm(prefix="info", obj=current_user)
    pwd_form = PasswordForm(prefix="pwd")
    if request.method == "POST":
        action = request.form.get("action")
        # 校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，
        # 否则错误提示会出现在另一个分类里，用户根本看不到。
        if action == "password":
            tab = "security"
        elif action == "info":
            tab = "info"
        if action == "info" and info_form.validate_on_submit
    ():
            current_user.display_name =
    (info_form.display_name.data or "").strip() or None
            current_user.gender = (info_form.gender.data or
    "").strip() or None
            current_user.email = (info_form.email.data or
    "").strip() or None
            current_user.phone = (info_form.phone.data or
    "").strip() or None
            db.session.commit()
            flash("个人资料已保存", "ok")
            return redirect(url_for("auth.profile",
    tab="info"))
        if action == "password" and pwd_form.validate_on_sub
    mit():
            if not check_password_hash(current_user.password
    _hash,

    pwd_form.old_password.data):
                pwd_form.old_password.errors.append("当前密码不正确
    ")
            else:
                current_user.password_hash =
    generate_password_hash(
                    pwd_form.new_password.data)
```

```
                db.session.commit()                                    2651
                flash("密码已修改", "ok")                               2652
                return redirect(url_for("auth.profile",                2653
    tab="security"))                                                   2654
    ctx = dict(info_form=info_form, pwd_form=pwd_form,                 2655
    avatar_form=EmptyForm(),                                           2656
                tab=tab, role_labels=ROLE_LABELS,                      2657
    genders=GENDERS)                                                   2658
    if tab == "users":                                                 2659
        ctx.update(_users_context())                                   2660
    elif tab == "logs":                                                2661
        ctx.update(_logs_context())                                    2662
    return render_template("auth/profile.html", **ctx)                 2663
def _users_context():                                                  2664
    """管理员分类「账号」所需的数据。"""                                  2665
    return {"users": User.query.order_by(User.id).all(),               2666
            "form": EmptyForm()}                                       2667
def _logs_context():                                                   2668
    """管理员分类「操作日志」所需的数据。"""                              2669
    page = request.args.get("page", 1, type=int)                       2670
    per_page = 50                                                      2671
    q = OperationLog.query.order_by(OperationLog.id.desc())            2672
    total = q.count()                                                  2673
    records = q.offset((page - 1) * per_page).limit(per_page2674
    ).all()                                                            2675
    users_map = {u.id: u.username for u in User.query.all()}2676
    pages = max(1, (total + per_page - 1) // per_page)                 2677
    return {"records": records, "page": page, "pages":                 2678
    pages,                                                             2679
            "total": total, "users_map": users_map}                    2680
# ---------------------------- 头像 ----------------------------2681
    --                                                                 2682
def _avatar_length_ok(file_storage):                                   2683
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""  2684
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *  2685
    1024 * 1024)                                                       2686
    stream = file_storage.stream                                       2687
    pos = stream.tell()                                                2688
    stream.seek(0, 2)                                                  2689
    size = stream.tell()                                               2690
    stream.seek(pos)                                                   2691
    return size <= limit, size                                         2692
def _save_avatar(file):                                                2693
    """保存头像文件，返回 (相对路径, 错误信息)。没有文件时返回 (None, None)。"""  2694
    if file is None or not file.filename:                              2695
        return None, None                                              2696
    ok, size = _avatar_length_ok(file)                                 2697
    if not ok:                                                         2698
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"             2699
                      % (current_app.config["MAX_AVATAR_LENG2700
```

```
    TH"] / 1024 / 1024,                                                    2701
                                size / 1024 / 1024))                       2702
    return up.save_upload(file, "avatars",                                 2703
                                current_app.config["ALLOWED_AVATAR         2704
    _EXT"],                                                                2705
                                current_app.config["UPLOAD_FOLDER"         2706
    ])                                                                     2707
@bp.route("/profile/avatar", methods=["POST"])                             2708
@login_required                                                            2709
def upload_avatar():                                                       2710
    form = EmptyForm()                                                     2711
    back = url_for("auth.profile", tab=request.form.get("tab               2712
    ") or "info")                                                          2713
    if not form.validate_on_submit():                                      2714
        flash("表单校验失败，请重试", "error")                                  2715
        return redirect(back)                                              2716
    rel, err = _save_avatar(request.files.get("avatar"))                   2717
    if err:                                                                2718
        flash(err, "error")                                                2719
        return redirect(back)                                              2720
    if rel is None:                                                        2721
        flash("请选择头像图片", "error")                                       2722
        return redirect(back)                                              2723
    # 换新头像时删掉旧文件，避免残留                                                2724
    up.remove_upload(current_user.avatar,                                  2725
    current_app.config["UPLOAD_FOLDER"])                                   2726
    current_user.avatar = rel                                              2727
    db.session.commit()                                                    2728
    flash("头像已更新", "ok")                                                 2729
    return redirect(back)                                                  2730
@bp.route("/profile/avatar/delete", methods=["POST"])                      2731
@login_required                                                            2732
def delete_avatar():                                                       2733
    form = EmptyForm()                                                     2734
    back = url_for("auth.profile", tab=request.form.get("tab               2735
    ") or "info")                                                          2736
    if form.validate_on_submit():                                          2737
        up.remove_upload(current_user.avatar,                              2738
    current_app.config["UPLOAD_FOLDER"])                                   2739
        current_user.avatar = None                                         2740
        db.session.commit()                                                2741
        flash("头像已移除", "ok")                                             2742
    return redirect(back)                                                  2743
# ---------------------------- 账号维护（仅管理员）                                2744
    ----------------------------                                           2745
@bp.route("/users")                                                        2746
@login_required                                                            2747
def users():                                                               2748
    """旧地址保留，统一并入「个人设置 · 账号」。"""                                      2749
    return redirect(url_for("auth.profile", tab="users"))                  2750
```

```
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", DEFAULT_ROLE)
        if role not in ROLE_LABELS:
            role = DEFAULT_ROLE
        if not username or not password:
            flash("账户名称和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("账户名称已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(
            username=username, role=role,
            password_hash=generate_password_hash(password),
            display_name=request.form.get("display_name",
    "").strip() or None,
            gender=request.form.get("gender", "").strip()
    or None,
            email=request.form.get("email", "").strip() or
    None,
            phone=request.form.get("phone", "").strip() or
    None))
        db.session.commit()
        flash("账号已添加", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/add_user.html",
    role_labels=ROLE_LABELS,
                           genders=GENDERS,
    default_role=DEFAULT_ROLE,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        if role not in ROLE_LABELS:
            role = user.role
        active = request.form.get("active") == "1"
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
```

```
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,
                                       User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
user_id=user.id))
        # 个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
user_id=user.id))
        user.role = role
        user.active = active
        user.display_name = request.form.get("display_name",
"").strip() or None
        user.gender = request.form.get("gender",
"").strip() or None
        user.email = email or None
        user.phone = request.form.get("phone", "").strip()
or None
        new_password = request.form.get("password",
"").strip()
        if new_password:
            user.password_hash = generate_password_hash(new_
password)
        db.session.commit()
        flash("账号信息已保存", "ok")
        return redirect(url_for("auth.profile",
tab="users"))
    return render_template("auth/edit_user.html", user=user,
                           role_labels=ROLE_LABELS,
genders=GENDERS,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/delete", methods=["POST"])
@login_required
def delete_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if user.id == current_user.id:
        flash("不能删除当前登录的账号", "error")
        return redirect(url_for("auth.profile",
tab="users"))
    if user.role == "admin":
        others = User.query.filter(User.role == "admin",
                                   User.id != user.id,
                                   User.active.is_(True)).co
unt()
```

```
        if others == 0:
            flash("系统至少需保留一名启用状态的管理员", "error")
            return redirect(url_for("auth.profile",
    tab="users"))
    db.session.delete(user)
    db.session.commit()
    flash("账号「%s」已删除" % user.username, "ok")
    return redirect(url_for("auth.profile", tab="users"))
# ---------------------------- 操作日志（仅管理员）
    ----------------------------
@bp.route("/logs")
@login_required
def logs():
    """旧地址保留，统一并入「个人设置 · 操作日志」。"""
    return redirect(url_for("auth.profile", tab="logs"))
# ==== app/views/project.py ====
# -*- coding: utf-8 -*-
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""
from datetime import datetime
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
    "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
```

```
                flash("项目名称不能为空", "error")                                2901
            else:                                                                 2902
                db.session.add(p)                                                 2903
                db.session.commit()                                               2904
                flash("项目已创建", "ok")                                          2905
                return redirect(url_for("project.index"))                         2906
        return render_template("project/edit.html", form=form,                    2907
        project=None)                                                             2908
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])                             2909
@login_required                                                                   2910
def edit(pid):                                                                    2911
        p = db.get_or_404(Project, pid)                                           2912
        form = EmptyForm()                                                        2913
        if request.method == "POST" and form.validate_on_submit(                  2914
        ):                                                                        2915
            p.name = request.form.get("name", "").strip()                         2916
            p.code = request.form.get("code", "").strip() or                      2917
        None                                                                      2918
            p.description = request.form.get("description",                       2919
        "").strip() or None                                                       2920
            p.status = request.form.get("status", "进行中")                        2921
            db.session.commit()                                                   2922
            flash("项目已保存", "ok")                                              2923
            return redirect(url_for("project.index"))                             2924
        return render_template("project/edit.html", form=form,                    2925
        project=p)                                                                2926
@bp.route("/<int:pid>/delete", methods=["POST"])                                  2927
@login_required                                                                   2928
def delete(pid):                                                                  2929
        p = db.get_or_404(Project, pid)                                           2930
        db.session.delete(p)                                                      2931
        db.session.commit()                                                       2932
        flash("项目已删除", "ok")                                                  2933
        return redirect(url_for("project.index"))                                 2934
# ------------------------------ 矿山图斑 ------------------------------2935
        ----                                                                      2936
@bp.route("/<int:pid>/patches")                                                   2937
@login_required                                                                   2938
def patches(pid):                                                                 2939
        p = db.get_or_404(Project, pid)                                           2940
        return render_template("project/patches.html",                            2941
        project=p,                                                                2942
                                    form=EmptyForm())                             2943
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])                       2944
@login_required                                                                   2945
def patch_delete(patch_id):                                                       2946
        """删除矿山图斑，及其微地质单元与指标数据。"""                                    2947
        from app.models import MicroUnit, IndicatorData,                          2948
        EvaluationResult, \                                                       2949
            ScoreDetail, ConstraintCheck, ReferenceEcosystem                      2950
```

```
    patch = db.get_or_404(MinePatch, patch_id)
    pid = patch.project_id
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patch_id=patch.id).all()]
    if unit_ids:
        IndicatorData.query.filter(IndicatorData.unit_id.in_(unit_ids)).delete(
            synchronize_session=False)
    for r in EvaluationResult.query.filter_by(patch_id=patch.id).all():
        ScoreDetail.query.filter_by(result_id=r.id).delete(
            synchronize_session=False)
    EvaluationResult.query.filter_by(patch_id=patch.id).delete(
        synchronize_session=False)
    ConstraintCheck.query.filter_by(patch_id=patch.id).delete(
        synchronize_session=False)
    MicroUnit.query.filter_by(patch_id=patch.id).delete(synchronize_session=False)
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).delete(
        synchronize_session=False)
    db.session.delete(patch)
    db.session.commit()
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")
    return redirect(url_for("project.patches", pid=pid))

@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
```

```
        else:                                                    3001
            db.session.add(patch)                                3002
            db.session.flush()                                   3003
            #  参照生态系统默认取论文值                          3004
            db.session.add(ReferenceEcosystem(                   3005
                patch_id=patch.id, coverage_benchmark=75.0,      3006
    shannon_benchmark=0.7))                                      3007
            db.session.commit()                                  3008
            flash("图斑已创建", "ok")                            3009
            return redirect(url_for("project.patches",           3010
    pid=p.id))                                                   3011
    return render_template("project/patch_edit.html",            3012
    form=form, project=p, patch=None)                            3013
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",          3014
    "POST"])                                                     3015
@login_required                                                  3016
def patch_edit(patch_id):                                        3017
    patch = db.get_or_404(MinePatch, patch_id)                   3018
    form = EmptyForm()                                           3019
    if request.method == "POST" and form.validate_on_submit(     3020
    ):                                                           3021
        def fv(name, cast=str):                                  3022
            v = request.form.get(name, "").strip()               3023
            return cast(v) if v != "" else None                  3024
        patch.name = request.form.get("name", "").strip()        3025
        patch.admin_region = fv("admin_region")                  3026
        patch.area_ha = fv("area_ha", float)                     3027
        patch.mineral_type = fv("mineral_type")                  3028
        patch.mining_method = fv("mining_method")                3029
        patch.stop_years = fv("stop_years", float)               3030
        patch.responsible_entity_missing =                       3031
    bool(request.form.get("responsible_entity_missing"))         3032
        #  参照生态系统                                         3033
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa     3034
    tch.id).first()                                              3035
        if ref is None:                                          3036
            ref = ReferenceEcosystem(patch_id=patch.id)          3037
            db.session.add(ref)                                  3038
        ref.coverage_benchmark = fv("coverage_benchmark",        3039
    float) or 75.0                                               3040
        ref.shannon_benchmark = fv("shannon_benchmark",          3041
    float) or 0.7                                                3042
        ref.dominant_species = fv("dominant_species")            3043
        patch.image_note = fv("image_note")                      3044
        db.session.commit()                                      3045
        flash("图斑已保存", "ok")                                3046
        return redirect(url_for("project.patches",               3047
    pid=patch.project_id))                                       3048
    return render_template("project/patch_edit.html",            3049
    form=form,                                                   3050
```

```
                                project=patch.project,                3051
    patch=patch,                                                      3052
                                reference=ReferenceEcosystem.quer     3053
    y.filter_by(                                                      3054
                                    patch_id=patch.id).first())       3055
# ---------------------------- 遥感影像挂接                           3056
    ----------------------------                                      3057
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])            3058
@login_required                                                       3059
def upload_image(patch_id):                                           3060
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                  3061
    from app.utils import uploads as up                               3062
    patch = db.get_or_404(MinePatch, patch_id)                        3063
    form = EmptyForm()                                                3064
    if not form.validate_on_submit():                                 3065
        flash("表单校验失败，请重试", "error")                        3066
        return redirect(url_for("project.patch_edit",                 3067
    patch_id=patch.id))                                               3068
    rel, err = up.save_upload(request.files.get("image"),             3069
    "patches",                                                        3070
                              current_app.config["ALLOWED_IM          3071
    AGE_EXT"],                                                        3072
                              current_app.config["UPLOAD_FOL          3073
    DER"])                                                            3074
    if err:                                                           3075
        flash(err, "error")                                           3076
        return redirect(url_for("project.patch_edit",                 3077
    patch_id=patch.id))                                               3078
    # 替换旧影像时清理原文件，避免残留                                3079
    if patch.image_path:                                              3080
        up.remove_upload(patch.image_path,                            3081
    current_app.config["UPLOAD_FOLDER"])                              3082
    patch.image_path = rel                                            3083
    patch.image_uploaded_at = datetime.utcnow()                       3084
    note = request.form.get("image_note", "").strip()                 3085
    if note:                                                          3086
        patch.image_note = note                                       3087
    db.session.commit()                                               3088
    flash("遥感影像已挂接", "ok")                                     3089
    return redirect(url_for("project.patch_edit",                     3090
    patch_id=patch.id))                                               3091
@bp.route("/patch/<int:patch_id>/image/delete",                       3092
    methods=["POST"])                                                 3093
@login_required                                                       3094
def delete_image(patch_id):                                           3095
    from app.utils import uploads as up                               3096
    patch = db.get_or_404(MinePatch, patch_id)                        3097
    form = EmptyForm()                                                3098
    if form.validate_on_submit():                                     3099
        up.remove_upload(patch.image_path,                            3100
```

```
    current_app.config["UPLOAD_FOLDER"])
        patch.image_path = None
        patch.image_uploaded_at = None
        db.session.commit()
        flash("遥感影像已移除", "ok")
    return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import UNIT_TYPES
bp = Blueprint("unit", __name__, url_prefix="/unit")
def recalc_ratio(patch_id):
    """按面积重算该图斑各单元面积占比。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    if total > 0:
        for u in units:
            u.ratio = round((u.area or 0.0) / total, 6)
    return total
@bp.route("/")
@login_required
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all
    ()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
```

```
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        new_type = request.form.get("unit_type",
    u.unit_type)
```

```
        if new_type not in UNIT_TYPES:                                  3201
            flash("微地质单元类型不正确", "error")                        3202
            return redirect(url_for("unit.edit",                        3203
    unit_id=u.id))                                                      3204
        # 改类型时同样不允许与本图斑已有的其他单元重复                    3205
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,            3206
                                                                        3207
    unit_type=new_type).first()                                         3208
        if dup is not None and dup.id != u.id:                          3209
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,    3210
    "error")                                                            3211
            return redirect(url_for("unit.edit",                        3212
    unit_id=u.id))                                                      3213
        u.unit_type = new_type                                          3214
        area = request.form.get("area", "").strip()                     3215
        u.area = float(area) if area else None                          3216
        u.description = request.form.get("description",                 3217
    "").strip() or None                                                 3218
        recalc_ratio(u.patch_id)                                        3219
        db.session.commit()                                             3220
        flash("微地质单元已保存", "ok")                                  3221
        return redirect(url_for("unit.index",                           3222
    patch_id=u.patch_id))                                               3223
    return render_template("unit/edit.html", unit=u,                    3224
    patch=u.patch,                                                      3225
                                unit_types=UNIT_TYPES, form=form)       3226
@bp.route("/delete/<int:unit_id>", methods=["POST"])                    3227
@login_required                                                         3228
def delete(unit_id):                                                    3229
    u = db.get_or_404(MicroUnit, unit_id)                               3230
    pid = u.patch_id                                                    3231
    name = u.unit_type                                                  3232
    # 一并删除该单元的指标数据，避免残留                                3233
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch            3234
    ronize_session=False)                                               3235
    db.session.delete(u)                                                3236
    db.session.flush()                                                  3237
    recalc_ratio(pid)                                                   3238
    db.session.commit()                                                 3239
    flash("微地质单元「%s」已删除" % name, "ok")                         3240
    return redirect(url_for("unit.index", patch_id=pid))                3241
# ==== app/views/standard.py ====                                       3242
# -*- coding: utf-8 -*-                                                 3243
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""      3244
from flask import Blueprint, render_template, request,                  3245
    redirect, url_for, flash                                            3246
from flask_login import login_required                                  3247
from app.forms import EmptyForm                                         3248
from app.models import db, Standard, StandardRule                       3249
bp = Blueprint("standard", __name__, url_prefix="/standard")            3250
```

```
@bp.route("/")
@login_required
def index():
    standards = Standard.query.order_by(Standard.category,
    Standard.province).all()
    local = [s for s in standards if s.category == "地方标准"]
    policy = [s for s in standards if s.category != "地方标准"]
    return render_template("standard/index.html",
    local=local, policy=policy,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s = Standard(
            province=request.form.get("province",
    "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
        else:
            db.session.add(s)
            db.session.commit()
            flash("标准已添加", "ok")
            return redirect(url_for("standard.index"))
    return render_template("standard/edit.html", form=form,
    standard=None)
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])
@login_required
def edit(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s.province = request.form.get("province",
    "").strip()
        s.name = request.form.get("name", "").strip()
        s.doc_number = request.form.get("doc_number",
```

```
        "").strip() or None                                              3301
            s.publish_date = request.form.get("publish_date",            3302
        "").strip() or None                                              3303
            s.implement_date = request.form.get("implement_date"         3304
        , "").strip() or None                                            3305
            s.category = request.form.get("category", "地方标准")          3306
            s.summary = request.form.get("summary", "").strip()          3307
        or None                                                          3308
            db.session.commit()                                          3309
            flash("已保存", "ok")                                          3310
            return redirect(url_for("standard.rules", sid=s.id))         3311
        return render_template("standard/edit.html", form=form,          3312
        standard=s)                                                      3313
@bp.route("/<int:sid>/delete", methods=["POST"])                         3314
@login_required                                                          3315
def delete(sid):                                                         3316
        s = db.get_or_404(Standard, sid)                                 3317
        StandardRule.query.filter_by(standard_id=sid).delete()           3318
        db.session.delete(s)                                             3319
        db.session.commit()                                              3320
        flash("已删除", "ok")                                              3321
        return redirect(url_for("standard.index"))                       3322
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])                   3323
@login_required                                                          3324
def rules(sid):                                                          3325
        s = db.get_or_404(Standard, sid)                                 3326
        form = EmptyForm()                                               3327
        if request.method == "POST" and form.validate_on_submit(         3328
        ):                                                               3329
            r = StandardRule(                                            3330
                standard_id=s.id,                                        3331
                item=request.form.get("item", "").strip(),               3332
                operator=request.form.get("operator", ">="),             3333
                threshold=request.form.get("threshold",                  3334
        "").strip() or None,                                             3335
                condition_type=request.form.get("condition_type"         3336
        , "必要条件"),                                                     3337
                region=request.form.get("region", "").strip()            3338
        or None,                                                         3339
                note=request.form.get("note", "").strip() or             3340
        None,                                                            3341
            )                                                            3342
            if not r.item:                                               3343
                flash("判定项不能为空", "error")                             3344
            else:                                                        3345
                db.session.add(r)                                        3346
                db.session.commit()                                      3347
                flash("规则已添加", "ok")                                    3348
            return redirect(url_for("standard.rules", sid=s.id))         3349
        rs = StandardRule.query.filter_by(standard_id=s.id).all(         3350
```

```
    )
    return render_template("standard/rules.html",
    form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
```

```
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch_id=p.id) \
            .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit():
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
        for field, _ in FIELDS:
            setattr(record, field, request.form.get(field) == "1")
        record.passed = all(getattr(record, f) for f, _ in FIELDS)
        failed = [name for f, name in FIELDS if not getattr(record, f)]
        record.reason = ("全部通过" if record.passed
                         else "未通过：" + "、".join(failed))
        db.session.commit()
        flash("判定结果已保存", "ok")
        return redirect(url_for("constraint.check", patch_id=patch.id))
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
    field_states = [(f, name, None if record is None else getattr(record, f))
                    for f, name in FIELDS]
    return render_template("constraint/check.html",
    patch=patch, form=form,
                           fields=FIELDS,
    field_states=field_states,
                           record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
```

```
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
# 一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 (指标名称列表, 默认权重列表)。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
```

```
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
```

```
range(n)] for i in range(n)]
def compute(upper):
    A = ahp.matrix_from_upper(upper)
    r = ahp.weights_and_consistency(A)
    bad = ahp.find_most_inconsistent(A)
    return {
        "weights": [round(v, 4) for v in r["weights"]],
        "lambda_max": round(r["lambda_max"], 4),
        "ci": r["ci"],
        "cr": r["cr"],
        "consistent": r["consistent"],
        "matrix": [[round(A[i, j], 4) for j in
range(n)] for i in range(n)],
        "most_inconsistent": bad,
    }
result = None
if request.method == "POST" and form.validate_on_submit(
):
    action = request.form.get("action", "calc")
    try:
        upper = [float(request.form.get("m_%d_%d" % (i,
j), "1"))
                 for i in range(n) for j in range(i +
1, n)]
        result = compute(upper)
    except (ValueError, ZeroDivisionError):
        flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
"error")
        return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                level=level, l1=l1))
    if action == "save":
        if not result["consistent"]:
            flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                  "请调整后再保存" % result["cr"], "error")
        else:
            pairs = to_code_pairs(unit_type, level, l1,
names,
                                  result["weights"])
            scheme, cnt = ws.save_weights(unit_type,
pairs)
            flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                  "此后该单元类型的评价将以此权重计算。"
                  % (unit_type, cnt, result["cr"]),
"ok")
            return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                    level=level, l1=l1))
elif request.values.get("calc") == "1":
    # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性
```

```
        result = compute([default[i][j] for i in range(n)
    for j in range(i + 1, n)])
    _, source_label, custom_scheme = ws.effective_weights(un
    it_type)
    return render_template(
        "ahp/matrix.html",
        form=form, unit_type=unit_type, level=level, l1=l1,
        names=names, vec=vec, default=default,
    result=result,
        source_label=source_label, custom_scheme=custom_sche
    me,
    )
@bp.route("/weights/reset", methods=["POST"])
def weights_reset():
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""
    unit_type = request.form.get("unit_type")
    form = EmptyForm()
    if form.validate_on_submit():
        if unit_type not in DEFAULT_WEIGHTS:
            flash("无效的微地质单元类型", "error")
        else:
            n = ws.reset_weights(unit_type)
            if n:
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,
    "ok")
            else:
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,
    "ok")
    return redirect(url_for("ahp.index"))
# ------------------------------ 指标维护 ------------------------------
    ----
@bp.route("/indicators")
def indicators():
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""
    if Indicator.query.count() == 0:
        ind_svc.seed_indicators()
    all_ind = Indicator.query.order_by(Indicator.level,
    Indicator.id).all()
    by_parent = {}
    for i in all_ind:
        by_parent.setdefault(i.parent_code, []).append(i)
    return render_template("ahp/indicators.html",
    by_parent=by_parent,
                           all_ind=all_ind,
    unit_types=UNIT_TYPES,
                           type_label=ind_svc.ideal_type_lab
    el,
                           form=EmptyForm())
@bp.route("/indicators/<code>/edit", methods=["GET",
    "POST"])
```

```
def indicator_edit(code):                                                3651
    ind = Indicator.query.filter_by(code=code).first_or_404(             3652
    )                                                                    3653
    form = EmptyForm()                                                   3654
    if request.method == "POST" and form.validate_on_submit(             3655
    ):                                                                   3656
        name = request.form.get("name", "").strip()                      3657
        if not name:                                                     3658
            flash("指标名称不能为空", "error")                              3659
            return redirect(url_for("ahp.indicator_edit",                3660
    code=code))                                                          3661
        ind.name = name                                                  3662
        ind.unit = request.form.get("unit", "").strip() or               3663
    None                                                                 3664
        ind.ideal_value = request.form.get("ideal_value",                3665
    "").strip() or None                                                  3666
        ind.ideal_type = request.form.get("ideal_type") or               3667
    ind.ideal_type                                                       3668
        units = request.form.getlist("enabled_units")                    3669
        ind.enabled_units = ",".join(units) if units else                3670
    None                                                                 3671
        ind.is_constraint = request.form.get("is_constraint"             3672
    ) == "1"                                                             3673
        ind.description = request.form.get("description",                3674
    "").strip() or None                                                  3675
        db.session.commit()                                              3676
        flash("指标「%s」已保存" % ind.name, "ok")                          3677
        return redirect(url_for("ahp.indicators"))                       3678
    return render_template("ahp/indicator_edit.html",                    3679
    ind=ind,                                                             3680
                           unit_types=UNIT_TYPES,                        3681
                           type_label=ind_svc.ideal_type_lab             3682
    el, form=form)                                                       3683
@bp.route("/indicators/reset", methods=["POST"])                         3684
def indicators_reset():                                                  3685
    """按论文取值重置指标定义。"""                                              3686
    form = EmptyForm()                                                   3687
    if form.validate_on_submit():                                        3688
        n = ind_svc.seed_indicators()                                    3689
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")                        3690
    return redirect(url_for("ahp.indicators"))                           3691
# ---------------------------- 多专家矩阵 ------------------------------3692
    -----                                                                3693
def expert_scope():                                                      3694
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""                              3695
    unit_type = request.values.get("unit_type") or                       3696
    UNIT_TYPES[0]                                                        3697
    if unit_type not in DEFAULT_WEIGHTS:                                 3698
        unit_type = UNIT_TYPES[0]                                        3699
    level = int(request.values.get("level", 1))                          3700
```

```
    l1 = normalize_l1(unit_type, level,                                3701
    request.values.get("l1"))                                          3702
    return unit_type, level, l1                                        3703
def expert_query(unit_type, level, l1):                                3704
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,              3705
    level=level)                                                       3706
    if level == 2:                                                     3707
        q = q.filter_by(l1_name=l1)                                    3708
    else:                                                              3709
        q = q.filter(ExpertMatrix.l1_name.is_(None))                   3710
    return q.order_by(ExpertMatrix.id)                                 3711
def default_scheme(unit_type, level, l1=None):                         3712
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""                3713
    name = "%s · %s%s" % (unit_type,                                   3714
                          "一级指标" if level == 1 else "二级指标",        3715
                          "(%s)" % l1 if l1 else "")                   3716
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ3717
    e,                                                                 3718
                                          level=level,                 3719
    name=name).first()                                                 3720
    if scheme is None:                                                 3721
        scheme = WeightScheme(unit_type=unit_type,                     3722
    level=level, name=name,                                            3723
                              is_default=False)                        3724
        db.session.add(scheme)                                         3725
        db.session.flush()                                             3726
    return scheme                                                      3727
@bp.route("/experts", methods=["GET", "POST"])                         3728
def experts():                                                         3729
    """多专家判断矩阵录入与汇总。"""                                       3730
    unit_type, level, l1 = expert_scope()                              3731
    if unit_type not in DEFAULT_WEIGHTS:                               3732
        flash("无效的微地质单元类型", "error")                             3733
        return redirect(url_for("ahp.index"))                          3734
    names, vec = indicator_list(unit_type, level, l1)                  3735
    n = len(names)                                                     3736
    form = EmptyForm()                                                 3737
    if request.method == "POST" and form.validate_on_submit(3738
    ):                                                                 3739
        action = request.form.get("action")                            3740
        if action == "add":                                            3741
            expert_name = request.form.get("expert_name",              3742
    "").strip() or "未具名专家"                                          3743
            try:                                                       3744
                upper = [float(request.form.get("m_%d_%d" %            3745
    (i, j), "1"))                                                      3746
                         for i in range(n) for j in range(i            3747
    + 1, n)]                                                           3748
            except ValueError:                                         3749
                flash("矩阵输入有误，请填入正数", "error")                    3750
```

```
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
            A = ahp.matrix_from_upper(upper)
            r = ahp.weights_and_consistency(A)
            scheme = default_scheme(unit_type, level, l1)
            db.session.add(ExpertMatrix(
                scheme_id=scheme.id,
                unit_type=unit_type, l1_name=(l1 if level
    == 2 else None),
                expert_name=expert_name, level=level,
                matrix_json=json.dumps([list(map(float,
    row)) for row in A]),
                cr=round(r["cr"], 6)))
            db.session.commit()
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
    r["cr"]), "ok")
        elif action == "save_agg":
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算
            weights = _aggregate(unit_type, level, l1)
            if weights is None:
                flash("暂无可汇总的专家矩阵", "error")
            elif not weights["consistent"]:
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"
                      "请先检查或调整专家矩阵" % weights["cr"],
    "error")
            else:
                names_, _ = indicator_list(unit_type,
    level, l1)
                pairs = to_code_pairs(unit_type, level, l1,
    names_,
                                      weights["weights"])
                _, cnt = ws.save_weights(unit_type, pairs)
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"
                      % (unit_type, cnt, weights["cr"]),
    "ok")
        return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
    unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
    default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
```

```
    unit_type, level, l1 = row.unit_type, row.level,          3801
    row.l1_name                                               3802
    form = EmptyForm()                                        3803
    if form.validate_on_submit():                             3804
        db.session.delete(row)                                3805
        db.session.commit()                                   3806
        flash("已删除专家矩阵", "ok")                          3807
    return redirect(url_for("ahp.experts",                    3808
    unit_type=unit_type,                                      3809
                                level=level, l1=l1))          3810
def _aggregate(unit_type, level, l1):                         3811
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""      3812
    rows = expert_query(unit_type, level, l1).all()           3813
    mats = []                                                 3814
    for r in rows:                                            3815
        try:                                                  3816
            mats.append([list(map(float, line))               3817
                         for line in json.loads(r.matrix_jso3818
    n)])                                                      3819
        except Exception:                                     3820
            continue                                          3821
    if not mats:                                              3822
        return None                                           3823
    agg = ahp.aggregate_matrices(mats)                        3824
    res = ahp.weights_and_consistency(agg)                    3825
    return {                                                  3826
        "count": len(mats),                                   3827
        "weights": res["weights"],                            3828
        "lambda_max": res["lambda_max"],                      3829
        "ci": res["ci"], "cr": res["cr"],                     3830
        "consistent": res["consistent"],                      3831
        "most_inconsistent": ahp.find_most_inconsistent(agg)3832
    ,                                                         3833
        "expert_crs": [r.cr for r in rows if r.cr is not      3834
    None],                                                    3835
    }                                                         3836
@bp.route("/experts/aggregate")                               3837
def experts_aggregate():                                      3838
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                3839
    unit_type, level, l1 = expert_scope()                     3840
    if unit_type not in DEFAULT_WEIGHTS:                      3841
        flash("无效的微地质单元类型", "error")                  3842
        return redirect(url_for("ahp.index"))                 3843
    names, vec = indicator_list(unit_type, level, l1)         3844
    rows = expert_query(unit_type, level, l1).all()           3845
    result = _aggregate(unit_type, level, l1)                 3846
    if result is not None:                                    3847
        result["weights"] = [round(v, 4) for v in             3848
    result["weights"]]                                        3849
        result["lambda_max"] = round(result["lambda_max"],    3850
```

```
    4)                                                                  3851
    return render_template("ahp/experts_aggregate.html",               3852
    unit_type=unit_type,                                                3853
                                level=level, l1=l1, names=names,        3854
    rows=rows,                                                          3855
                                result=result, form=EmptyForm())        3856
# ==== app/views/indicator.py ====                                      3857
# -*- coding: utf-8 -*-                                                 3858
"""指标数据采集与管理 —— 路由与页面逻辑。                                3859
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标              3860
（岩质边坡无土壤指标、平台无地形指标）。                                3861
"""                                                                     3862
import io                                                               3863
import json                                                             3864
from flask import (Blueprint, render_template, request,                 3865
    redirect, url_for,                                                  3866
                      flash, send_file)                                 3867
from app.forms import EmptyForm                                         3868
from app.models import db, MinePatch, MicroUnit,                        3869
    IndicatorData                                                       3870
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,             3871
    UNIT_ENABLED_INDICATORS                                             3872
from app.services import import_service as imp                          3873
bp = Blueprint("indicator", __name__,                                   3874
    url_prefix="/indicator")                                            3875
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)                      3876
SOIL_FIELDS = [                                                         3877
    ("soil_ratio", "土石比", ""),                                       3878
    ("thickness", "土层厚度", "cm"),                                    3879
    ("ph", "pH", ""),                                                   3880
    ("organic_matter", "有机质", "g/kg"),                               3881
]                                                                       3882
VEG_FIELDS = [                                                          3883
    ("coverage", "植被覆盖度", "%"),                                    3884
    ("shannon", "Shannon-Wiener 多样性指数", ""),                       3885
]                                                                       3886
TERRAIN_FIELDS = [                                                      3887
    ("slope", "坡度", "°"),                                             3888
    ("aspect", "坡向", ""),                                             3889
]                                                                       3890
def unit_field_config(unit_type):                                       3891
    """根据单元类型返回需录入的字段分组。"""                            3892
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())                3893
    groups = []                                                         3894
    if "土壤指标" in enabled:                                           3895
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True        3896
    = 含重金属                                                          3897
    if "植被指标" in enabled:                                           3898
        groups.append(("植被指标", VEG_FIELDS, False))                  3899
    if "地形指标" in enabled:                                           3900
```

```
        groups.append(("地形指标", TERRAIN_FIELDS, False))
    return groups
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    return render_template("indicator/index.html",
    patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir
    st()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit(
    ):
        def f(name, cast=float):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        heavy = {}
        for m in HEAVY_METALS:
            v = f("hm_" + m)
            if v is not None:
                heavy[m] = v
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        ind.soil_ratio = f("soil_ratio")
        ind.thickness = f("thickness")
        ind.ph = f("ph")
        ind.organic_matter = f("organic_matter")
        ind.heavy_metals = json.dumps(heavy,
    ensure_ascii=False) if heavy else None
        ind.coverage = f("coverage")
        ind.shannon = f("shannon")
        ind.slope = f("slope")
        ind.aspect = request.form.get("aspect", "").strip()
    or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness":
    ind.thickness,
            "ph": ind.ph, "organic_matter":
    ind.organic_matter,
```

```
            "coverage": ind.coverage, "shannon":
    ind.shannon,
            "slope": ind.slope, "aspect": ind.aspect,
        }
        current["heavy"] = ind.heavy_metals_dict()
    return render_template("indicator/edit.html",
                           unit=unit, form=form,
    groups=groups,
                           heavy_metals=HEAVY_METALS,
                           aspect_types=ASPECT_TYPES,
    current=current)
# ---------------------------- 批量导入 ----------------------------
    ----
@bp.route("/import", methods=["GET", "POST"])
def import_data():
    """批量导入指标数据（CSV / Excel）。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    form = EmptyForm()
    report = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        patch_id = request.form.get("patch_id", type=int)
        patch = db.session.get(MinePatch, patch_id) if
    patch_id else None
        if patch is None:
            flash("请选择矿山图斑", "error")
            return redirect(url_for("indicator.import_data")
    )
        file = request.files.get("datafile")
        if file is None or not file.filename:
            flash("请选择要导入的文件", "error")
            return redirect(url_for("indicator.import_data")
    )
        ext = file.filename.rsplit(".", 1)[-1].lower()
        if ext not in ("csv", "xlsx", "xls"):
            flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")
            return redirect(url_for("indicator.import_data")
    )
        units = MicroUnit.query.filter_by(patch_id=patch.id)
    .all()
        unit_map = {}
        for u in units:
            unit_map.setdefault(u.unit_type, u)
        if not unit_map:
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
            return redirect(url_for("indicator.import_data")
    )
        try:
            parsed, errors = imp.parse_upload(file.filename,
```

```
    file.stream, unit_map)                                          4001
        except Exception as e:                                      4002
            flash("文件解析失败：%s" % e, "error")                    4003
            return redirect(url_for("indicator.import_data")        4004
    )                                                               4005
        saved = 0                                                   4006
        for rec in parsed:                                          4007
            unit = unit_map[rec["unit_type"]]                       4008
            ind = IndicatorData.query.filter_by(unit_id=unit        4009
    .id).first()                                                    4010
            if ind is None:                                         4011
                ind = IndicatorData(unit_id=unit.id)                4012
                db.session.add(ind)                                 4013
            f = rec["fields"]                                       4014
            for key in ("soil_ratio", "thickness", "ph",            4015
    "organic_matter",                                               4016
                        "coverage", "shannon", "slope",             4017
    "aspect"):                                                      4018
                if key in f:                                        4019
                    setattr(ind, key, f[key])                       4020
            if rec["heavy_metals"]:                                 4021
                ind.heavy_metals = json.dumps(rec["heavy_met        4022
    als"],                                                          4023
                                                                    4024
    ensure_ascii=False)                                             4025
            saved += 1                                              4026
        db.session.commit()                                         4027
        report = {"patch": patch.name, "saved": saved,              4028
    "errors": errors,                                               4029
                  "total": len(parsed) + len(                       4030
                      [e for e in errors if e.startswith("第        4031
    ")])}                                                           4032
        if saved:                                                   4033
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")      4034
        else:                                                       4035
            flash("没有可导入的数据行，请检查文件内容", "error")        4036
    return render_template("indicator/import.html",                 4037
    patches=patches,                                                4038
                           form=form, report=report,                4039
    headers=imp.HEADERS)                                            4040
@bp.route("/import/template")                                       4041
def import_template():                                              4042
    """下载导入模板（CSV，含一行示例）。"""                             4043
    csv_text = imp.build_template_csv()                             4044
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于        4045
    Excel 正确识别中文                                                4046
    return send_file(io.BytesIO(data), as_attachment=True,          4047
                     download_name="指标数据导入模板.csv",             4048
                     mimetype="text/csv")                           4049
# ==== app/views/evaluate.py ====                                   4050
```

```
# -*- coding: utf-8 -*-
"""适宜性定量评价引擎 —— 路由与页面逻辑。
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。
"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import EmptyForm
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail)
from app.services import evaluation_service as ev
from app.services import workflow as wf
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else
    0.0,
            "data": data,
        })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
```

```
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
    str(d["raw"]),
                "std_value": _round(d["std_value"]),
                "weight": _round(d["weight"]),
                "contribution": _round(d["contribution"]),
            } for d in u["details"]],
        })
    return out
def save_evaluation(patch_id, units):
    """执行评价并把结果与逐层明细落库，返回评价记录。
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    res = ev.score_mine(units, weights)
    saved = EvaluationResult(
        patch_id=patch_id,
```

```
            final_score=round(res["final_score"], 4),
            grade=res["grade"],
            measures=res["grade_info"]["measure"],
            unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                                   ensure_ascii=False),
    )
    db.session.add(saved)
    db.session.flush()        # 先取得 result.id，再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
                result_id=saved.id,
                unit_id=unit_ids.get(u["unit_type"]),
                l1_name=d["l1"], l2_name=d["l2"],
                raw_value=None if d["raw"] is None else
    str(d["raw"]),
                std_value=d["std_value"],
    weight=d["weight"],
                contribution=d["contribution"]))
    db.session.commit()
    return saved
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
```

```
                flash("无法执行评价 —— " + reason, "error")
                return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                                progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                                unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
from app.utils.timefmt import now_utc
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
```

```
        return patch, None, None, None, prog
    rec = prog["result"]
    result = {
        "final_score": rec.final_score,
        "grade": rec.grade,
        "grade_info": grade_of(rec.final_score),
        "unit_results": json.loads(rec.unit_scores) if
    rec.unit_scores else [],
    }
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    progress = {p.id: wf.report_progress(p.id) for p in
    patches}
    return render_template("report/index.html",
    patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
```

```
    if result is None:                                                  4301
        return _deny(prog, patch_id)                                    4302
    chart_units, chart_weight = _charts(result)                         4303
    return render_template("report/detail.html",                        4304
    patch=patch, result=result,                                         4305
                                record=record,                          4306
    reference=reference,                                                4307
                                chart_units=chart_units,                4308
    chart_weight=chart_weight)                                          4309
@bp.route("/<int:patch_id>/word")                                       4310
def word(patch_id):                                                     4311
    patch, reference, result, record, prog =                            4312
    _prepare(patch_id)                                                  4313
    if result is None:                                                  4314
        return _deny(prog, patch_id)                                    4315
    buf = rs.build_report(patch, result)                                4316
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name              4317
    return send_file(buf, as_attachment=True,                           4318
    download_name=filename,                                             4319
                        mimetype="application/vnd.openxmlformat4320
    s-officedocument."                                                  4321
                            "wordprocessingml.document")                4322
@bp.route("/<int:patch_id>/pdf")                                        4323
def pdf(patch_id):                                                      4324
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""                4325
    patch, reference, result, record, prog =                            4326
    _prepare(patch_id)                                                  4327
    if result is None:                                                  4328
        return _deny(prog, patch_id)                                    4329
    chart_units, chart_weight = _charts(result)                         4330
    html = render_template(                                             4331
        "report/print.html", patch=patch,                               4332
    reference=reference, result=result,                                 4333
        chart_units=chart_units, chart_weight=chart_weight,             4334
        generated_at=now_utc())                                         4335
    data, err = browser.html_to_pdf(html)                               4336
    if err:                                                             4337
        flash("PDF 导出失败：%s" % err, "error")                          4338
        return redirect(url_for("report.detail",                        4339
    patch_id=patch_id))                                                 4340
    return send_file(io.BytesIO(data), as_attachment=True,              4341
                        download_name="自然恢复适宜性评价报告_%s.pdf" %    4342
    patch.name,                                                         4343
                        mimetype="application/pdf")                     4344
# ==== app/views/search.py ====                                         4345
# -*- coding: utf-8 -*-                                                 4346
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。                    4347
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，                             4348
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。                        4349
"""                                                                     4350
```

```
from sqlalchemy import or_
from flask import Blueprint, render_template, request
from app.models import db, Project, MinePatch, Standard
bp = Blueprint("search", __name__)
LIMIT = 50              #  每类结果上限，避免一次渲染过多
@bp.route("/search")
def index():
    q = (request.args.get("q") or "").strip()
    projects, patches, standards = [], [], []
    if q:
        like = "%" + q + "%"
        projects = Project.query.filter(or_(
            Project.name.like(like),
            Project.code.like(like),
            Project.description.like(like),
        )).order_by(Project.id.desc()).limit(LIMIT).all()
        patches = MinePatch.query.filter(or_(
            MinePatch.name.like(like),
            MinePatch.admin_region.like(like),
            MinePatch.mineral_type.like(like),
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()
        standards = Standard.query.filter(or_(
            Standard.name.like(like),
            Standard.doc_number.like(like),
            Standard.province.like(like),
            Standard.summary.like(like),
        )).order_by(Standard.id).limit(LIMIT).all()
    total = len(projects) + len(patches) + len(standards)
    return render_template("search.html", q=q,
    projects=projects,
                           patches=patches,
    standards=standards, total=total)
#  ==== app/templates/base.html ====
{% import "_macros.html" as ui %}<!DOCTYPE html>
<html lang="zh-CN">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <link rel="icon" href="{{ url_for('static',
    filename='favicon.ico') }}">
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock
    %}</title>
  <style>
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */
    :root {
      --accent: #2c6b5a;                /* 主色：青绿，千里江山石绿偏青 */
      --accent-hover: #1e5548;
      --accent-soft: #e6f0ec;
      --side-1: #1a4a4a;                /* 侧边栏：深青墨 */
```

```
    --side-2: #0f3338;                                                  4401
    --danger: #a0392a;                  /* 朱砂红 */                    4402
    --danger-soft: #fbeae6;                                             4403
    --ok: #2c6b5a;                                                      4404
    --ok-soft: #e6f0ec;                                                 4405
    --todo: #c08a3e;                    /* 赭石色待办 */                4406
    --ink: #2a2520;                     /* 深褐墨色 */                  4407
    --ink-soft: #6b5d4f;                /* 褐灰 */                      4408
    --line: #e0d8c8;                    /* 绢本边线 */                  4409
    --surface: #fdfaf2;                 /* 暖白绢本 */                  4410
    --canvas: #f5efe0;                  /* 绢本米黄背景 */              4411
    --radius: 10px;                                                     4412
    --shadow: 0 2px 8px rgba(42, 37, 32, .06);                          4413
    --header-h: 60px;                                                   4414
    --nav-w: 226px;                                                     4415
}                                                                       4416
* { box-sizing: border-box; }                                           4417
html, body { margin: 0; padding: 0; }                                   4418
body {                                                                  4419
    font-family: "Microsoft YaHei", "PingFang SC", "Noto                4420
Serif SC", "SimSun", serif;                                             4421
    font-size: 14px;                                                    4422
    line-height: 1.6;                                                   4423
    color: var(--ink);                                                  4424
    background: var(--canvas);                                          4425
}                                                                       4426
/* ---------- 跳到主要内容（键盘可达性） ---------- */                   4427
.skip-link {                                                            4428
    position: absolute;                                                 4429
    left: 12px;                                                         4430
    top: -60px;                                                         4431
    z-index: 30;                                                        4432
    padding: 8px 14px;                                                  4433
    background: var(--accent);                                          4434
    color: #fff;                                                        4435
    border-radius: var(--radius);                                       4436
    text-decoration: none;                                              4437
    transition: top .15s ease-out;                                      4438
}                                                                       4439
.skip-link:focus { top: 12px; }                                         4440
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------              4441
   固定为一屏高，页面滚动交给右侧内容区，                               4442
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */      4443
.app-shell {                                                            4444
    display: flex;                                                      4445
    align-items: stretch;                                               4446
    height: 100vh;                                                      4447
    height: 100dvh;                                                     4448
    overflow: hidden;                                                   4449
}                                                                       4450
```

```
/* ---------- 侧边栏 ---------- */
.app-side {
  flex: 0 0 var(--nav-w);
  width: var(--nav-w);
  display: flex;
  flex-direction: column;
  background: linear-gradient(180deg, var(--side-1) 0%, var(--side-2) 100%);
  color: #fff;
}
.app-brand {
  display: flex;
  align-items: center;
  gap: 11px;
  padding: 17px 16px 15px;
  border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
  flex: none;
  display: grid;
  place-items: center;
  width: 36px;
  height: 36px;
  border-radius: 10px;
  background: rgba(255, 255, 255, .15);
}
.app-brand__name {
  font-size: 14.5px;
  font-weight: 600;
  line-height: 1.35;
  letter-spacing: .2px;
}
.app-brand__sub {
  font-size: 11.5px;
  line-height: 1.35;
  color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y: auto; }
.app-nav__group {
  margin: 12px 0 5px;
  padding: 0 10px;
  font-size: 11.5px;
  letter-spacing: .05em;
  color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
  display: flex;
  align-items: center;
```

```
    gap: 10px;
    padding: 9px 11px;
    margin-bottom: 2px;
    border-radius: 8px;
    color: rgba(255, 255, 255, .80);
    text-decoration: none;
    transition: background-color .16s ease, color .16s
ease;
}
.app-nav a:hover { background: rgba(255, 255, 255,
.10); color: #fff; }
.app-nav a:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
    background: rgba(255, 255, 255, .17);
    color: #fff;
    font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
    margin-left: auto;
    min-width: 20px;
    padding: 0 6px;
    font-size: 11px;
    font-weight: 600;
    line-height: 18px;
    text-align: center;
    border-radius: 9px;
    background: var(--todo);
    color: #fff;
    font-variant-numeric: tabular-nums;
}
.app-user {
    display: flex;
    align-items: center;
    gap: 8px;
    padding: 10px 14px;
    border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
    display: flex;
    flex: 1;
    min-width: 0;
    align-items: center;
    gap: 10px;
    padding: 4px 6px;
    border-radius: 8px;
    color: inherit;
```

```
    text-decoration: none;
    transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255, .10); }
.app-user__me:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-user__avatar {
    flex: none;
    display: grid;
    place-items: center;
    width: 32px;
    height: 32px;
    border-radius: 50%;
    background: rgba(255, 255, 255, .16);
    font-size: 13px;
    font-weight: 600;
    object-fit: cover;
}
.app-user__name {
    display: block;
    font-size: 13px;
    line-height: 1.3;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}
.app-user__state {
    display: flex;
    align-items: center;
    gap: 5px;
    font-size: 11.5px;
    line-height: 1.3;
    color: rgba(255, 255, 255, .60);
}
.app-user__dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #6ed49a;
}
.app-user__out {
    flex: none;
    padding: 6px;
    border-radius: 6px;
    color: rgba(255, 255, 255, .68);
    text-decoration: none;
}
```

```
.app-user__out:hover { color: #fff; background:
rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ----------
   整个骨架固定一屏高，滚动只发生在内容区内部：
   左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */
.app-content {
  flex: 1;
  min-width: 0;
  display: flex;
  flex-direction: column;
  overflow-y: auto;
  overflow-x: hidden;
  /* 细滚动条：默认半透明，悬停时加深，不抢视线 */
  scrollbar-width: thin;
  scrollbar-color: rgba(31, 107, 69, .28) transparent;
}
.app-content::-webkit-scrollbar { width: 12px; }
.app-content::-webkit-scrollbar-track { background:
transparent; }
.app-content::-webkit-scrollbar-thumb {
  background: rgba(31, 107, 69, .26);
  border: 3px solid transparent;
  background-clip: content-box;
  border-radius: 999px;
}
.app-content::-webkit-scrollbar-thumb:hover {
  background: rgba(31, 107, 69, .46);
  background-clip: content-box;
}
/* ---------- 顶栏：面包屑 + 搜索 ----------
   粘在内容区顶部，滚动时面包屑与搜索框始终可用 */
.app-header {
  position: sticky;
  top: 0;
  z-index: 20;
  display: flex;
  align-items: center;
  gap: 14px;
  height: var(--header-h);
  padding: 0 22px;
  background: var(--surface);
  border-bottom: 1px solid var(--line);
}
.crumb {
  display: flex;
  align-items: center;
  gap: 7px;
  font-size: 13px;
  color: var(--ink-soft);
  white-space: nowrap;
```

```
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
  display: flex;
  align-items: center;
  gap: 8px;
  width: 268px;
  padding: 7px 12px;
  background: var(--canvas);
  border: 1px solid var(--line);
  border-radius: 9px;
  color: var(--ink-soft);
  transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
  flex: 1;
  min-width: 0;
  border: none;
  outline: none;
  background: transparent;
  font-family: inherit;
  font-size: 13px;
  color: var(--ink);
}
.search-box button {
  display: grid;
  place-items: center;
  padding: 0;
  border: none;
  background: transparent;
  color: inherit;
  cursor: pointer;
}
.icon-btn {
  display: grid;
  place-items: center;
  width: 34px;
  height: 34px;
  border-radius: 9px;
  border: 1px solid transparent;
  color: var(--ink-soft);
  text-decoration: none;
```

```
    transition: background-color .16s ease, color .16s ease,
                border-color .16s ease;
}
.icon-btn:hover {
    color: var(--accent);
    background: var(--accent-soft);
    border-color: rgba(31, 107, 69, .18);
}
.app-main {
    flex: 1;
    min-width: 0;
    padding: 22px 26px 26px;
}
.container { max-width: none; width: 100%; }
/* ---------- 页脚 ---------- */
.app-foot {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 14px;
    padding: 0 26px 18px;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 未登录时的裸页面（登录页） ---------- */
.login-shell {
    display: flex;
    flex-direction: column;
    min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 16px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
}
.card > h3 {
    margin: 20px 0 10px;
    font-size: 14px;
```

```
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0 16px;
    font-variant-numeric: tabular-nums;
}
caption {
    text-align: left;
    color: var(--ink-soft);
    font-size: 12px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
    padding: 8px 12px;
    text-align: center;
    vertical-align: middle;
}
th {
    background: var(--accent-soft);
    font-weight: 600;
    white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
    display: block;
    margin-bottom: 4px;
    color: var(--ink-soft);
    font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
    padding: 6px 9px;
    font-family: inherit;
    font-size: 14px;
    color: var(--ink);
    background: #fff;
    border: 1px solid #c6ccd4;
    border-radius: 4px;
    max-width: 100%;
}
input[type="number"] {
```

```
  width: 104px;
  text-align: center;
  font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize: vertical; }
input:focus-visible, select:focus-visible, textarea:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
  border-color: var(--accent);
}
.checkbox {
  display: inline-flex;
  align-items: center;
  gap: 7px;
  cursor: pointer;
  user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0; cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 18px;
  font-family: inherit;
  font-size: 14px;
  line-height: 1.4;
  color: #fff;
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
  cursor: pointer;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.btn:hover {
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
```

```
.btn:focus-visible { outline: 2px solid var(--accent);
outline-offset: 2px; }
.btn--ghost {
  color: var(--accent);
  background: rgba(31, 107, 69, .06);
  border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
}
.btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 13px;
border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
  background: rgba(31, 107, 69, .80);
  border-color: rgba(31, 107, 69, .30);
  box-shadow: none;
}
.btn--ghost:disabled:hover { background: rgba(31, 107,
69, .06); color: var(--accent); }
/* ---------- 表格操作列的行内小按钮 ---------- */
   取代原来「文字 | 文字 | 文字」的写法：
   圆角矩形 + 半透明底色，悬停时变实色。 */
.row-actions {
  display: inline-flex;
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */
  align-items: center;
  gap: 6px;
  white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
  display: inline-flex;
  align-items: center;
  padding: 4px 11px;
  font-family: inherit;
  font-size: 13px;
```

```
    line-height: 1.5;
    color: var(--accent);
    background: rgba(31, 107, 69, .07);
    border: 1px solid rgba(31, 107, 69, .20);
    border-radius: 7px;
    cursor: pointer;
    text-decoration: none;
    white-space: nowrap;
    transition: background-color .16s ease, border-color .16s ease,
                color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.link-btn:hover {
    color: #fff;
    background: var(--accent);
    border-color: var(--accent);
    box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
}
.link-btn:active { transform: translateY(1px); box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.link-btn--danger {
    color: var(--danger);
    background: rgba(179, 38, 30, .07);
    border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
    color: #fff;
    background: var(--danger);
    border-color: var(--danger);
    box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
    padding: 10px 14px;
    margin-bottom: 14px;
    border-radius: var(--radius);
    border-left: 4px solid;
}
.flash.error {
    color: var(--danger);
    background: var(--danger-soft);
    border-left-color: var(--danger);
}
.flash.ok {
```

```
    color: var(--ok);                                                    4951
    background: var(--ok-soft);                                          4952
    border-left-color: var(--ok);                                        4953
}                                                                        4954
/* ---------- 浮层提示（登录成功、各类操作完成、登录失败） ---------- 4955
   外层 toast-stack 固定在页面中上方，内层每条各自淡入 → 停留 → 淡出，  4956
   多条提示会依次排开而不重叠。                                         4957
   底色是浅绿半透明：既有一层底色把文字托住，又能透出后面的内容，      4958
   不用高饱和色块，提示不该比页面内容更抢眼。                           4959
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */      4960
.toast-stack {                                                           4961
  position: fixed;                                                       4962
  top: 16px;                                                             4963
  left: 50%;                                                             4964
  z-index: 100;                                                          4965
  display: flex;                                                         4966
  flex-direction: column;                                                4967
  align-items: center;                                                   4968
  gap: 6px;                                                              4969
  transform: translateX(-50%);                                           4970
  pointer-events: none;                                                  4971
}                                                                        4972
.toast-center {                                                          4973
  max-width: min(620px, 82vw);                                           4974
  padding: 8px 22px;                                                     4975
  border-radius: 999px;                                                  4976
  /* 浅绿、半透明：不遮住背后的内容，也比纯透明更托得住文字 */          4977
  background: rgba(223, 241, 231, .76);                                  4978
  border: 1px solid rgba(31, 107, 69, .14);                              4979
  color: #2f4a3c;                                                        4980
  font-size: 16px;                                                       4981
  font-weight: 500;                                                      4982
  line-height: 1.65;                                                     4983
  letter-spacing: .03em;                                                 4984
  text-align: center;                                                    4985
  text-wrap: pretty;                                                     4986
  backdrop-filter: blur(6px) saturate(1.1);                              4987
  -webkit-backdrop-filter: blur(6px) saturate(1.1);                      4988
  box-shadow: 0 4px 16px rgba(23, 32, 27, .07);                          4989
  animation: toast-fade 2.6s ease-in-out forwards;                       4990
}                                                                        4991
/* 失败类提示：同一套观感，换成浅红底，一眼能区分成败 */               4992
.toast-center--error {                                                   4993
  background: rgba(252, 233, 230, .78);                                  4994
  border-color: rgba(163, 43, 34, .16);                                  4995
  color: #9c2b23;                                                        4996
}                                                                        4997
@keyframes toast-fade {                                                  4998
  0%          { opacity: 0; transform: translateY(-6px); }               4999
  14%, 74%    { opacity: 1; transform: none; }                           5000
```

```
  100%      { opacity: 0; transform: none; }
}
/* 系统设置里要求「减少动态效果」时，去掉位移，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
  .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
  @keyframes toast-fade-plain {
    0%        { opacity: 0; }
    14%, 74%  { opacity: 1; }
    100%      { opacity: 0; }
  }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 12.5px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background:
var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */
.body--login {
  background:
    radial-gradient(1100px 640px at 50% 8%,
                    #edf6f1 0%, rgba(237, 246, 241, 0)
72%),
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);
}
```

```
.body--login .app-main { padding: 0; }
.body--login .container { max-width: none; width: 100%;
}
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/* ---------- 工作台：页首横幅 ---------- */
.hero {
  position: relative;
  overflow: hidden;
  min-height: 176px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /* 左侧压深保证文字可读，右侧透出远山插画；
     背景按底部对齐，避免远山被居中裁掉 */
  background:
    linear-gradient(100deg, rgba(15, 40, 45, .90) 0%,
                            rgba(20, 55, 58, .55) 45%,
                            rgba(20, 55, 58, .12) 100%),
    url("{{ url_for('static', filename='hero_landscape.j
pg') }}")
    center center / cover no-repeat;
  color: #fff;
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
```

```
    display: flex;
    flex-wrap: wrap;
    gap: 10px;
    margin-top: 18px;
}
.hero-btn {
    display: inline-flex;
    align-items: center;
    gap: 6px;
    padding: 7px 15px;
    font-size: 13px;
    border-radius: 8px;
    text-decoration: none;
    transition: background-color .16s ease, border-color .16s ease,
                color .16s ease;
}
.hero-btn--solid {
    color: #14432f;
    background: rgba(255, 255, 255, .92);
    border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
    color: #fff;
    background: rgba(255, 255, 255, .14);
    border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
    gap: 12px;
    margin-bottom: 22px;
}
.stat-card {
    display: block;
    padding: 15px 17px;
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    color: var(--ink);
    text-decoration: none;
    transition: border-color .16s ease, box-shadow .16s
```

```
ease;
}
.stat-card:hover {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
  display: block;
  font-size: 12.5px;
  color: var(--ink-soft);
}
.stat-card__value {
  display: block;
  margin: 3px 0 2px;
  font-size: 26px;
  font-weight: 700;
  line-height: 1.25;
  color: var(--accent);
  font-variant-numeric: tabular-nums;
}
.stat-card__unit {
  margin-left: 3px;
  font-size: 13px;
  font-weight: 500;
  color: var(--ink-soft);
}
.stat-card__note {
  display: block;
  font-size: 12px;
  color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 16px; }
.section__desc { margin: 0; font-size: 12.5px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
  gap: 12px;
}
```

```
.step-card {
  position: relative;
  display: block;
  padding: 16px 16px 15px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s ease,
              transform .16s ease;
}
.step-card:hover {
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 5px 16px rgba(19, 60, 43, .10);
  transform: translateY(-1px);
}
.step-card:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.step-card__no {
  position: absolute;
  top: 13px;
  right: 15px;
  font-size: 17px;
  font-weight: 700;
  color: #d7e3db;
  font-variant-numeric: tabular-nums;
}
.step-card__icon {
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  margin-bottom: 10px;
  border-radius: 9px;
  background: var(--accent-soft);
  color: var(--accent);
}
.step-card__title {
  display: block;
  margin-bottom: 5px;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
.step-card__desc {
  display: block;
  font-size: 12.5px;
```

```
    line-height: 1.65;
    color: var(--ink-soft);
    text-wrap: pretty;
}
.step-card__badge {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #fdf1e2;
    color: #9a5a12;
    border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
    background: #fafbfa;
    border-color: #eceff0;
    box-shadow: none;
    cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
    background: #f1f3f2;
    color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #f1f3f2;
    color: #8b948f;
}
/* ---------- 结果摘要 ---------- */
.stat-row {
    display: flex;
    flex-wrap: wrap;
    gap: 28px;
    margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
```

```
margin: 4px 0 18px; }
.step {
  display: flex; align-items: center; gap: 7px;
  padding: 8px 12px; font-size: 13px;
  background: #fff; border: 1px solid var(--line);
  border-radius: var(--radius);
}
.step__no {
  display: inline-flex; align-items: center;
justify-content: center;
  width: 20px; height: 20px; border-radius: 50%;
  background: #eef1f5; color: var(--ink-soft);
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块（中性底色，不用整块警示色压页面） ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-left: 3px solid var(--todo);
  border-radius: var(--radius);
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
/* ---------- 前置条件清单（缺什么 + 去哪补，逐条一行） ---------- */
.prereq {
  margin: 0 0 16px;
  padding: 16px 18px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
}
.prereq__head { display: flex; align-items: flex-start;
```

```
gap: 12px; }
.prereq__icon {
  display: grid;
  place-items: center;
  width: 34px;
  height: 34px;
  flex: none;
  border-radius: 10px;
  background: var(--accent-soft);
  color: var(--accent);
}
.prereq__title { margin: 0; font-size: 14px;
font-weight: 600; }
.prereq__desc {
  margin: 4px 0 0;
  font-size: 12.5px;
  line-height: 1.7;
  color: var(--ink-soft);
}
.prereq__list { margin: 14px 0 0; padding: 0;
list-style: none; }
.prereq__item {
  display: flex;
  align-items: center;
  gap: 12px;
  padding: 10px 0;
  border-top: 1px solid var(--line);
  font-size: 13px;
}
.prereq__name { font-weight: 600; }
.prereq__need { flex: 1; min-width: 0; color:
var(--ink-soft); }
.prereq__item .btn { flex: none; }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
  padding-bottom: 9px;
  border-bottom: 1px solid var(--line);
```

```
    }
    .patch-panel__name {
      margin: 0;
      font-size: 15px;
      font-weight: 600;
    }
    .patch-panel__meta {
      font-size: 12.5px;
      color: var(--ink-soft);
    }
    .patch-panel__meta a { color: var(--accent); }
    .patch-panel__tail { margin-left: auto; }
    .patch-panel .notice { margin: 0; }
    .patch-panel table { margin: 0; }
    /* ---------- 窄屏适配 ---------- */
    @media (max-width: 960px) {
      /* 上下布局时取消「一屏高 + 内容区内部滚动」，
         否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */
      .app-shell {
        flex-direction: column;
        height: auto;
        min-height: 100vh;
        min-height: 100dvh;
        overflow: visible;
      }
      .app-content { overflow: visible; }
      .app-header { position: static; }
      .app-side { width: auto; flex: none; }
      .app-brand { padding: 12px 14px; }
      .app-nav { padding: 4px 10px 10px; }
      .app-nav ul { display: flex; flex-wrap: wrap; gap:
    4px; }
      .app-nav a { margin-bottom: 0; }
      .app-nav__group { margin: 8px 0 4px; }
      .app-user { padding: 10px 14px; }
      .app-header { padding: 0 14px; gap: 10px; }
      .search-box { width: 180px; }
      .app-main { padding: 16px 14px 20px; }
      .app-foot { padding: 0 14px 14px; }
    }
    @media (max-width: 640px) {
      .crumb { display: none; }
      .search-box { width: auto; flex: 1; }
    }
  </style>
</head>
<body class="{% if not current_user.is_authenticated
    %}body--login{% endif %}">
  <a class="skip-link" href="#main">跳到主要内容</a>
  {# 取一次即可（同一请求内重复调用返回相同内容）。
```

```
    类别为 toast / ok / toast-error 的渲染成中上方浮层，出现后自动淡出；
    其余仍是页面顶部的提示条 ——
    需要对照着改表单的错误，留在页面上比一闪而过更有用。 #}
{% set flash_msgs = get_flashed_messages(with_categories=true) %}
<div class="toast-stack">
  {% for category, message in flash_msgs if category in ("toast", "ok", "toast-error") %}
  <div class="toast-center{% if category == 'toast-error' %} toast-center--error{% endif %}"
       role="status" aria-live="polite">{{ message }}</div>
  {% endfor %}
</div>
{% if current_user.is_authenticated %}
<div class="app-shell">
  <aside class="app-side">
    <div class="app-brand">
      <span class="app-brand__mark">{{ ui.icon("mountain", 21) }}</span>
      <span>
        <span class="app-brand__name">矿山自然恢复评价</span><br>
        <span class="app-brand__sub">适宜性评价系统 V1.0</span>
      </span>
    </div>
    <nav class="app-nav" aria-label="功能导航">
      <ul>
        <li>
          <a href="/"{% if home_active %} aria-current="page"{% endif %}>
            {{ ui.icon("home") }}<span>工作台</span>
          </a>
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for item in g.links %}
        <li>
          <a href="{{ item.href }}"{% if item.active %} aria-current="page"{% endif %}>
            {{ ui.icon(item.icon) }}<span>{{ item.label }}</span>
            {% if item.badge %}<span class="app-nav__badge">{{ item.badge }}</span>{% endif %}
          </a>
        </li>
        {% endfor %}
      </ul>
```

```
      {% endfor %}                                                    5501
    </nav>                                                            5502
    <div class="app-user">                                            5503
      <a class="app-user__me" href="/profile"                         5504
title="个人设置">                                                     5505
        {% if current_user.avatar %}                                  5506
        <img class="app-user__avatar"                                 5507
             src="{{ url_for('static',                                5508
filename=current_user.avatar) }}" alt="">                             5509
        {% else %}                                                    5510
        <span class="app-user__avatar"                                5511
aria-hidden="true">{{ current_user.initial }}</span>                  5512
        {% endif %}                                                   5513
        <span style="min-width:0;">                                   5514
          <span class="app-user__name">{{                             5515
current_user.shown_name }}</span>                                     5516
          <span class="app-user__state">                              5517
            <span class="app-user__dot"                               5518
aria-hidden="true"></span>                                            5519
            {{ role_labels.get(current_user.role,                     5520
current_user.role) }}                                                 5521
          </span>                                                     5522
        </span>                                                       5523
      </a>                                                            5524
      <a class="app-user__out" href="/logout" title="退出登录"        5525
         aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>        5526
    </div>                                                            5527
</aside>                                                              5528
<div class="app-content">                                             5529
  <header class="app-header">                                         5530
    <nav class="crumb" aria-label="当前位置">                         5531
      <a href="/">首页</a>                                            5532
      {{ ui.icon("chevron", 13) }}                                    5533
      <strong>{{ page_title }}</strong>                               5534
    </nav>                                                            5535
    <span class="app-header__spacer"></span>                          5536
    <form class="search-box" method="get"                             5537
action="/search" role="search">                                       5538
      {{ ui.icon("search", 16) }}                                     5539
      <label for="q" class="visually-hidden">搜索项目、图斑、标准<    5540
/label>                                                               5541
      <input id="q" type="search" name="q"                            5542
autocomplete="off"                                                    5543
             placeholder="搜索项目、图斑、标准" value="{{             5544
search_q }}">                                                         5545
      <button type="submit" aria-label="搜索">{{                      5546
ui.icon("chevron", 15) }}</button>                                    5547
    </form>                                                           5548
    {% if current_user.is_admin %}                                    5549
    <a class="icon-btn" href="/profile?tab=logs"                      5550
```

```
      title="操作日志"                                                    5551
              aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>        5552
          {% endif %}                                                    5553
        </header>                                                        5554
    {% else %}                                                           5555
    <div class="login-shell">                                            5556
    {% endif %}                                                          5557
      <main id="main" class="app-main">                                  5558
        <div class="container">                                          5559
          {% for category, message in flash_msgs if category             5560
      not in ("toast", "ok", "toast-error") %}                           5561
          <p class="flash {{ category }}" role="status">{{               5562
      message }}</p>                                                     5563
          {% endfor %}                                                   5564
          {% block content %}{% endblock %}                              5565
        </div>                                                           5566
      </main>                                                            5567
    {% if current_user.is_authenticated %}                               5568
        <footer class="app-foot">                                        5569
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0</span>          5570
          <span>数据截止 {{ today }}</span>                              5571
        </footer>                                                        5572
      </div>                                                             5573
    </div>                                                               5574
    {% else %}                                                           5575
    </div>                                                               5576
    {% endif %}                                                          5577
    {% block scripts %}{% endblock %}                                    5578
</body>                                                                  5579
</html>                                                                  5580
#  ==== app/templates/_macros.html ====                                  5581
{#  共用宏：图标、流程状态面板 #}                                        5582
{#  ================================================================== 5583
      ==================                                                 5584
    图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）               5585
    不引图标字体或外部图片，避免额外请求与资源依赖；                    5586
    stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。        5587
    ================================================================== 5588
      ================== #}                                             5589
{% set ICONS = {                                                         5590
  "dot": '<circle cx="12" cy="12" r="3"/>',                              5591
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2              5592
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',                          5593
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2             5594
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',                  5595
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8                             5596
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4                       5597
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',                                  5598
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6              5599
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',                     5600
```

```
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
    17h5"/>',
  "database": '<ellipse cx="12" cy="5.5" rx="8"
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
    8-3.2"/>',
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
  "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
    21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
    4.4V21"/>',
  "mail": '<rect x="3" y="5" width="18" height="14"
    rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>',
  "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0
    0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6
    16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>',
  "lock": '<rect x="4" y="10.4" width="16" height="10.6"
    rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>',
  "briefcase": '<rect x="3" y="7.4" width="18"
    height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1
    10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>',
  "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
    20.5-4-4"/>',
  "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
    6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
    1-3.6 0"/>',
  "plus": '<path d="M12 5v14M5 12h14"/>',
  "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
    4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
    21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
  "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
    2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
    7.5"/><path d="M5.5 12H15"/>',
  "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
  "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
    10v4.2"/><path d="M12 17.4h.01"/>',
  "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
    20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
```

```
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
    </span>
  </div>
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">2</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step {% if progress.data.ready %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">3</span>
    <span class="step__name">录入指标数据</span>
    <span class="step__state">
      {% if progress.data.ready %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step step--next">
    <span class="step__no">4</span>
    <span class="step__name">执行适宜性评价</span>
    <span class="step__state">当前步骤</span>
  </div>
</div>
{% endmacro %}
{# 前置条件的统一外壳：一张中性卡片 + 一个清单。
```

```
    缺什么、去哪补都写在每一行里，不用整块警示色压页面。  #}
{% macro prereq_card(title, desc, icon_name="warn") %}
<section class="prereq">
  <div class="prereq__head">
    <span class="prereq__icon">{{ icon(icon_name, 18)
    }}</span>
    <div>
      <p class="prereq__title">{{ title }}</p>
      {% if desc %}<p class="prereq__desc">{{ desc }}</p>{%
    endif %}
    </div>
  </div>
  <ul class="prereq__list">{{ caller() }}</ul>
</section>
{% endmacro %}
{# 前置条件未满足时的提示：只列缺什么，并给出处理入口 #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
{% call prereq_card("前置条件未满足，本步骤暂不能执行",
                    "按论文 4.1 节的流程顺序，先把下面这些补上。") %}
  {% for b in progress.blocking %}
  <li class="prereq__item">
    <span class="prereq__name">{{ b.label }}</span>
    <span class="prereq__need">{{ b.reason }}</span>
    <a class="btn btn--sm" href="{{ b.fix_url }}">{{
    b.fix_label }}</a>
  </li>
  {% endfor %}
{% endcall %}
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label
    }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
    下游各页按图斑组织数据，故只要存在「还没建图斑的项目」，就逐条列出并给出入口。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
{% call prereq_card("前置条件未满足",
                    "本页按矿山图斑组织数据：项目下先建立图斑，才会出现在这里。",
                    "mountain") %}
  {% for p in projects_without_patch %}
  <li class="prereq__item">
```

```
      <span class="prereq__name">项目「{{ p.name }}」</span>
      <span class="prereq__need">缺少：矿山图斑</span>
      <a class="btn btn--sm" href="/project/{{ p.id
      }}/patch/add">
        {{ icon("plus", 14) }}去新建图斑
      </a>
    </li>
    {% endfor %}
{% endcall %}
{% endif %}
{% endmacro %}
# ==== app/templates/_login_badge.svg ====
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg
      " viewBox="0 0 240 240" role="img"
      aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
    <title>系统徽标</title>
    <g class="lp-ring" style="--k: 2.7">
      <circle cx="162.00" cy="120.00" r="2.60"
      fill="#c4f5ab"/>
      <circle cx="157.84" cy="138.22" r="2.97"
      fill="#b8f5ab"/>
      <circle cx="146.19" cy="152.84" r="3.06"
      fill="#acf5ab"/>
      <circle cx="129.35" cy="160.95" r="2.80"
      fill="#abf5b7"/>
      <circle cx="110.65" cy="160.95" r="2.40"
      fill="#abf5c2"/>
      <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce"/>
      <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da"/>
      <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6"/>
      <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da"/>
      <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce"/>
      <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2"/>
      <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7"/>
      <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab"/>
      <circle cx="157.84" cy="101.78" r="2.23"
      fill="#b8f5ab"/>
    </g>
    <g class="lp-ring" style="--k: 2.15">
      <circle cx="177.29" cy="129.07" r="3.92"
      fill="#a4e28d"/>
      <circle cx="171.68" cy="146.33" r="4.01"
      fill="#9be28d"/>
      <circle cx="161.01" cy="161.01" r="3.87"
      fill="#91e28d"/>
      <circle cx="146.33" cy="171.68" r="3.55"
      fill="#8de292"/>
      <circle cx="129.07" cy="177.29" r="3.17"
      fill="#8de29c"/>
      <circle cx="110.93" cy="177.29" r="2.88"
```

```
    fill="#8de2a6" />                                                    5801
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af" /> 5802
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9" /> 5803
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2" /> 5804
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc" /> 5805
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc" /> 5806
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2" /> 5807
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9" /> 5808
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af" /> 5809
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6" /> 5810
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c" /> 5811
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292" /> 5812
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d" /> 5813
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d" /> 5814
    <circle cx="177.29" cy="110.93" r="3.63"                             5815
    fill="#a4e28d" />                                                    5816
  </g>                                                                   5817
  <g class="lp-ring" style="--k: 1.85">                                  5818
    <circle cx="194.00" cy="120.00" r="4.94"                             5819
    fill="#91cf72" />                                                    5820
    <circle cx="191.85" cy="137.71" r="4.92"                             5821
    fill="#88cf72" />                                                    5822
    <circle cx="185.52" cy="154.39" r="4.73"                             5823
    fill="#80cf72" />                                                    5824
    <circle cx="175.39" cy="169.07" r="4.42"                             5825
    fill="#78cf72" />                                                    5826
    <circle cx="162.04" cy="180.90" r="4.06"                             5827
    fill="#72cf73" />                                                    5828
    <circle cx="146.24" cy="189.19" r="3.73"                             5829
    fill="#72cf7b" />                                                    5830
    <circle cx="128.92" cy="193.46" r="3.51"                             5831
    fill="#72cf83" />                                                    5832
    <circle cx="111.08" cy="193.46" r="3.45"                             5833
    fill="#72cf8b" />                                                    5834
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93" /> 5835
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c" /> 5836
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4" /> 5837
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac" /> 5838
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4" /> 5839
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc" /> 5840
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4" /> 5841
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac" /> 5842
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4" /> 5843
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c" /> 5844
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93" /> 5845
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b" /> 5846
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83" /> 5847
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b" /> 5848
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73" /> 5849
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72" /> 5850
```

```
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72" />
    <circle cx="191.85" cy="102.29" r="4.80"
    fill="#88cf72" />
  </g>
  <g class="lp-ring" style="--k: 1.62">
    <circle cx="209.57" cy="128.82" r="5.68"
    fill="#76bb59" />
    <circle cx="206.12" cy="146.13" r="5.40"
    fill="#6fbb59" />
    <circle cx="199.37" cy="162.43" r="5.06"
    fill="#68bb59" />
    <circle cx="189.57" cy="177.10" r="4.71"
    fill="#61bb59" />
    <circle cx="177.10" cy="189.57" r="4.41"
    fill="#5abb59" />
    <circle cx="162.43" cy="199.37" r="4.19"
    fill="#59bb5f" />
    <circle cx="146.13" cy="206.12" r="4.10"
    fill="#59bb66" />
    <circle cx="128.82" cy="209.57" r="4.15"
    fill="#59bb6d" />
    <circle cx="111.18" cy="209.57" r="4.32"
    fill="#59bb73" />
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a" />
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81" />
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88" />
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f" />
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96" />
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d" />
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4" />
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4" />
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d" />
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96" />
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f" />
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88" />
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81" />
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a" />
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73" />
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d" />
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66" />
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f" />
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59" />
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59" />
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59" />
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59" />
    <circle cx="209.57" cy="111.18" r="5.85"
    fill="#76bb59" />
  </g>
  <g class="lp-ring" style="--k: 1.46">
    <circle cx="226.00" cy="120.00" r="5.94"
```

```
    fill="#65a843" />                                                                5901
    <circle cx="224.55" cy="137.45" r="5.62"                                         5902
    fill="#5fa843" />                                                                5903
    <circle cx="220.26" cy="154.42" r="5.29"                                         5904
    fill="#59a843" />                                                                5905
    <circle cx="213.22" cy="170.45" r="5.00"                                         5906
    fill="#53a843" />                                                                5907
    <circle cx="203.65" cy="185.11" r="4.77"                                         5908
    fill="#4da843" />                                                                5909
    <circle cx="191.79" cy="197.99" r="4.63"                                         5910
    fill="#47a843" />                                                                5911
    <circle cx="177.98" cy="208.74" r="4.59"                                         5912
    fill="#43a846" />                                                                5913
    <circle cx="162.58" cy="217.07" r="4.67"                                         5914
    fill="#43a84c" />                                                                5915
    <circle cx="146.02" cy="222.76" r="4.85"                                         5916
    fill="#43a852" />                                                                5917
    <circle cx="128.75" cy="225.64" r="5.10"                                         5918
    fill="#43a858" />                                                                5919
    <circle cx="111.25" cy="225.64" r="5.42"                                         5920
    fill="#43a85e" />                                                                5921
    <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864" />                        5922
    <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a" />                        5923
    <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870" />                        5924
    <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876" />                        5925
    <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c" />                        5926
    <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882" />                        5927
    <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888" />                        5928
    <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e" />                        5929
    <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894" />                        5930
    <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e" />                        5931
    <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888" />                         5932
    <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882" />                         5933
    <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c" />                         5934
    <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876" />                         5935
    <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870" />                         5936
    <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a" />                         5937
    <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864" />                         5938
    <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e" />                        5939
    <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858" />                        5940
    <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852" />                        5941
    <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c" />                        5942
    <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846" />                        5943
    <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843" />                        5944
    <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843" />                        5945
    <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843" />                        5946
    <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843" />                        5947
    <circle cx="224.55" cy="102.55" r="6.23"                                         5948
    fill="#5fa843" />                                                                5949
  </g>                                                                               5950
```

```
    <g class="lp-badge__mark">
      <circle cx="120" cy="120" r="27" fill="#14432f"/>
      <g transform="translate(120 120) scale(1.5)
      translate(-12 -12)" fill="none" stroke="#ffffff"
      stroke-width="1.9" stroke-linecap="round"
      stroke-linejoin="round">
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>
        <circle cx="17" cy="6" r="2.2"/>
      </g>
    </g>
</svg>
# ==== app/templates/error.html ====
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
    <h2>{{ code }} · {{ message }}</h2>
    <p class="muted">
      请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/">返回首页</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
    /* ================== 登录页（仿 Apple 账户页） ==================
       纯白底、居中、大量留白；色彩只用在徽标与按钮上。
       首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
       随滚动逐条弹入、往回滚时弹回。 */
    .lp {
      color: #1d1d1f;
      background:
        linear-gradient(180deg, rgba(245, 239, 224, .88) 0%,
                                rgba(245, 239, 224, .92)
      100%),
        url("../static/hero_qinglv.jpg")
        center top / cover no-repeat fixed;
    }
    /* ---------- 首屏 ---------- */
    .lp-hero {
      display: flex;
      flex-direction: column;
      align-items: center;
      justify-content: center;
```

```
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
    position: relative;
    background:
      linear-gradient(180deg, rgba(15, 40, 45, .72) 0%,
                              rgba(20, 55, 58, .50) 50%,
                              rgba(15, 40, 45, .76) 100%),
      url("../static/hero_qinglv.jpg")
      center center / cover no-repeat;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
  .lp-badge {
    width: 212px;
    height: 212px;
    margin-bottom: 36px;
    --p: 0;
    animation: lp-in .7s ease both;
  }
  .lp-badge__svg {
    display: block;
    width: 100%;
    height: 100%;
    overflow: visible;                    /* 散开时圆点不被画布裁掉 */
    opacity: calc(1 - var(--p) * .94);
    filter: blur(calc(var(--p) * 2.6px));
  }
  .lp-ring {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
  }
  .lp-badge__mark {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + .18 * var(--p)));
  }
  .lp-title {
    margin: 0 0 18px;
    font-size: clamp(28px, 4.2vw, 42px);
    font-weight: 700;
    line-height: 1.3;
    letter-spacing: -.4px;
    color: #fff;
    text-shadow: 0 2px 12px rgba(0, 0, 0, .3);
    animation: lp-in .7s ease .06s both;
  }
  .lp-cta {
```

```
    padding: 13px 52px;
    font-family: inherit;
    font-size: 17px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    animation: lp-in .7s ease .18s both;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
}
.lp-cta:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 26px rgba(44, 107, 90, .4);
}
.lp-cta:active { transform: translateY(1px); box-shadow:
    none; }
.lp-cta:focus-visible { outline: 3px solid rgba(31, 107,
    69, .40); outline-offset: 3px; }
.lp-more {
    margin-top: 38px;
    font-size: 13px;
    color: rgba(255, 255, 255, .7);
    text-decoration: none;
    animation: lp-in .7s ease .24s both;
}
.lp-more:hover { color: #fff; }
@keyframes lp-in {
    from { opacity: 0; transform: translateY(12px); }
    to   { opacity: 1; transform: none; }
}
/* ---------- 滚动进场 ----------
    进入视口时从屏外飞入，离开视口时飞回，可反复。
    交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
.lp-reveal {
    opacity: 0;
    transition: opacity .7s ease,
                transform 1.05s cubic-bezier(.18, 1.04, .3,
    1),
                border-color .62s ease, background-color
    .62s ease;
    transition-delay: 0s;
    will-change: opacity, transform;
}
/* 起始位置整体在屏外：横向按视口宽度取位移，纵向按视口高度取位移，
    再带一点缩小，落位时按贝塞尔曲线的过冲轻微回弹。 */
.lp-reveal[data-dir="left"]  { transform:
```

```
    translate3d(-56vw, 5vh, 0) scale(.9); }
.lp-reveal[data-dir="right"] { transform:
    translate3d(56vw, 5vh, 0) scale(.9); }
.lp-reveal[data-dir="up"]    { transform: translate3d(0,
    24vh, 0) scale(.94); }
.lp-reveal.is-in {
    opacity: 1;
    transform: none;
    transition-delay: calc(var(--i, 0) * 110ms);
}
/* 标题颜色由淡到深，强化「浮现」的层次感 */
.lp-reveal h2,
.lp-reveal h3,
.lp-reveal .lp-mod__name { color: #c2d2c8; transition:
    color .7s ease; }
.lp-reveal.is-in h3,
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }
.lp-reveal.is-in h2 { color: #1d1d1f; }
/* ---------- 系统介绍 ---------- */
.lp-intro {
    max-width: none;
    margin: 0;
    padding: 56px 24px 72px;
    background: transparent;
    /* 卡片从屏外飞入，横向超出的部分裁掉，免得出现横向滚动条；
       用 clip 而不是 hidden，纵向滚动仍由页面本身负责 */
    overflow-x: clip;
}
.lp-intro > * { max-width: 1040px; margin-left: auto;
    margin-right: auto; }
.lp-intro__title {
    margin: 0 auto 26px;
    font-size: clamp(22px, 2.6vw, 30px);
    text-align: center;
    letter-spacing: -.3px;
}
/* 三张特点卡与九个模块卡都按三等分排版，并整体居中 */
.lp-points {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 16px;
    margin-bottom: 46px;
}
.lp-point {
    padding: 30px 26px;
    background: rgba(253, 250, 242, .82);
    backdrop-filter: blur(12px);
    -webkit-backdrop-filter: blur(12px);
    border: 1px solid rgba(224, 216, 200, .6);
    border-radius: 22px;
```

```
    box-shadow: 0 24px 60px rgba(42, 37, 32, .10), 0 8px 20px rgba(42, 37, 32, .05);
    transition: transform .25s ease, box-shadow .25s ease;
}
.lp-point:hover {
    transform: translateY(-4px);
    box-shadow: 0 32px 72px rgba(42, 37, 32, .14), 0 12px 28px rgba(42, 37, 32, .07);
}
.lp-point__icon {
    display: grid;
    place-items: center;
    width: 40px;
    height: 40px;
    margin-bottom: 16px;
    border-radius: 12px;
    background: rgba(44, 107, 90, .1);
    color: var(--accent);
}
.lp-point h3 { margin: 0 0 8px; font-size: 16px; }
.lp-point p {
    margin: 0;
    font-size: 13.5px;
    line-height: 1.85;
    color: #6e6e73;
    text-wrap: pretty;
}
.lp-sub-title {
    margin: 0 auto 20px;
    font-size: 17px;
    text-align: center;
}
.lp-mods {
    display: grid;
    grid-template-columns: repeat(3, minmax(0, 1fr));
    gap: 12px;
    margin-bottom: 46px;
    align-items: start;
}
/*  模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明  */
.lp-mod {
    position: relative;
    padding: 17px 18px 17px 48px;
    background: rgba(253, 250, 242, .75);
    backdrop-filter: blur(8px);
    -webkit-backdrop-filter: blur(8px);
    border: 1px solid rgba(224, 216, 200, .6);
    border-radius: 14px;
    outline: none;
    transition: border-color .3s ease, box-shadow .3s ease;
```

```
}
.lp-mod:hover,
.lp-mod:focus-within,
.lp-mod:focus {
  border-color: rgba(44, 107, 90, .5);
  box-shadow: 0 8px 22px rgba(42, 37, 32, .08);
}
.lp-mod__no {
  position: absolute;
  left: 17px;
  top: 17px;
  font-size: 13px;
  font-weight: 700;
  color: #c8d6cd;
  font-variant-numeric: tabular-nums;
}
.lp-mod__name {
  display: block;
  font-size: 14px;
  font-weight: 600;
}
/* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
.lp-mod__body {
  display: grid;
  grid-template-rows: 0fr;
  transition: grid-template-rows .34s ease, margin-top
  .34s ease;
}
.lp-mod:hover .lp-mod__body,
.lp-mod:focus-within .lp-mod__body {
  grid-template-rows: 1fr;
  margin-top: 7px;
}
.lp-mod__desc {
  overflow: hidden;
  margin: 0;
  font-size: 12.5px;
  line-height: 1.7;
  color: #6e6e73;
  opacity: 0;
  transition: opacity .3s ease;
  text-wrap: pretty;
}
.lp-mod:hover .lp-mod__desc,
.lp-mod:focus-within .lp-mod__desc { opacity: 1; }
/* 触摸设备没有悬停，直接把说明摊开 */
@media (hover: none) {
  .lp-mod__body { grid-template-rows: 1fr; margin-top:
  7px; }
  .lp-mod__desc { opacity: 1; }
```

```
}                                                                    6251
.lp-foot {                                                           6252
  margin: 0;                                                         6253
  padding-top: 26px;                                                 6254
  border-top: 1px solid #e6e6e8;                                     6255
  text-align: center;                                                6256
  font-size: 12.5px;                                                 6257
  color: #86868b;                                                    6258
}                                                                    6259
/* ---------- 登录面板：点击后渐变浮现 ---------- */                    6260
.lp-auth {                                                           6261
  position: fixed;                                                   6262
  inset: 0;                                                          6263
  z-index: 60;                                                       6264
  display: grid;                                                     6265
  place-items: center;                                               6266
  padding: 24px;                                                     6267
  opacity: 0;                                                        6268
  visibility: hidden;                                                6269
  transition: opacity .42s ease, visibility 0s linear                6270
  .42s;                                                              6271
}                                                                    6272
.lp-auth.is-open {                                                   6273
  opacity: 1;                                                        6274
  visibility: visible;                                               6275
  transition: opacity .42s ease, visibility 0s;                      6276
}                                                                    6277
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */                        6278
.lp-auth__scrim {                                                    6279
  position: absolute;                                                6280
  inset: 0;                                                          6281
  background:                                                        6282
    radial-gradient(1000px 640px at 50% 0%,                          6283
                    rgba(31, 107, 69, .22), rgba(31, 107,            6284
  69, 0) 70%),                                                       6285
    linear-gradient(180deg, rgba(255, 255, 255, .84),                6286
                            rgba(246, 250, 247, .95));               6287
  backdrop-filter: blur(16px) saturate(1.25);                        6288
  -webkit-backdrop-filter: blur(16px) saturate(1.25);                6289
}                                                                    6290
.lp-auth__panel {                                                    6291
  position: relative;                                                6292
  width: 100%;                                                       6293
  max-width: 396px;                                                  6294
  padding: 36px 36px 26px;                                           6295
  background: #fff;                                                  6296
  border-radius: 20px;                                               6297
  box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                     6298
  opacity: 0;                                                        6299
  transform: translateY(18px) scale(.975);                           6300
```

```
    transition: transform .55s cubic-bezier(.16, .84, .34,
    1) .05s,
                opacity .55s ease .05s;
}
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:
    none; }
.lp-auth__close {
    position: absolute;
    top: 14px;
    right: 14px;
    display: grid;
    place-items: center;
    width: 30px;
    height: 30px;
    color: #86868b;
    background: transparent;
    border: none;
    border-radius: 50%;
    cursor: pointer;
    transition: background-color .18s ease, color .18s ease;
}
.lp-auth__close:hover { background: #f5f5f7; color:
    #1d1d1f; }
.lp-auth__title {
    margin: 0 0 6px;
    text-align: center;
    font-size: 24px;
    font-weight: 700;
    letter-spacing: -.2px;
}
.lp-auth__sub {
    margin: 0 0 26px;
    text-align: center;
    font-size: 13px;
    color: #86868b;
}
/* 字段校验错误：贴着输入框显示，不做成会消失的浮层 ——
   这类消息要对着输入框看，一闪而过会让人以为「点了没反应」 */
.lp-field__error {
    margin: 7px 0 0;
    font-size: 12.5px;
    color: #9c2b23;
}
.lp-field input[aria-invalid="true"] {
    background: #fdeeec;
    border-color: rgba(156, 43, 35, .35);
}
.lp-field { margin-bottom: 15px; }
.lp-field label {
    display: block;
```

```
    margin-bottom: 7px;                                                    6351
    font-size: 13px;                                                       6352
    color: #6e6e73;                                                        6353
  }                                                                        6354
  .lp-field input {                                                        6355
    width: 100%;                                                           6356
    padding: 13px 16px;                                                    6357
    font-family: inherit;                                                  6358
    font-size: 16px;                /* 不小于 16px，避免手机端聚焦时自动放大 */    6359
    color: #1d1d1f;                                                        6360
    background: #f5f5f7;                                                   6361
    border: 1px solid transparent;                                         6362
    border-radius: 12px;                                                   6363
    outline: none;                                                         6364
    transition: background-color .18s ease, border-color                   6365
    .18s ease,                                                             6366
                box-shadow .18s ease;                                      6367
  }                                                                        6368
  .lp-field input::placeholder { color: #a1a1a6; }                         6369
  .lp-field input:focus {                                                  6370
    background: #fff;                                                      6371
    border-color: rgba(31, 107, 69, .55);                                  6372
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                          6373
  }                                                                        6374
  .lp-remember {                                                           6375
    display: flex;                                                         6376
    align-items: center;                                                   6377
    gap: 8px;                                                              6378
    margin: 4px 0 20px;                                                    6379
    font-size: 13px;                                                       6380
    color: #6e6e73;                                                        6381
    cursor: pointer;                                                       6382
  }                                                                        6383
  .lp-remember input {                                                     6384
    width: 16px;                                                           6385
    height: 16px;                                                          6386
    margin: 0;                                                             6387
    accent-color: var(--accent);                                           6388
    cursor: pointer;                                                       6389
  }                                                                        6390
  .lp-submit {                                                             6391
    width: 100%;                                                           6392
    padding: 13px 20px;                                                    6393
    font-family: inherit;                                                  6394
    font-size: 16px;                                                       6395
    font-weight: 500;                                                      6396
    color: #fff;                                                           6397
    background: var(--accent);                                             6398
    border: none;                                                          6399
    border-radius: 980px;                                                  6400
```

```
    cursor: pointer;                                                    6401
    transition: background-color .2s ease, box-shadow .2s               6402
    ease,                                                               6403
                transform .2s ease;                                     6404
}                                                                       6405
.lp-submit:hover {                                                      6406
    background: var(--accent-hover);                                    6407
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                     6408
}                                                                       6409
.lp-submit:active { transform: translateY(1px);                         6410
    box-shadow: none; }                                                 6411
.lp-submit:focus-visible { outline: 3px solid rgba(31,                  6412
    107, 69, .40); outline-offset: 3px; }                               6413
.lp-auth__hint {                                                        6414
    margin: 18px 0 0;                                                   6415
    text-align: center;                                                 6416
    font-size: 12.5px;                                                  6417
    color: #a1a1a6;                                                     6418
}                                                                       6419
.lp-auth__reg {                                                         6420
    color: var(--accent);                                               6421
    font-weight: 500;                                                   6422
    text-decoration: none;                                              6423
    border-bottom: 1px solid rgba(31, 107, 69, .35);                    6424
    transition: border-color .18s ease;                                 6425
}                                                                       6426
.lp-auth__reg:hover { border-bottom-color: var(--accent);               6427
    }                                                                   6428
/* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */                6429
@media (prefers-reduced-motion: reduce) {                               6430
    .lp-badge, .lp-title, .lp-cta, .lp-more { animation:                6431
    none; }                                                             6432
    .lp-badge__svg { opacity: 1; filter: none; }                        6433
    .lp-ring, .lp-badge__mark { transform: none; }                      6434
    .lp-reveal { opacity: 1; transform: none !important; }              6435
    .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name              6436
    { color: inherit; }                                                 6437
    .lp-reveal.is-in h2 { color: #1d1d1f; }                             6438
    .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {               6439
    color: var(--accent); }                                             6440
    .lp-auth, .lp-auth__panel { transition: opacity .2s                 6441
    linear; }                                                           6442
    .lp-auth__panel { transform: none; }                                6443
}                                                                       6444
/* 窄屏退成两列、一列，避免卡片被挤变形 */                                  6445
@media (max-width: 900px) {                                             6446
    .lp-points, .lp-mods { grid-template-columns: repeat(2,             6447
    minmax(0, 1fr)); }                                                  6448
}                                                                       6449
@media (max-width: 620px) {                                             6450
```

```
    .lp-points, .lp-mods { grid-template-columns: minmax(0, 1fr); }
  }
  @media (max-width: 560px) {
    .lp-hero { padding: 36px 18px 48px; min-height: auto; }
    .lp-badge { width: 160px; height: 160px; margin-bottom: 26px; }
    .lp-auth__panel { padding: 30px 24px 22px; }
  }
</style>
<div class="lp">
  {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口 ---------------- #}
  <section class="lp-hero">
    <div class="lp-badge" id="lp-badge" aria-hidden="true">
      {% include "_login_badge.svg" %}
    </div>
    <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>
    <button type="button" class="lp-cta" id="lp-open"
            aria-expanded="false" aria-controls="lp-auth">登录</button>
    <a class="lp-more" href="#intro">
      向下滑动，了解系统 <span aria-hidden="true">↓</span>
    </a>
  </section>
  {# ---------------- 系统介绍：随滚动逐条弹入 ---------------- #}
  <section class="lp-intro" id="intro">
    <h2 class="lp-intro__title lp-reveal"
    data-dir="up">一套把评价流程固定下来的工具</h2>
    <div class="lp-points">
      {% set pdirs = ['left', 'up', 'right'] %}
      {% set points = [
        ('sliders', '方法可溯',
         '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回看。'),
        ('shield', '流程可控',
         '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重建或人工辅助的结论。'),
        ('file', '结果可复算',
         '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
      ] %}
      {% for ic, title, text in points %}
      <div class="lp-point lp-reveal" data-dir="{{
    pdirs[loop.index0 % 3] }}"
           style="--i: {{ loop.index0 }}">
        <span class="lp-point__icon">{{ icon(ic, 20)
    }}</span>
        <h3>{{ title }}</h3>
        <p>{{ text }}</p>
      </div>
```

```
    {% endfor %}                                                    6501
  </div>                                                            6502
  <h3 class="lp-sub-title lp-reveal"                                6503
  data-dir="up">九个功能模块</h3>                                    6504
  <div class="lp-mods">                                             6505
    {% set mdirs = ['left', 'up', 'right'] %}                       6506
    {% for no, name, desc in modules %}                             6507
    <div class="lp-mod lp-reveal" data-dir="{{                      6508
  mdirs[loop.index0 % 3] }}"                                        6509
        style="--i: {{ loop.index0 % 3 }}" tabindex="0">            6510
      <span class="lp-mod__no">{{ no }}</span>                      6511
      <span class="lp-mod__name">{{ name }}</span>                  6512
      <div class="lp-mod__body"><p class="lp-mod__desc">{{          6513
  desc }}</p></div>                                                 6514
    </div>                                                          6515
    {% endfor %}                                                    6516
  </div>                                                            6517
  <p class="lp-foot lp-reveal" data-dir="up">                       6518
    历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0 ·                  6519
    运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库                  6520
  </p>                                                              6521
</section>                                                          6522
{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}  6523
<div class="lp-auth" id="lp-auth" aria-hidden="true">               6524
  <div class="lp-auth__scrim" data-lp-close></div>                  6525
  <div class="lp-auth__panel" role="dialog"                         6526
  aria-modal="true"                                                 6527
      aria-labelledby="lp-auth-title">                              6528
    <button type="button" class="lp-auth__close"                    6529
  data-lp-close                                                     6530
            aria-label="关闭登录框">                                6531
      <svg width="16" height="16" viewBox="0 0 24 24"               6532
  fill="none"                                                       6533
          stroke="currentColor" stroke-width="2"                    6534
  stroke-linecap="round"                                            6535
          aria-hidden="true"><path d="M6 6l12 12M18 6L6             6536
  18"/></svg>                                                       6537
    </button>                                                       6538
    <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>         6539
    <p class="lp-auth__sub">使用系统账号登录</p>                    6540
    <form method="post" novalidate>                                 6541
      {{ form.hidden_tag() }}                                       6542
      <div class="lp-field">                                        6543
        <label for="{{ form.username.id }}">账户名称</label>        6544
        {{ form.username(id=form.username.id,                       6545
  autocomplete="username",                                          6546
                         placeholder="请输入账户名称",              6547
                         **({'aria-invalid': 'true'} if             6548
  form.username.errors else {})) }}                                 6549
        {% for e in form.username.errors %}                         6550
```

```
            <p class="lp-field__error">{{ e }}</p>                    6551
            {% endfor %}                                              6552
          </div>                                                      6553
          <div class="lp-field">                                      6554
            <label for="{{ form.password.id }}">密码</label>          6555
            {{ form.password(id=form.password.id,                     6556
    autocomplete="current-password",                                  6557
                             placeholder="请输入密码",                 6558
                             **({'aria-invalid': 'true'} if           6559
    form.password.errors else {})) }}                                 6560
            {% for e in form.password.errors %}                       6561
            <p class="lp-field__error">{{ e }}</p>                    6562
            {% endfor %}                                              6563
          </div>                                                      6564
          <label class="lp-remember" for="{{ form.remember.id       6565
    }}">                                                              6566
            {{ form.remember(id=form.remember.id) }}                  6567
            <span>记住我（下次自动登录，一个月内有效）</span>                6568
          </label>                                                    6569
          {{ form.submit(class_="lp-submit") }}                       6570
        </form>                                                       6571
        <p class="lp-auth__hint">                                     6572
          还没有账号？<a class="lp-auth__reg"                          6573
                        href="{{ url_for('auth.register')             6574
    }}">点我注册</a>                                                  6575
        </p>                                                          6576
        <p class="lp-auth__hint" style="margin-top:10px;">            6577
          演示账号：admin 密码：admin123                               6578
        </p>                                                          6579
      </div>                                                          6580
    </div>                                                            6581
</div>                                                                6582
{% endblock %}                                                        6583
{% block scripts %}                                                   6584
<script>                                                              6585
(function () {                                                        6586
  var reduce = window.matchMedia &&                                   6587
    window.matchMedia('(prefers-reduced-motion:                       6588
    reduce)').matches;                                                6589
  /* ---------- 登录面板 ---------- */                                 6590
  var btn = document.getElementById('lp-open');                       6591
  var auth = document.getElementById('lp-auth');                      6592
  var lastFocus = null;                                               6593
  function openPanel() {                                              6594
    lastFocus = document.activeElement;                               6595
    auth.classList.add('is-open');                                    6596
    auth.setAttribute('aria-hidden', 'false');                        6597
    btn.setAttribute('aria-expanded', 'true');                        6598
    document.body.style.overflow = 'hidden';                          6599
    // 记到地址栏，直接访问 /login#login 也能展开登录框                    6600
```

```
    if (location.hash !== '#login') {
      history.replaceState(null, '', location.pathname +
    '#login');
    }
    var first = auth.querySelector('input');
    if (first) window.setTimeout(function () {
    first.focus(); }, 300);
  }
  function closePanel() {
    auth.classList.remove('is-open');
    auth.setAttribute('aria-hidden', 'true');
    btn.setAttribute('aria-expanded', 'false');
    document.body.style.overflow = '';
    if (location.hash === '#login') {
      history.replaceState(null, '', location.pathname);
    }
    if (lastFocus && lastFocus.focus) lastFocus.focus();
  }
  if (btn && auth) {
    btn.addEventListener('click', openPanel);
    Array.prototype.forEach.call(
      auth.querySelectorAll('[data-lp-close]'),
      function (el) { el.addEventListener('click',
    closePanel); }
    );
    document.addEventListener('keydown', function (e) {
      if (e.key === 'Escape' && auth.classList.contains('is-
    open')) closePanel();
    });
    // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框。
    // 失败可能落在两处：字段校验错误在表单里，凭据错误在浮层里。
    if (location.hash === '#login'
        || document.querySelector('.lp-field__error')
        || document.querySelector('.toast-center')) {
      openPanel();
    }
  }
  /* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */
     原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
     内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
     与徽标用同一次滚动回调，行为可预期、也便于排查。 */
  var items = document.querySelectorAll('.lp-reveal');
  var badge = document.getElementById('lp-badge');
  var ticking = false;
  function applyScroll() {
    ticking = false;
    var vh = window.innerHeight || 800;
    /* 徽标：滚动前约三分之一屏内完成散开淡出 */
    if (badge) {
      var range = vh * 0.34;
```

```
        var p = range > 0 ? window.scrollY / range : 0;
        badge.style.setProperty('--p', Math.min(1,
      Math.max(0, p)).toFixed(3));
      }
      /* 内容：先批量测量，再改类名，避免读写交替触发反复重排 */
      if (reduce || !items.length) return;
      var box = [];
      for (var i = 0; i < items.length; i++) {
        box.push(items[i].getBoundingClientRect());
      }
      for (var j = 0; j < items.length; j++) {
        var r = box[j];
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内
        var shown = r.top < vh * 0.88 && r.bottom > vh * 0.06;
        if (shown) items[j].classList.add('is-in');
        else items[j].classList.remove('is-in');
      }
    }
    function onScroll() {
      if (!ticking) {
        ticking = true;
        window.requestAnimationFrame(applyScroll);
      }
    }
    // 减少动态效果时不做滚动特效，内容直接显示
    if (reduce) {
      Array.prototype.forEach.call(items, function (el) {
        el.classList.add('is-in');
      });
      if (badge) badge.style.setProperty('--p', '0');
    } else {
      window.addEventListener('scroll', onScroll, { passive:
      true });
      window.addEventListener('resize', onScroll);
      applyScroll();
    }
  })();
</script>
{% endblock %}
# ==== app/templates/auth/register.html ====
{% extends "base.html" %}
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 注册页 ==================
     与登录页同一套观感：浅绿渐变底 + 白色卡片。
     必填只有账户名称与密码，其余收在「选填」分隔线以下，
     这样首次注册不会被一堆输入框挡住。 */
  .rg {
    min-height: 100dvh;
```

```
    display: grid;                                                    6701
    place-items: center;                                              6702
    padding: 40px 20px 56px;                                          6703
    background:                                                       6704
      radial-gradient(900px 520px at 50% 0%,                          6705
                      rgba(31, 107, 69, .14), rgba(31, 107,           6706
    69, 0) 72%),                                                      6707
      #f6faf7;                                                        6708
  }                                                                   6709
  .rg-card {                                                          6710
    width: 100%;                                                      6711
    max-width: 470px;                                                 6712
    padding: 38px 40px 30px;                                          6713
    background: #fff;                                                 6714
    border-radius: 20px;                                              6715
    box-shadow: 0 26px 70px rgba(14, 46, 32, .13);                    6716
    animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)               6717
    both;                                                             6718
  }                                                                   6719
  @keyframes rg-in {                                                  6720
    from { opacity: 0; transform: translateY(16px)                    6721
    scale(.98); }                                                     6722
    to   { opacity: 1; transform: none; }                             6723
  }                                                                   6724
  .rg-title {                                                         6725
    margin: 0 0 6px;                                                  6726
    text-align: center;                                               6727
    font-size: 25px;                                                  6728
    font-weight: 700;                                                 6729
    letter-spacing: -.2px;                                            6730
  }                                                                   6731
  .rg-sub {                                                           6732
    margin: 0 0 28px;                                                 6733
    text-align: center;                                               6734
    font-size: 13px;                                                  6735
    color: #86868b;                                                   6736
  }                                                                   6737
  .rg-field { margin-bottom: 15px; }                                  6738
  .rg-field > label {                                                 6739
    display: block;                                                   6740
    margin-bottom: 7px;                                               6741
    font-size: 13px;                                                  6742
    color: #6e6e73;                                                   6743
  }                                                                   6744
  .rg-field > label .rg-opt {                                         6745
    margin-left: 5px;                                                 6746
    font-size: 12px;                                                  6747
    color: #a1a1a6;                                                   6748
  }                                                                   6749
  .rg-field input[type="text"],                                       6750
```

```
.rg-field input[type="password"] {
  width: 100%;
  padding: 12px 15px;
  font-family: inherit;
  font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */
  color: #1d1d1f;
  background: #f5f5f7;
  border: 1px solid transparent;
  border-radius: 12px;
  outline: none;
  transition: background-color .18s ease, border-color .18s ease,
              box-shadow .18s ease;
}
.rg-field input::placeholder { color: #a1a1a6; }
.rg-field input:focus {
  background: #fff;
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
.rg-field input[aria-invalid="true"] {
  border-color: rgba(180, 60, 50, .55);
  background: #fff;
}
.rg-field select {
  width: 100%;
  padding: 12px 15px;
  font-family: inherit;
  font-size: 15px;
  color: #1d1d1f;
  background: #f5f5f7;
  border: 1px solid transparent;
  border-radius: 12px;
  outline: none;
}
.rg-field select:focus {
  background: #fff;
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
.rg-err {
  margin: 6px 0 0;
  font-size: 12.5px;
  color: var(--danger);
}
/* 「选填」分隔线 */
.rg-split {
  display: flex;
  align-items: center;
  gap: 12px;
```

```
    margin: 24px 0 18px;
    font-size: 12.5px;
    color: #a1a1a6;
  }
  .rg-split::before,
  .rg-split::after {
    content: "";
    flex: 1;
    height: 1px;
    background: #ececee;
  }
  .rg-pair {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 0 14px;
  }
  @media (max-width: 480px) { .rg-pair {
    grid-template-columns: 1fr; } }
  /* 头像：圆形预览 + 选择文件 */
  .rg-avatar {
    display: flex;
    align-items: center;
    gap: 14px;
    margin-bottom: 15px;
  }
  .rg-avatar__face {
    display: grid;
    place-items: center;
    flex: none;
    width: 54px;
    height: 54px;
    border-radius: 50%;
    background: #eef2f0;
    color: #9aa8a1;
    font-size: 12px;
    overflow: hidden;
    background-size: cover;
    background-position: center;
  }
  .rg-avatar__pick { min-width: 0; }
  .rg-avatar__pick input[type="file"] {
    max-width: 100%;
    font-family: inherit;
    font-size: 12.5px;
    color: #6e6e73;
  }
  .rg-avatar__tip {
    margin: 4px 0 0;
    font-size: 12px;
    color: #a1a1a6;
```

```
  }                                                                    6851
  .rg-submit {                                                         6852
    width: 100%;                                                       6853
    margin-top: 24px;                                                  6854
    padding: 13px 20px;                                                6855
    font-family: inherit;                                              6856
    font-size: 16px;                                                   6857
    font-weight: 500;                                                  6858
    color: #fff;                                                       6859
    background: var(--accent);                                         6860
    border: none;                                                      6861
    border-radius: 980px;                                              6862
    cursor: pointer;                                                   6863
    transition: background-color .2s ease, box-shadow .2s              6864
    ease,                                                              6865
                transform .2s ease;                                    6866
  }                                                                    6867
  .rg-submit:hover {                                                   6868
    background: var(--accent-hover);                                   6869
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                    6870
  }                                                                    6871
  .rg-submit:active { transform: translateY(1px);                      6872
    box-shadow: none; }                                                6873
  .rg-submit:focus-visible { outline: 3px solid rgba(31,               6874
    107, 69, .40); outline-offset: 3px; }                              6875
  .rg-foot {                                                           6876
    margin: 20px 0 0;                                                  6877
    text-align: center;                                                6878
    font-size: 13px;                                                   6879
    color: #86868b;                                                    6880
  }                                                                    6881
  .rg-foot a { color: var(--accent); text-decoration: none;            6882
    }                                                                  6883
  .rg-foot a:hover { text-decoration: underline; }                     6884
 @media (prefers-reduced-motion: reduce) {                             6885
    .rg-card { animation: none; }                                      6886
  }                                                                    6887
</style>                                                               6888
<div class="rg">                                                       6889
  <div class="rg-card">                                                6890
    <h1 class="rg-title">创建账号</h1>                                  6891
    <p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>          6892
    <form method="post" enctype="multipart/form-data"                  6893
    novalidate>                                                        6894
      {{ form.hidden_tag() }}                                          6895
      <div class="rg-field">                                           6896
        <label for="{{ form.username.id }}">账户名称</label>           6897
        {{ form.username(id=form.username.id,                          6898
    autocomplete="username",                                           6899
                              placeholder="2–64 个字符",               6900
```

```
                        **({'aria-invalid': 'true'} if
form.username.errors else {})) }}
      {% for e in form.username.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <div class="rg-field">
      <label for="{{ form.password.id }}">密码</label>
      {{ form.password(id=form.password.id,
autocomplete="new-password",
                       placeholder="至少 6 位",
                       **({'aria-invalid': 'true'} if
form.password.errors else {})) }}
      {% for e in form.password.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <div class="rg-field">
      <label for="{{ form.confirm.id }}">确认密码</label>
      {{ form.confirm(id=form.confirm.id,
autocomplete="new-password",
                      placeholder="再次输入密码",
                      **({'aria-invalid': 'true'} if
form.confirm.errors else {})) }}
      {% for e in form.confirm.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
    </div>
    <p class="rg-split">以下均为选填</p>
    <div class="rg-avatar">
      <span class="rg-avatar__face" id="rg-face"
aria-hidden="true">头像</span>
      <div class="rg-avatar__pick">
        <label class="visually-hidden"
for="rg-avatar">选择头像图片</label>
        <input id="rg-avatar" type="file" name="avatar"
               accept=".png,.jpg,.jpeg,.gif,.webp">
        <p class="rg-avatar__tip">PNG / JPG / GIF /
WebP，不超过 4 MB</p>
      </div>
    </div>
    <div class="rg-pair">
      <div class="rg-field">
        <label for="{{ form.display_name.id }}">昵称</label>
        {{ form.display_name(id=form.display_name.id,
                             placeholder="留空则显示账户名称") }}
      </div>
      <div class="rg-field">
        <label for="{{ form.gender.id }}">性别</label>
        {{ form.gender(id=form.gender.id) }}
      </div>
    </div>
    <div class="rg-field">
```

```
          <label for="{{ form.email.id }}">邮箱</label>
          {{ form.email(id=form.email.id,
      autocomplete="email",
                        placeholder="用于联系，可留空",
                        **({'aria-invalid': 'true'} if
      form.email.errors else {})) }}
          {% for e in form.email.errors %}<p
      class="rg-err">{{ e }}</p>{% endfor %}
        </div>
        <div class="rg-field" style="margin-bottom:0;">
          <label for="{{ form.phone.id }}">手机号</label>
          {{ form.phone(id=form.phone.id, autocomplete="tel",
      placeholder="可留空") }}
        </div>
        {{ form.submit_reg(class_="rg-submit") }}
      </form>
      <p class="rg-foot">已有账号？<a href="{{
      url_for('auth.login') }}">返回登录</a></p>
    </div>
</div>
{% endblock %}
{% block scripts %}
<script>
//  选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了
(function () {
  var input = document.getElementById('rg-avatar');
  var face = document.getElementById('rg-face');
  if (!input || !face || !window.FileReader) return;
  input.addEventListener('change', function () {
    var file = input.files && input.files[0];
    if (!file) return;
    var reader = new FileReader();
    reader.onload = function (e) {
      face.textContent = '';
      face.style.backgroundImage = 'url("' +
    e.target.result + '")';
    };
    reader.readAsDataURL(file);
  });
})();
</script>
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
```

```
      <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
      <p class="hero__desc">
        以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
        建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，
        系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
      </p>
      <div class="hero__actions">
        <a class="hero-btn hero-btn--ghost"
      href="/indicator/import">
          {{ icon("upload", 16) }}批量导入
        </a>
        <a class="hero-btn hero-btn--solid"
      href="/project/add">
          {{ icon("plus", 16) }}新建评价项目
        </a>
      </div>
    </div>
  </section>
  {# ---------- 数据概览 ---------- #}
  <section class="stat-cards">
    <a class="stat-card" href="/project/">
      <span class="stat-card__label">评价项目</span>
      <span class="stat-card__value">{{ dash.stats.projects
      }}<span class="stat-card__unit">个</span></span>
      <span class="stat-card__note">
        {%- if projects_without_patch -%}
          其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
        {%- elif dash.progress.patches -%}
          已建立 {{ dash.progress.patches }} 个矿山图斑
        {%- else -%}
          尚未建立矿山图斑
        {%- endif -%}
      </span>
    </a>
    <a class="stat-card" href="/project/">
      <span class="stat-card__label">矿山图斑</span>
      <span class="stat-card__value">{{ dash.stats.patches
      }}<span class="stat-card__unit">个</span></span>
      <span class="stat-card__note">
        {% if dash.progress.ready %}其中 {{ dash.progress.ready
      }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
      </span>
    </a>
    <a class="stat-card" href="/unit/">
      <span class="stat-card__label">微地质单元</span>
      <span class="stat-card__value">{{ dash.stats.unit_types
      }}<span class="stat-card__unit">类</span></span>
      <span class="stat-card__note">
        {% if dash.progress.units %}共划分 {{
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}
```

```
      </span>                                                                  7051
    </a>                                                                       7052
    <a class="stat-card" href="/standard/">                                    7053
      <span class="stat-card__label">标准与政策</span>                          7054
      <span class="stat-card__value">{{ dash.stats.standards                   7055
      }}<span class="stat-card__unit">项</span></span>                         7056
      <span class="stat-card__note">地方标准与政策文件</span>                    7057
    </a>                                                                       7058
</section>                                                                     7059
{# ---------- 评价流程 ---------- #}                                          7060
<section class="section">                                                      7061
    <div class="section__head">                                                7062
      <h2 class="section__title">评价流程</h2>                                 7063
      <p class="section__desc">                                                7064
        按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算                7065
      </p>                                                                     7066
    </div>                                                                     7067
    <div class="step-grid">                                                    7068
      {% for s in dash.steps %}                                                7069
      {% if s.available %}                                                     7070
      <a class="step-card" href="{{ s.href }}">                                7071
        <span class="step-card__no">{{ s.no }}</span>                          7072
        <span class="step-card__icon">{{ icon(s.icon, 18)                      7073
      }}</span>                                                                7074
        <span class="step-card__title">{{ s.title }}</span>                    7075
        <span class="step-card__desc">{{ s.desc }}</span>                      7076
        {% if s.pending %}                                                     7077
        <span class="step-card__badge">待处理 {{ s.pending                     7078
      }}</span>                                                                7079
        {% endif %}                                                            7080
      </a>                                                                     7081
      {% else %}                                                               7082
      <div class="step-card step-card--locked"                                 7083
      aria-disabled="true">                                                    7084
        <span class="step-card__no">{{ s.no }}</span>                          7085
        <span class="step-card__icon">{{ icon(s.icon, 18)                      7086
      }}</span>                                                                7087
        <span class="step-card__title">{{ s.title }}</span>                    7088
        <span class="step-card__desc">{{ s.desc }}</span>                      7089
        <span class="step-card__lock">{{ s.note }}</span>                      7090
      </div>                                                                   7091
      {% endif %}                                                              7092
      {% endfor %}                                                             7093
    </div>                                                                     7094
</section>                                                                     7095
{# ---------- 系统管理 ---------- #}                                          7096
{# 个人设置、账号与操作日志的入口都在左侧导航栏，工作台不再重复列出 #}        7097
{% endblock %}                                                                 7098
# ==== app/templates/auth/add_user.html ====                                   7099
{% extends "base.html" %}                                                      7100
```

```
{% block title %}新增账号{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>新增账号</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="username">账户名称</label>
      <input id="username" type="text" name="username" required
             maxlength="64" autocomplete="off" style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="password">密码</label>
      <input id="password" type="password" name="password" required
             autocomplete="new-password" style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="role">职务</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if code == default_role %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
      <p class="muted" style="margin:6px 0 0;">
        管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。
      </p>
    </div>
    <hr style="margin:22px 0;border:none;border-top:1px solid var(--line);">
    <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3>
    <div class="field">
      <label class="field__label" for="display_name">昵称</label>
      <input id="display_name" type="text" name="display_name" maxlength="64"
             placeholder="留空则显示账户名称">
    </div>
    <div class="field">
      <label class="field__label" for="gender">性别</label>
      <select id="gender" name="gender">
        <option value="">不填</option>
        {% for g in genders %}<option value="{{ g }}">{{ g }}</option>{% endfor %}
```

```
        </select>                                                    7151
      </div>                                                         7152
      <div class="field">                                            7153
        <label class="field__label" for="email">邮箱</label>        7154
        <input id="email" type="text" name="email"                   7155
      maxlength="128">                                               7156
      </div>                                                         7157
      <div class="field">                                            7158
        <label class="field__label" for="phone">手机号</label>      7159
        <input id="phone" type="text" name="phone"                   7160
      maxlength="32">                                                7161
      </div>                                                         7162
      <p class="actions" style="margin-top:18px;">                   7163
        <button type="submit" class="btn">保存</button>              7164
        <a class="btn btn--ghost" href="{{                           7165
      url_for('auth.profile', tab='users') }}">返回</a>              7166
      </p>                                                           7167
    </form>                                                          7168
</div>                                                               7169
{% endblock %}                                                       7170
# ==== app/templates/auth/edit_user.html ====                        7171
{% extends "base.html" %}                                            7172
{% block title %}编辑账号 · {{ user.username }}{% endblock %}        7173
{% block content %}                                                  7174
<div class="card" style="max-width:560px;">                          7175
  <h2>编辑账号 · {{ user.username }}</h2>                            7176
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>          7177
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"         7178
    >                                                                7179
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。    7180
  </p>                                                               7181
  <form method="post">                                               7182
    {{ form.hidden_tag() }}                                          7183
    <div class="field">                                              7184
      <label class="field__label" for="display_name">昵称</lab       7185
    el>                                                              7186
      <input id="display_name" type="text"                           7187
    name="display_name" maxlength="64"                               7188
           value="{{ user.display_name or '' }}"                     7189
           placeholder="留空则显示账户名称">                         7190
    </div>                                                           7191
    <div class="field">                                              7192
      <label class="field__label" for="gender">性别</label>          7193
      <select id="gender" name="gender">                             7194
        <option value="">不填</option>                               7195
        {% for g in genders %}                                       7196
        <option value="{{ g }}" {% if user.gender == g               7197
    %}selected{% endif %}>{{ g }}</option>                           7198
        {% endfor %}                                                 7199
      </select>                                                      7200
```

```
</div>
<div class="field">
  <label class="field__label" for="email">邮箱</label>
  <input id="email" type="text" name="email"
maxlength="128"
         value="{{ user.email or '' }}">
</div>
<div class="field">
  <label class="field__label" for="phone">手机号</label>
  <input id="phone" type="text" name="phone"
maxlength="32"
         value="{{ user.phone or '' }}">
</div>
<hr style="margin:22px 0;border:none;border-top:1px
solid var(--line);">
<h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>
<p class="muted" style="margin:0 0
14px;font-size:12.5px;">
  职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志；
  操作员可使用全部业务功能。
</p>
<div class="field">
  <label class="field__label" for="role">职务</label>
  <select id="role" name="role">
    {% for code, label in role_labels.items() %}
    <option value="{{ code }}" {% if user.role == code
%}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="checkbox" for="active">
    <input id="active" type="checkbox" name="active"
value="1"
           {% if user.active %}checked{% endif %}>
    <span>启用该账号</span>
  </label>
  <p class="muted" style="margin:6px 0 0;">
    停用后该账号无法登录，但其历史操作记录会保留。
    系统至少需保留一名启用状态的管理员。
  </p>
</div>
<div class="field">
  <label class="field__label" for="password">重置密码（留空则不修改
) </label>
  <input id="password" type="password" name="password"
         autocomplete="new-password"
style="width:280px;"
         placeholder="留空表示保持原密码">
</div>
```

```
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="{{
      url_for('auth.profile', tab='users') }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/profile.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 个人设置（仿 Apple 账户页） ==================
     左栏是身份与分类，右栏是该分类下的功能卡。
     每张卡默认只显示当前值，点开才展开对应的修改表单 ——
     不点就不会误改，页面也不会一上来堆满输入框。 */
  .acct {
    display: grid;
    grid-template-columns: 244px 1fr;
    gap: 40px;
    align-items: start;
  }
  @media (max-width: 900px) {
    .acct { grid-template-columns: 1fr; gap: 24px; }
  }
  /* ---------- 左栏 ---------- */
  .acct-rail__avatar {
    display: grid;
    place-items: center;
    width: 88px;
    height: 88px;
    border-radius: 50%;
    background: #e9edeb;
    color: #97a09b;
    font-size: 32px;
    font-weight: 500;
    object-fit: cover;
  }
  .acct-rail__name {
    margin: 16px 0 2px;
    font-size: 16px;
    font-weight: 600;
  }
  .acct-rail__sub {
    margin: 0 0 24px;
    font-size: 13px;
    color: #86868b;
    word-break: break-all;
```

```
}
.acct-tabs {
  display: flex;
  flex-direction: column;
  gap: 2px;
  margin-bottom: 24px;
}
.acct-tabs a {
  padding: 7px 12px;
  border: 1px solid transparent;
  border-radius: 7px;
  font-size: 15px;
  color: #1d1d1f;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease;
}
.acct-tabs a:hover { background: #f5f5f7; }
.acct-tabs a.is-active {
  font-weight: 600;
  border-color: #bcc9c1;
  background: #fff;
}
/* 更换头像：折叠在左栏，点开才出现文件选择 */
.acct-avatar-edit > summary {
  display: inline-block;
  padding: 5px 12px;
  font-size: 13px;
  color: var(--accent);
  background: var(--accent-soft);
  border-radius: 7px;
  cursor: pointer;
  list-style: none;
}
.acct-avatar-edit > summary::-webkit-details-marker {
  display: none; }
.acct-avatar-edit > summary:hover { background: #dcebe2; }
.acct-avatar-edit__body { margin-top: 12px; }
.acct-avatar-edit__body .btn { width: 100%;
  margin-bottom: 8px; }
.acct-avatar-edit__file {
  margin: 0 0 10px;
  font-size: 12px;
  color: var(--ink-soft);
  word-break: break-all;
}
/* ---------- 右栏 ---------- */
.acct-title {
  margin: 0 0 8px;
  font-size: 26px;
```

```
    font-weight: 700;
    letter-spacing: -.3px;
  }
  .acct-lead {
    margin: 0 0 26px;
    font-size: 14px;
    color: #6e6e73;
  }
  /* 固定两列，与 Apple 账户页的 2×2 一致；
     卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
  .acct-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 16px;
    align-items: start;
  }
  @media (max-width: 760px) {
    .acct-grid { grid-template-columns: 1fr; }
  }
  /* 功能卡：details/summary 实现，点开才展开表单 */
  .acct-card {
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    transition: border-color .2s ease, box-shadow .2s ease;
  }
  .acct-card[open] {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
  }
  .acct-card > summary {
    position: relative;
    display: block;
    padding: 18px 56px 18px 20px;
    border-radius: 14px;
    cursor: pointer;
    list-style: none;
  }
  .acct-card > summary::-webkit-details-marker { display:
    none; }
  .acct-card > summary:hover { background: #fafbfa; }
  .acct-card > summary:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
  .acct-card__title {
    display: block;
    font-size: 15px;
    font-weight: 600;
  }
```

```
.acct-card__value {
  display: block;
  margin-top: 3px;
  font-size: 14px;
  color: #6e6e73;
  overflow-wrap: anywhere;
}
.acct-card__icon {
  position: absolute;
  top: 17px;
  right: 17px;
  display: grid;
  place-items: center;
  width: 27px;
  height: 27px;
  border-radius: 8px;
  background: var(--accent-soft);
  color: var(--accent);
}
.acct-card__body {
  padding: 4px 20px 20px;
  border-top: 1px solid #f0f1f2;
  margin-top: 2px;
}
.acct-card__body .field:first-child { margin-top: 16px; }
.acct-card__body .btn { width: 100%; }
/* 只读信息列表 */
.acct-kv {
  display: grid;
  grid-template-columns: 88px 1fr;
  gap: 7px 12px;
  margin: 16px 0 0;
  font-size: 13.5px;
}
.acct-kv dt { color: var(--ink-soft); }
.acct-kv dd { margin: 0; }
.field__error {
  margin: 5px 0 0;
  font-size: 12.5px;
  color: var(--danger);
}
/* 管理员分类（账号 / 操作日志）里的表格面板 */
.acct-panel {
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  overflow: hidden;
}
.acct-panel__head {
  display: flex;
```

```
      align-items: center;                                                    7451
      justify-content: space-between;                                         7452
      gap: 12px;                                                              7453
      padding: 13px 20px;                                                     7454
      border-bottom: 1px solid #f0f1f2;                                       7455
    }                                                                         7456
    .acct-panel__count {                                                      7457
      margin: 0;                                                              7458
      font-size: 13.5px;                                                      7459
      color: #6e6e73;                                                         7460
    }                                                                         7461
    .acct-panel table { margin: 0; }                                          7462
    .acct-panel th,                                                           7463
    .acct-panel td { border: 0; border-bottom: 1px solid                      7464
      #f0f1f2; }                                                              7465
    .acct-panel thead th { background: #f7f9f8; }                             7466
    .acct-panel tbody tr:last-child td { border-bottom: 0; }                  7467
    .acct-panel tbody tr:hover { background: #fafbfa; }                       7468
</style>                                                                      7469
<div class="acct">                                                            7470
    {# ---------------- 左栏：身份 + 分类 ---------------- #}                  7471
    <aside class="acct-rail">                                                 7472
      {% if current_user.avatar %}                                            7473
      <img class="acct-rail__avatar"                                          7474
           src="{{ url_for('static', filename=current_user.ava                7475
      tar) }}"                                                                7476
           alt="{{ current_user.shown_name }} 的头像">                        7477
      {% else %}                                                              7478
      <span class="acct-rail__avatar" aria-hidden="true">{{                   7479
      current_user.initial }}</span>                                          7480
      {% endif %}                                                             7481
      <p class="acct-rail__name">{{ current_user.shown_name                   7482
      }}</p>                                                                  7483
      <p class="acct-rail__sub">                                              7484
        {{ role_labels.get(current_user.role,                                 7485
      current_user.role) }}                                                   7486
        {% if current_user.email %} · {{ current_user.email                   7487
      }}{% endif %}                                                           7488
      </p>                                                                    7489
      <nav class="acct-tabs" aria-label="个人设置分类">                       7490
        <a href="{{ url_for('auth.profile', tab='info') }}"                   7491
           {% if tab == 'info' %}class="is-active"                            7492
      aria-current="page"{% endif %}>个人信息</a>                             7493
        <a href="{{ url_for('auth.profile', tab='security')                   7494
      }}"                                                                     7495
           {% if tab == 'security' %}class="is-active"                        7496
      aria-current="page"{% endif %}>登录与安全性</a>                         7497
        {% if current_user.is_admin %}                                        7498
        <a href="{{ url_for('auth.profile', tab='users') }}"                  7499
           {% if tab == 'users' %}class="is-active"                           7500
```

```
    aria-current="page"{% endif %}>账号</a>                          7501
      <a href="{{ url_for('auth.profile', tab='logs') }}"            7502
         {% if tab == 'logs' %}class="is-active"                     7503
    aria-current="page"{% endif %}>操作日志</a>                      7504
      {% endif %}                                                    7505
    </nav>                                                           7506
    <details class="acct-avatar-edit">                               7507
      <summary>更换头像</summary>                                    7508
      <div class="acct-avatar-edit__body">                           7509
        <form method="post" action="/profile/avatar"                 7510
              enctype="multipart/form-data">                         7511
          {{ avatar_form.hidden_tag() }}                             7512
          <input type="hidden" name="tab" value="{{ tab }}">         7513
          <label class="visually-hidden"                             7514
    for="avatar-file">选择头像图片</label>                           7515
          <input id="avatar-file" type="file" name="avatar"          7516
                 accept=".png,.jpg,.jpeg,.gif,.webp">                7517
          <p class="acct-avatar-edit__file">PNG / JPG / GIF          7518
    / WebP，不超过 4 MB</p>                                          7519
          <button type="submit" class="btn                           7520
    btn--sm">上传头像</button>                                       7521
        </form>                                                      7522
        {% if current_user.avatar %}                                 7523
        <form method="post" action="/profile/avatar/delete"          7524
              onsubmit="return confirm('确定移除当前头像吗？');">    7525
          {{ avatar_form.hidden_tag() }}                             7526
          <input type="hidden" name="tab" value="{{ tab }}">         7527
          <button type="submit" class="btn btn--sm                   7528
    btn--ghost">移除头像</button>                                    7529
        </form>                                                      7530
        {% endif %}                                                  7531
      </div>                                                         7532
    </details>                                                       7533
  </aside>                                                           7534
  {# ---------------- 右栏：功能卡 ---------------- #}               7535
  <div class="acct-main">                                            7536
    {% if tab == 'security' %}                                       7537
    <h1 class="acct-title">登录与安全性</h1>                         7538
    <p class="acct-lead">管理与登录账号、账号安全有关的设置。</p>   7539
    <div class="acct-grid">                                          7540
      <details class="acct-card"                                     7541
               {% if pwd_form.old_password.errors or                 7542
    pwd_form.new_password.errors or pwd_form.confirm.errors          7543
    %}open{% endif %}>                                               7544
        <summary>                                                    7545
          <span class="acct-card__title">密码</span>                 7546
          <span class="acct-card__value">修改登录密码</span>         7547
          <span class="acct-card__icon">{{ icon("lock", 15)          7548
    }}</span>                                                        7549
        </summary>                                                   7550
```

```
      <div class="acct-card__body">                                    7551
        <form method="post">                                           7552
          {{ pwd_form.hidden_tag() }}                                  7553
          <input type="hidden" name="action"                           7554
value="password">                                                      7555
          <input type="hidden" name="tab"                              7556
value="security">                                                      7557
          <div class="field">                                          7558
            <label class="field__label" for="{{                        7559
pwd_form.old_password.id }}">当前密码</label>                          7560
            {{ pwd_form.old_password(id=pwd_form.old_passw             7561
ord.id,                                                                7562
                                                                       7563
autocomplete="current-password") }}                                    7564
            {% for e in pwd_form.old_password.errors %}                7565
            <p class="field__error">{{ e }}</p>                        7566
            {% endfor %}                                               7567
          </div>                                                       7568
          <div class="field">                                          7569
            <label class="field__label" for="{{                        7570
pwd_form.new_password.id }}">新密码</label>                            7571
            {{ pwd_form.new_password(id=pwd_form.new_passw             7572
ord.id,                                                                7573
                                                                       7574
autocomplete="new-password") }}                                        7575
            {% for e in pwd_form.new_password.errors %}                7576
            <p class="field__error">{{ e }}</p>                        7577
            {% endfor %}                                               7578
          </div>                                                       7579
          <div class="field">                                          7580
            <label class="field__label" for="{{                        7581
pwd_form.confirm.id }}">确认新密码</label>                             7582
            {{ pwd_form.confirm(id=pwd_form.confirm.id,                7583
                                autocomplete="new-password             7584
") }}                                                                  7585
            {% for e in pwd_form.confirm.errors %}                     7586
            <p class="field__error">{{ e }}</p>                        7587
            {% endfor %}                                               7588
          </div>                                                       7589
          <p class="actions">{{ pwd_form.submit_pwd(class_             7590
="btn") }}</p>                                                         7591
        </form>                                                        7592
      </div>                                                           7593
    </details>                                                         7594
    <details class="acct-card">                                        7595
      <summary>                                                        7596
        <span class="acct-card__title">登录账号</span>                 7597
        <span class="acct-card__value">{{                              7598
current_user.username }}</span>                                        7599
        <span class="acct-card__icon">{{ icon("user", 15)              7600
```

```
}}</span>                                                                7601
      </summary>                                                         7602
      <div class="acct-card__body">                                      7603
        <dl class="acct-kv">                                             7604
          <dt>登录账号</dt><dd class="mono">{{                           7605
current_user.username }}</dd>                                            7606
          <dt>职务</dt>                                                  7607
          <dd>{{ role_labels.get(current_user.role,                      7608
current_user.role) }}</dd>                                               7609
          <dt>状态</dt>                                                  7610
          <dd>                                                           7611
            {% if current_user.active %}                                 7612
            <span class="tag tag--ok">启用</span>                        7613
            {% else %}                                                   7614
            <span class="tag tag--warn">停用</span>                      7615
            {% endif %}                                                  7616
          </dd>                                                          7617
          <dt>创建时间</dt>                                              7618
          <dd class="mono">{{ current_user.created_at |                  7619
localdt }}</dd>                                                          7620
        </dl>                                                            7621
      </div>                                                             7622
    </details>                                                           7623
</div>                                                                   7624
{% elif tab == 'users' %}                                                7625
<h1 class="acct-title">账号</h1>                                         7626
<p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p 7627
>                                                                        7628
{% include "auth/_users.html" %}                                         7629
{% elif tab == 'logs' %}                                                 7630
<h1 class="acct-title">操作日志</h1>                                     7631
<p class="acct-lead">记录各账号在本系统中的关键操作。</p>               7632
{% include "auth/_logs.html" %}                                          7633
{% else %}                                                               7634
<h1 class="acct-title">个人信息</h1>                                     7635
<p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>          7636
<form method="post">                                                     7637
  {{ info_form.hidden_tag() }}                                           7638
  <input type="hidden" name="action" value="info">                       7639
  <input type="hidden" name="tab" value="info">                          7640
  <div class="acct-grid">                                                7641
    <details class="acct-card" {% if                                     7642
info_form.display_name.errors %}open{% endif %}>                         7643
      <summary>                                                          7644
        <span class="acct-card__title">昵称</span>                       7645
        <span class="acct-card__value">                                  7646
          {{ current_user.display_name or "未设置" }}                    7647
        </span>                                                          7648
        <span class="acct-card__icon">{{ icon("user",                    7649
15) }}</span>                                                            7650
```

```
        </summary>                                                    7651
        <div class="acct-card__body">                                 7652
          <div class="field">                                         7653
            <label class="field__label" for="{{                       7654
info_form.display_name.id }}">                                        7655
              昵称                                                    7656
            </label>                                                  7657
            {{ info_form.display_name(id=info_form.display            7658
_name.id,                                                             7659
                                                                      7660
placeholder="留空则显示账户名称") }}                                  7661
            {% for e in info_form.display_name.errors %}              7662
            <p class="field__error">{{ e }}</p>                       7663
            {% endfor %}                                              7664
          </div>                                                      7665
          <p class="actions">{{ info_form.submit_info(clas            7666
s_="btn") }}</p>                                                      7667
        </div>                                                        7668
      </details>                                                      7669
      <details class="acct-card" {% if                                7670
info_form.gender.errors %}open{% endif %}>                            7671
        <summary>                                                     7672
          <span class="acct-card__title">性别</span>                  7673
          <span class="acct-card__value">{{                           7674
current_user.gender or "未设置" }}</span>                             7675
          <span class="acct-card__icon">{{ icon("user",               7676
15) }}</span>                                                         7677
        </summary>                                                    7678
        <div class="acct-card__body">                                 7679
          <div class="field">                                         7680
            <label class="field__label" for="{{                       7681
info_form.gender.id }}">性别</label>                                  7682
            {{ info_form.gender(id=info_form.gender.id) }}            7683
            {% for e in info_form.gender.errors %}                    7684
            <p class="field__error">{{ e }}</p>                       7685
            {% endfor %}                                              7686
          </div>                                                      7687
          <p class="actions">{{ info_form.submit_info(clas            7688
s_="btn") }}</p>                                                      7689
        </div>                                                        7690
      </details>                                                      7691
      <details class="acct-card" {% if                                7692
info_form.email.errors %}open{% endif %}>                             7693
        <summary>                                                     7694
          <span class="acct-card__title">邮箱</span>                  7695
          <span class="acct-card__value">{{                           7696
current_user.email or "未设置" }}</span>                              7697
          <span class="acct-card__icon">{{ icon("mail",               7698
15) }}</span>                                                         7699
        </summary>                                                    7700
```

```
            <div class="acct-card__body">
              <div class="field">
                <label class="field__label" for="{{ info_form.email.id }}">邮箱</label>
                {{ info_form.email(id=info_form.email.id) }}
                {% for e in info_form.email.errors %}
                <p class="field__error">{{ e }}</p>
                {% endfor %}
              </div>
              <p class="actions">{{ info_form.submit_info(class_="btn") }}</p>
            </div>
          </details>
          <details class="acct-card" {% if info_form.phone.errors %}open{% endif %}>
            <summary>
              <span class="acct-card__title">手机号</span>
              <span class="acct-card__value">{{ current_user.phone or "未设置" }}</span>
              <span class="acct-card__icon">{{ icon("phone", 15) }}</span>
            </summary>
            <div class="acct-card__body">
              <div class="field">
                <label class="field__label" for="{{ info_form.phone.id }}">手机号</label>
                {{ info_form.phone(id=info_form.phone.id) }}
                {% for e in info_form.phone.errors %}
                <p class="field__error">{{ e }}</p>
                {% endfor %}
              </div>
              <p class="actions">{{ info_form.submit_info(class_="btn") }}</p>
            </div>
          </details>
        </div>
      </form>
      {% endif %}
    </div>
</div>
{% endblock %}
{% block scripts %}
<script>
// 同时只展开一张卡，避免点开多张后页面被撑得很长
(function () {
  var cards = document.querySelectorAll('.acct-card');
  Array.prototype.forEach.call(cards, function (card) {
    card.addEventListener('toggle', function () {
      if (!card.open) return;
      Array.prototype.forEach.call(cards, function (other) {
```

```
            if (other !== card && other.open) other.open =
      false;
         });
      });
   });
})();
</script>
{% endblock %}
# ==== app/templates/auth/_users.html ====
{# 管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}
<div class="acct-panel">
   <div class="acct-panel__head">
      <p class="acct-panel__count">共 {{ users|length }}
      个账号</p>
      <a class="btn btn--sm" href="{{ url_for('auth.add_user')
      }}">新增账号</a>
   </div>
   <table>
      <thead>
         <tr>
            <th scope="col">ID</th>
            <th scope="col">账户名称</th>
            <th scope="col">昵称</th>
            <th scope="col">职务</th>
            <th scope="col">状态</th>
            <th scope="col">操作</th>
         </tr>
      </thead>
      <tbody>
         {% for u in users %}
         <tr>
            <td class="num">{{ u.id }}</td>
            <td class="mono">
               {{ u.username }}
               {% if u.id == current_user.id %}<span
      class="muted">（当前登录）</span>{% endif %}
            </td>
            <td style="text-align:left;">
               {% if u.display_name %}{{ u.display_name }}
               {% else %}<span class="muted">—</span>{% endif %}
            </td>
            <td>
               <span class="tag {{ 'tag--ok' if u.role ==
      'admin' else 'tag--muted' }}">
                  {{ role_labels.get(u.role, u.role) }}
               </span>
            </td>
            <td>
               {% if u.active %}<span class="tag
      tag--ok">启用</span>
```

```
            {% else %}<span class="tag tag--muted">停用</span>{% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="{{ url_for('auth.edit_user', user_id=u.id) }}">编辑</a>
              <form method="post"
                    action="{{ url_for('auth.delete_user', user_id=u.id) }}"
                    onsubmit="return confirm('确定删除账号「{{ u.username }}」吗？\n该操作无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn link-btn--danger"
                        {% if u.id == current_user.id %}disabled
                        title="不能删除当前登录的账号"{% endif %}>删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
</div>
# ==== app/templates/auth/_logs.html ====
{# 管理员的「操作日志」分类。由 auth/profile.html 内联引入，不单独成页。 #}
<div class="acct-panel">
  <div class="acct-panel__head">
    <p class="acct-panel__count">
      共 {{ total }} 条记录，当前第 {{ page }} / {{ pages }} 页
    </p>
    {% if pages > 1 %}
    <nav class="row-actions" aria-label="日志分页">
      {% if page > 1 %}
      <a class="link-btn"
         href="{{ url_for('auth.profile', tab='logs', page=page - 1) }}">上一页</a>
      {% endif %}
      {% if page < pages %}
      <a class="link-btn"
         href="{{ url_for('auth.profile', tab='logs', page=page + 1) }}">下一页</a>
      {% endif %}
    </nav>
    {% endif %}
  </div>
  {% if not records %}
```

```
  <p class="muted" style="padding:18px 20px;">暂无操作记录。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">时间</th>
        <th scope="col">操作人</th>
        <th scope="col">操作</th>
        <th scope="col">对象</th>
        <th scope="col">说明</th>
      </tr>
    </thead>
    <tbody>
      {% for r in records %}
      <tr>
        <td class="mono nowrap">{{ r.created_at |
    localdt("%Y-%m-%d %H:%M:%S") }}</td>
        <td>{{ users_map.get(r.user_id, "—") }}</td>
        <td style="text-align:left;">{{ r.action }}</td>
        <td>
          {% if r.target_type %}{{ r.target_type }}{% if
    r.target_id %}#{{ r.target_id }}{% endif %}
          {% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ r.detail or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
</div>
# ==== app/templates/project/index.html ====
{% extends "base.html" %}
{% block title %}项目与矿山档案{% endblock %}
{% block content %}
<div class="card">
  <h2>项目与矿山档案</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/add">新建项目</a>
  </p>
  {% if not projects %}
  <p class="muted">
    暂无项目。点击上方「新建项目」开始建立评价项目，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  <table>
    <caption class="muted">共 {{ projects|length }}
```

```
      个项目</caption>                                                    7901
      <thead>                                                             7902
        <tr>                                                              7903
          <th scope="col">项目名称</th>                                   7904
          <th scope="col">编号</th>                                       7905
          <th scope="col">状态</th>                                       7906
          <th scope="col">图斑数</th>                                     7907
          <th scope="col">操作</th>                                       7908
        </tr>                                                             7909
      </thead>                                                            7910
      <tbody>                                                             7911
        {% for p in projects %}                                           7912
        <tr>                                                              7913
          <td style="text-align:left;">{{ p.name }}</td>                  7914
          <td class="mono">{{ p.code or "—" }}</td>                       7915
          <td><span class="tag tag--muted">{{ p.status                    7916
      }}</span></td>                                                      7917
          <td class="num">                                                7918
            {% if p.patches.count() %}                                    7919
              {{ p.patches.count() }}                                     7920
            {% else %}                                                    7921
              <span class="tag tag--warn">0</span>                        7922
            {% endif %}                                                   7923
          </td>                                                           7924
          <td class="nowrap">                                             7925
            <span class="row-actions">                                    7926
              <a class="link-btn" href="/project/{{ p.id                  7927
      }}/patches">{% if p.patches.count() %}矿山图斑{% else %}+           7928
      新建图斑{% endif %}</a>                                             7929
              <a class="link-btn" href="/project/{{ p.id                  7930
      }}/edit">编辑</a>                                                   7931
              <form method="post" action="/project/{{ p.id                7932
      }}/delete"                                                          7933
                    onsubmit="return confirm('确定删除项目「{{            7934
      p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  7935
                {{ form.hidden_tag() }}                                   7936
                <button type="submit" class="link-btn                     7937
      link-btn--danger">删除</button>                                     7938
              </form>                                                     7939
            </span>                                                       7940
          </td>                                                           7941
        </tr>                                                             7942
        {% endfor %}                                                      7943
      </tbody>                                                            7944
    </table>                                                              7945
    {% endif %}                                                           7946
</div>                                                                    7947
{% endblock %}                                                            7948
# ==== app/templates/project/edit.html ====                               7949
{% extends "base.html" %}                                                 7950
```

```
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%
  endblock %}
{% block content %}
<div class="card" style="max-width:640px;">
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="name">项目名称</label>
      <input id="name" type="text" name="name"
    style="width:100%;" required
             value="{{ project.name if project else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="code">项目编号</label>
      <input id="code" type="text" name="code"
    style="width:240px;"
             value="{{ project.code or '' if project else
    '' }}">
    </div>
    {% if project %}
    <div class="field">
      <label class="field__label" for="status">状态</label>
      <select id="status" name="status">
        {% for s in ["进行中", "已完成", "已归档"] %}
        <option value="{{ s }}" {% if project.status == s
    %}selected{% endif %}>{{ s }}</option>
        {% endfor %}
      </select>
    </div>
    {% endif %}
    <div class="field">
      <label class="field__label" for="description">项目说明</la
    bel>
      <textarea id="description" name="description">{{
    project.description or '' if project else ''
    }}</textarea>
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/project/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/project/patches.html ====
{% extends "base.html" %}
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}
{% block content %}
<div class="card">
```

```
<h2>矿山图斑 · {{ project.name }}</h2>
<p class="actions" style="margin-bottom:12px;">
  <a class="btn btn--sm" href="/project/{{ project.id
  }}/patch/add">新建图斑</a>
  <a class="btn btn--sm btn--ghost"
  href="/project/">返回项目列表</a>
</p>
{% if not project.patches.count() %}
<p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
{% else %}
<table>
  <caption class="muted">共 {{ project.patches.count() }}
  个图斑</caption>
  <thead>
    <tr>
      <th scope="col">图斑名称</th>
      <th scope="col">行政区</th>
      <th scope="col">面积（公顷）</th>
      <th scope="col">开采矿种</th>
      <th scope="col">微地质单元</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in project.patches %}
    <tr>
      <td style="text-align:left;">{{ p.name }}</td>
      <td>{{ p.admin_region or "—" }}</td>
      <td class="num">{{ p.area_ha if p.area_ha is not
  none else "—" }}</td>
      <td>{{ p.mineral_type or "—" }}</td>
      <td class="num">{{ p.units.count() }}</td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/unit/{{ p.id
  }}">微地质单元</a>
          <a class="link-btn" href="/constraint/{{ p.id
  }}">限制性判定</a>
          <a class="link-btn" href="/project/patch/{{
  p.id }}/edit">编辑</a>
          <form method="post" action="/project/patch/{{
  p.id }}/delete"
                onsubmit="return confirm('确定删除图斑「{{
  p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">
            {{ form.hidden_tag() }}
            <button type="submit" class="link-btn
  link-btn--danger">删除</button>
          </form>
        </span>
      </td>
```

```
        </tr>                                                                  8051
        {% endfor %}                                                           8052
      </tbody>                                                                 8053
    </table>                                                                   8054
    {% endif %}                                                                8055
</div>                                                                         8056
{% endblock %}                                                                 8057
# ==== app/templates/project/patch_edit.html ====                              8058
{% extends "base.html" %}                                                      8059
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%                    8060
    endblock %}                                                                8061
{% block content %}                                                            8062
<div class="card" style="max-width:720px;">                                    8063
  <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name              8064
    }}</h2>                                                                    8065
  <form method="post">                                                         8066
    {{ form.hidden_tag() }}                                                    8067
    <div class="field">                                                        8068
      <label class="field__label" for="patch-name">图斑名称</lab               8069
    el>                                                                        8070
      <input id="patch-name" type="text" name="name"                           8071
    style="width:100%;" required                                               8072
             value="{{ patch.name if patch else '' }}">                        8073
    </div>                                                                     8074
    <div class="field">                                                        8075
      <label class="field__label" for="admin-region">所在行政区</              8076
    label>                                                                     8077
      <input id="admin-region" type="text"                                     8078
    name="admin_region" style="width:100%;"                                    8079
             value="{{ patch.admin_region or '' if patch                       8080
    else '' }}">                                                               8081
    </div>                                                                     8082
    <div class="field">                                                        8083
      <label class="field__label" for="area-ha">图斑面积（公顷）</la           8084
    bel>                                                                       8085
      <input id="area-ha" type="number" step="any"                             8086
    name="area_ha"                                                             8087
             value="{{ patch.area_ha if patch and                              8088
    patch.area_ha is not none else '' }}">                                     8089
    </div>                                                                     8090
    <div class="field">                                                        8091
      <label class="field__label" for="mineral-type">开采矿种</l               8092
    abel>                                                                      8093
      <input id="mineral-type" type="text"                                     8094
    name="mineral_type" style="width:240px;"                                   8095
             value="{{ patch.mineral_type or '' if patch                       8096
    else '' }}">                                                               8097
    </div>                                                                     8098
    <div class="field">                                                        8099
      <label class="field__label" for="mining-method">开采方式</               8100
```

```
label>                                                                  8101
  <input id="mining-method" type="text"                                 8102
name="mining_method" style="width:240px;"                               8103
          value="{{ patch.mining_method or '' if patch                  8104
else '' }}">                                                            8105
</div>                                                                  8106
<div class="field">                                                     8107
  <label class="field__label" for="stop-years">停产年限</lab            8108
el>                                                                     8109
  <input id="stop-years" type="number" step="any"                       8110
name="stop_years"                                                       8111
          value="{{ patch.stop_years if patch and                       8112
patch.stop_years is not none else '' }}">                               8113
</div>                                                                  8114
<div class="field">                                                     8115
  <label class="checkbox" for="responsible-missing">                    8116
    <input id="responsible-missing" type="checkbox"                     8117
name="responsible_entity_missing" value="1"                             8118
            {% if not patch or patch.responsible_entity_m               8119
issing %}checked{% endif %}>                                            8120
    <span>责任主体缺失</span>                                           8121
  </label>                                                              8122
</div>                                                                  8123
<h3>参照生态系统（指标标准化基准）</h3>                                 8124
<p class="muted">                                                       8125
  论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。   8126
  该基准与图斑绑定管理，可按不同参照生态系统分别设定。                 8127
</p>                                                                    8128
<div class="field">                                                     8129
  <label class="field__label" for="coverage-benchmark">植               8130
被覆盖度基准（%）</label>                                               8131
  <input id="coverage-benchmark" type="number"                          8132
step="any" name="coverage_benchmark"                                    8133
          value="{{ reference.coverage_benchmark if                     8134
reference else 75 }}">                                                  8135
</div>                                                                  8136
<div class="field">                                                     8137
  <label class="field__label" for="shannon-benchmark">Sh                8138
annon-Wiener 多样性指数基准</label>                                     8139
  <input id="shannon-benchmark" type="number"                           8140
step="any" name="shannon_benchmark"                                     8141
          value="{{ reference.shannon_benchmark if                      8142
reference else 0.7 }}">                                                 8143
</div>                                                                  8144
<div class="field">                                                     8145
  <label class="field__label" for="dominant-species">建群种             8146
</label>                                                                8147
  <input id="dominant-species" type="text"                              8148
name="dominant_species" style="width:100%;"                             8149
          value="{{ reference.dominant_species or '' if                8150
```

```
      reference else '' }}">                                                8151
      </div>                                                                8152
      <p class="actions" style="margin-top:18px;">                          8153
        <button type="submit" class="btn">保存</button>                     8154
        <a class="btn btn--ghost" href="/project/{{                         8155
      project.id }}/patches">返回</a>                                       8156
      </p>                                                                  8157
    </form>                                                                 8158
</div>                                                                      8159
{% if patch %}                                                              8160
<div class="card" style="max-width:720px;">                                 8161
  <h2>遥感影像挂接</h2>                                                     8162
  <p class="muted">                                                         8163
    挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。 8164
    支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。            8165
  </p>                                                                      8166
  {% if patch.image_path %}                                                 8167
  <figure style="margin:0 0 16px;">                                         8168
    <img src="{{ url_for('static', filename=patch.image_path8169
    ) }}"                                                                   8170
         alt="图斑「{{ patch.name }}」已挂接的遥感影像"                     8171
         style="max-width:100%;border:1px solid                             8172
    var(--line);border-radius:6px;">                                        8173
    <figcaption class="muted" style="margin-top:6px;">                      8174
      {{ patch.image_note or "遥感影像" }}                                  8175
      {% if patch.image_uploaded_at %}                                      8176
      · 上传于 {{ patch.image_uploaded_at | localdt }}                      8177
      {% endif %}                                                           8178
    </figcaption>                                                           8179
  </figure>                                                                 8180
  <form method="post" action="/project/patch/{{ patch.id                    8181
    }}/image/delete"                                                        8182
        onsubmit="return confirm('确定移除该遥感影像吗？');">               8183
    {{ form.hidden_tag() }}                                                 8184
    <button type="submit" class="btn btn--sm                                8185
    btn--danger">移除影像</button>                                          8186
  </form>                                                                   8187
  <hr style="margin:16px 0;border:none;border-top:1px solid                 8188
    var(--line);">                                                          8189
  {% endif %}                                                               8190
  <form method="post" action="/project/patch/{{ patch.id                    8191
    }}/image"                                                               8192
        enctype="multipart/form-data">                                      8193
    {{ form.hidden_tag() }}                                                 8194
    <div class="field">                                                     8195
      <label class="field__label" for="image-file">                         8196
        {{ "替换影像文件" if patch.image_path else "选择影像文件" }}        8197
      </label>                                                              8198
      <input id="image-file" type="file" name="image"                       8199
             accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t8200
```

```
      iff" required>                                                    8201
      </div>                                                            8202
      <div class="field">                                               8203
        <label class="field__label" for="image-note">影像说明（如来源、8204
      时相）</label>                                                    8205
        <input id="image-note" type="text" name="image_note"            8206
      style="width:100%;"                                               8207
               placeholder="如：2025 年 6 月无人机正射影像"             8208
               value="{{ patch.image_note or '' }}">                    8209
      </div>                                                            8210
      <p class="actions">                                               8211
        <button type="submit" class="btn">{{ "替换影像" if             8212
      patch.image_path else "上传并挂接" }}</button>                    8213
      </p>                                                              8214
    </form>                                                             8215
</div>                                                                  8216
{% endif %}                                                             8217
{% endblock %}                                                          8218
# ==== app/templates/unit/list.html ====                                8219
{% extends "base.html" %}                                               8220
{% from "_macros.html" import patch_guide %}                            8221
{% block title %}微地质单元{% endblock %}                               8222
{% block content %}                                                     8223
<div class="card">                                                      8224
  <h2>微地质单元</h2>                                                   8225
  {{ patch_guide(projects_without_patch) }}                             8226
  {% if not rows %}                                                     8227
  <p class="muted">                                                     8228
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。                   8229
  </p>                                                                  8230
  {% else %}                                                            8231
  <table>                                                               8232
    <thead>                                                             8233
      <tr>                                                              8234
        <th scope="col">所属项目</th>                                   8235
        <th scope="col">矿山图斑</th>                                   8236
        <th scope="col">已划分单元</th>                                 8237
        <th scope="col">占比合计</th>                                   8238
        <th scope="col">操作</th>                                       8239
      </tr>                                                             8240
    </thead>                                                            8241
    <tbody>                                                             8242
      {% for r in rows %}                                               8243
      <tr>                                                              8244
        <td>{{ r.patch.project.name if r.patch.project else             8245
      "—" }}</td>                                                       8246
        <td style="text-align:left;">{{ r.patch.name }}</td>            8247
        <td class="num">{{ r.count }} 个</td>                           8248
        <td class="num">                                                8249
          {% if r.count %}                                              8250
```

```
                {{ "%.2f%%"|format(r.ratio_sum * 100) }}                8251
                {% if r.ratio_sum < 0.999 or r.ratio_sum >              8252
    1.001 %}                                                            8253
                <span class="warn">（不等于 100%）</span>                 8254
                {% endif %}                                             8255
              {% else %}—{% endif %}                                    8256
            </td>                                                       8257
            <td class="nowrap">                                         8258
              <span class="row-actions">                                8259
                <a class="link-btn" href="/unit/{{ r.patch.id           8260
    }}">划分与维护</a>                                                  8261
                <a class="link-btn" href="/indicator/">录入指标数据</a  8262
    >                                                                   8263
              </span>                                                   8264
            </td>                                                       8265
          </tr>                                                         8266
          {% endfor %}                                                  8267
        </tbody>                                                        8268
      </table>                                                          8269
      {% endif %}                                                       8270
</div>                                                                  8271
{% endblock %}                                                          8272
# ==== app/templates/unit/index.html ====                               8273
{% extends "base.html" %}                                               8274
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}            8275
{% block content %}                                                     8276
<div class="card">                                                      8277
  <h2>微地质单元 · {{ patch.name }}</h2>                                8278
  {% if units %}                                                        8279
  <table>                                                               8280
    <caption class="muted">                                             8281
      共 {{ units|length }} 个单元；                                    8282
      面积合计 <span class="num">{{ "%.4f"|format(total_area)           8283
    }}</span>；                                                         8284
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *         8285
    100) }}</span>                                                      8286
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >             8287
    1.001) %}                                                           8288
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>  8289
      {% endif %}                                                       8290
    </caption>                                                          8291
    <thead>                                                             8292
      <tr>                                                              8293
        <th scope="col">微地质单元</th>                                 8294
        <th scope="col">面积</th>                                       8295
        <th scope="col">面积占比</th>                                   8296
        <th scope="col">说明</th>                                       8297
        <th scope="col">操作</th>                                       8298
      </tr>                                                             8299
    </thead>                                                            8300
```

```
    <tbody>
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none
    else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or
    "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/edit/{{ u.id
    }}">编辑</a>
            <form method="post" action="/unit/delete/{{
    u.id }}"
                  onsubmit="return confirm('确定删除微地质单元「{{
    u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">尚未划分微地质单元。请在下方添加。</p>
  {% endif %}
  <h3>添加微地质单元</h3>
  <form method="post" action="/unit/{{ patch.id }}/add">
    {{ form.hidden_tag() }}
    <div class="actions">
      <label class="field__label" for="add-unit-type"
    style="margin:0;">类型</label>
      <select id="add-unit-type" name="unit_type">
        {% for t in unit_types %}<option value="{{ t }}">{{
    t }}</option>{% endfor %}
      </select>
      <label class="field__label" for="add-unit-area"
    style="margin:0;">面积</label>
      <input id="add-unit-area" type="number" step="any"
    name="area">
      <label class="field__label" for="add-unit-desc"
    style="margin:0;">说明</label>
      <input id="add-unit-desc" type="text"
```

```
      name="description" style="width:220px;">                    8351
        <button type="submit" class="btn btn--sm">添加</button>  8352
      </div>                                                      8353
    </form>                                                       8354
    <p class="actions" style="margin-top:20px;">                  8355
      <a class="btn btn--sm btn--ghost" href="/project/{{         8356
      patch.project_id }}/patches">返回图斑列表</a>               8357
      <a class="btn btn--sm btn--ghost"                           8358
      href="/indicator/">去录入指标数据</a>                       8359
    </p>                                                          8360
</div>                                                            8361
{% endblock %}                                                    8362
# ==== app/templates/unit/edit.html ====                          8363
{% extends "base.html" %}                                         8364
{% block title %}编辑微地质单元{% endblock %}                     8365
{% block content %}                                               8366
<div class="card">                                                8367
    <h2>编辑微地质单元 · {{ patch.name }}</h2>                    8368
    <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>  8369
    <form method="post">                                          8370
      {{ form.hidden_tag() }}                                     8371
      <div class="field">                                         8372
        <label class="field__label" for="unit-type">微地质单元类型</l 8373
      abel>                                                       8374
        <select id="unit-type" name="unit_type">                  8375
          {% for t in unit_types %}                               8376
          <option value="{{ t }}" {% if unit.unit_type == t       8377
      %}selected{% endif %}>{{ t }}</option>                      8378
          {% endfor %}                                            8379
        </select>                                                 8380
      </div>                                                      8381
      <div class="field">                                         8382
        <label class="field__label" for="unit-area">面积</label>  8383
        <input id="unit-area" type="number" step="any"            8384
      name="area"                                                 8385
               value="{{ unit.area if unit.area is not none       8386
      else '' }}">                                                8387
      </div>                                                      8388
      <div class="field">                                         8389
        <label class="field__label" for="unit-desc">说明</label>  8390
        <input id="unit-desc" type="text" name="description"      8391
      style="width:420px;"                                        8392
               value="{{ unit.description or '' }}">              8393
      </div>                                                      8394
      <p class="actions" style="margin-top:16px;">                8395
        <button type="submit" class="btn">保存</button>           8396
        <a class="btn btn--ghost" href="/unit/{{                  8397
      unit.patch_id }}">返回</a>                                  8398
      </p>                                                        8399
    </form>                                                       8400
```

```
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>  <p class="actions"
    style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
  </p>
  <h3>地方标准</h3>
  {% if local %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">编号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in local %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
```

```
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无地方标准。运行 <span class="mono">python seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
  {% endif %}
  <h3>政策文件</h3>
  {% if policy %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">文号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in policy %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id }}/delete"
                  onsubmit="return confirm('确定删除「{{ s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无政策文件。</p>
  {% endif %}
```

```
</div>                                                                  8501
{% endblock %}                                                          8502
# ==== app/templates/standard/edit.html ====                            8503
{% extends "base.html" %}                                               8504
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%          8505
   endblock %}                                                          8506
{% block content %}                                                     8507
<div class="card" style="max-width:800px;">                             8508
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                 8509
  <form method="post">                                                  8510
    {{ form.hidden_tag() }}                                             8511
    <div class="field">                                                 8512
      <label class="field__label" for="province">省份</label>           8513
      <input id="province" type="text" name="province"                  8514
    style="width:160px;" required                                       8515
            value="{{ standard.province if standard else                8516
    '' }}">                                                             8517
    </div>                                                              8518
    <div class="field">                                                 8519
      <label class="field__label" for="std-name">名称</label>           8520
      <input id="std-name" type="text" name="name"                      8521
    style="width:100%;" required                                        8522
            value="{{ standard.name if standard else ''                 8523
    }}">                                                                8524
    </div>                                                              8525
    <div class="field">                                                 8526
      <label class="field__label" for="doc-number">标准编号 /           8527
    文号</label>                                                        8528
      <input id="doc-number" type="text" name="doc_number"              8529
    style="width:320px;"                                                8530
            value="{{ standard.doc_number or '' if                      8531
    standard else '' }}">                                               8532
    </div>                                                              8533
    <div class="field">                                                 8534
      <label class="field__label" for="category">类别</label>           8535
      <select id="category" name="category">                            8536
        {% for c in ["地方标准", "政策文件"] %}                         8537
        <option value="{{ c }}" {% if standard and                      8538
    standard.category == c %}selected{% endif %}>{{ c                   8539
    }}</option>                                                         8540
        {% endfor %}                                                    8541
      </select>                                                         8542
    </div>                                                              8543
    <div class="field">                                                 8544
      <label class="field__label" for="publish-date">发布时间</l        8545
    abel>                                                               8546
      <input id="publish-date" type="text"                              8547
    name="publish_date" style="width:200px;"                            8548
            placeholder="如 2023-03-30"                                 8549
            value="{{ standard.publish_date or '' if                    8550
```

```
      standard else '' }}">                                                    8551
      </div>                                                                   8552
      <div class="field">                                                      8553
        <label class="field__label" for="implement-date">实施时间<             8554
      /label>                                                                  8555
        <input id="implement-date" type="text"                                 8556
      name="implement_date" style="width:200px;"                               8557
              placeholder="如 2023-04-30（未明确可留空）"                        8558
              value="{{ standard.implement_date or '' if                       8559
      standard else '' }}">                                                    8560
      </div>                                                                   8561
      <div class="field">                                                      8562
        <label class="field__label" for="summary">特点 /                       8563
      判定标准</label>                                                         8564
        <textarea id="summary" name="summary"                                  8565
      style="min-height:96px;">{{ standard.summary or '' if                    8566
      standard else '' }}</textarea>                                           8567
      </div>                                                                   8568
      <p class="actions" style="margin-top:18px;">                             8569
        <button type="submit" class="btn">保存</button>                        8570
        {% if standard %}                                                      8571
        <form method="post" action="/standard/{{ standard.id                   8572
      }}/delete" style="display:inline;"                                       8573
              onsubmit="return confirm('确定删除「{{                           8574
      standard.province }} {{ standard.name                                    8575
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">                    8576
          {{ form.hidden_tag() }}                                              8577
          <button type="submit" class="btn                                     8578
      btn--danger">删除该标准</button>                                         8579
        </form>                                                                8580
        {% endif %}                                                            8581
        <a class="btn btn--ghost" href="/standard/">返回</a>                   8582
      </p>                                                                     8583
    </form>                                                                    8584
</div>                                                                         8585
{% endblock %}                                                                 8586
# ==== app/templates/standard/rules.html ====                                  8587
{% extends "base.html" %}                                                      8588
{% block title %}结构化规则 · {{ standard.province }}{% endblock               8589
    %}                                                                         8590
{% block content %}                                                            8591
<div class="card">                                                             8592
    <h2>结构化规则 · {{ standard.province }} {{ standard.name                  8593
      }}</h2>                                                                  8594
    {% if rules %}                                                             8595
    <table>                                                                    8596
      <caption class="muted">共 {{ rules|length }}                             8597
      条规则</caption>                                                         8598
      <thead>                                                                  8599
        <tr>                                                                   8600
```

```
          <th scope="col">判定项</th>                                    8601
          <th scope="col">条件</th>                                      8602
          <th scope="col">条件类型</th>                                  8603
          <th scope="col">适用分区</th>                                  8604
          <th scope="col">备注</th>                                      8605
          <th scope="col">操作</th>                                      8606
        </tr>                                                            8607
      </thead>                                                           8608
      <tbody>                                                            8609
        {% for r in rules %}                                             8610
        <tr>                                                             8611
          <td style="text-align:left;">{{ r.item }}</td>                 8612
          <td class="mono nowrap">{{ r.operator }} {{                    8613
      r.threshold or "" }}</td>                                          8614
          <td>{{ r.condition_type }}</td>                                8615
          <td>{{ r.region or "—" }}</td>                                 8616
          <td style="text-align:left;">{{ r.note or "—"                  8617
      }}</td>                                                            8618
          <td>                                                           8619
            <form method="post" action="/standard/rule/{{               8620
      r.id }}/delete"                                                    8621
                  onsubmit="return confirm('确定删除规则「{{ r.item      8622
      }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">             8623
              {{ form.hidden_tag() }}                                    8624
              <button type="submit" class="btn btn--sm                   8625
      btn--danger">删除</button>                                         8626
            </form>                                                      8627
          </td>                                                          8628
        </tr>                                                            8629
        {% endfor %}                                                     8630
      </tbody>                                                           8631
    </table>                                                             8632
    {% else %}                                                           8633
    <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>                  8634
    {% endif %}                                                          8635
    <h3>添加规则</h3>                                                    8636
    <form method="post">                                                 8637
      {{ form.hidden_tag() }}                                            8638
      <div class="actions">                                              8639
        <div>                                                            8640
          <label class="field__label" for="r-item">判定项</label>        8641
          <input id="r-item" type="text" name="item"                     8642
      placeholder="如：植被覆盖度" required>                             8643
        </div>                                                           8644
        <div>                                                            8645
          <label class="field__label" for="r-op">比较符</label>          8646
          <select id="r-op" name="operator">                             8647
            {% for o in [">=", ">", "<=", "<", "=="]                     8648
      %}<option value="{{ o }}">{{ o }}</option>{% endfor %}             8649
          </select>                                                      8650
```

```
        </div>
        <div>
          <label class="field__label" for="r-thr">阈值</label>
          <input id="r-thr" type="text" name="threshold"
    placeholder="如：50" style="width:110px;">
        </div>
        <div>
          <label class="field__label" for="r-cond">条件类型</label
    >
          <select id="r-cond" name="condition_type">
            <option value="必要条件">必要条件</option>
            <option value="充分条件">充分条件</option>
          </select>
        </div>
        <div>
          <label class="field__label" for="r-region">适用分区</lab
    el>
          <input id="r-region" type="text" name="region"
    placeholder="如宁夏分区" style="width:130px;">
        </div>
        <div>
          <label class="field__label" for="r-note">备注</label>
          <input id="r-note" type="text" name="note"
    style="width:220px;">
        </div>
        <div>
          <button type="submit" class="btn">添加规则</button>
        </div>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost"
      href="/standard/">返回标准列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
    <h2>省际口径比对</h2>
    {% if not groups %}
    <p class="muted">
      暂无规则数据。请先在「标准知识库」中为标准录入规则，
      或运行 <span class="mono">python seed_demo.py</span> 载入论文表
      1 数据。
    </p>
```

```
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name
    }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
  <h2>限制性指标判定</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
```

```
            <th scope="col">矿山图斑</th>                                    8751
            <th scope="col">判定结果</th>                                    8752
            <th scope="col">操作</th>                                        8753
          </tr>                                                              8754
        </thead>                                                             8755
        <tbody>                                                              8756
          {% for p in patches %}                                             8757
          {% set r = latest.get(p.id) %}                                     8758
          <tr>                                                               8759
            <td>{{ p.project.name if p.project else "—" }}</td>              8760
            <td style="text-align:left;">{{ p.name }}</td>                   8761
            <td style="text-align:left;">                                    8762
              {% if r %}                                                     8763
                {% if r.passed %}                                            8764
                <span class="tag tag--ok">通过</span>                        8765
                {% else %}                                                   8766
                <span class="tag tag--warn">不通过</span>                    8767
                {% endif %}                                                  8768
                <span class="muted">{{ r.reason }}</span>                    8769
              {% else %}                                                     8770
              <span class="tag tag--muted">未判定</span>                     8771
              {% endif %}                                                    8772
            </td>                                                            8773
            <td class="nowrap">                                              8774
              <span class="row-actions">                                     8775
                <a class="link-btn" href="/constraint/{{ p.id                8776
        }}">执行判定</a>                                                     8777
              </span>                                                        8778
            </td>                                                            8779
          </tr>                                                              8780
          {% endfor %}                                                       8781
        </tbody>                                                             8782
      </table>                                                               8783
      {% endif %}                                                            8784
    </div>                                                                   8785
    {% endblock %}                                                           8786
    # ==== app/templates/constraint/check.html ====                          8787
    {% extends "base.html" %}                                                8788
    {% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}         8789
    {% block content %}                                                      8790
    <div class="card" style="max-width:760px;">                              8791
      <h2>限制性条件判定 · {{ patch.name }}</h2>                             8792
      <form method="post">                                                   8793
        {{ form.hidden_tag() }}                                              8794
        {% for field, name, val in field_states %}                           8795
        <fieldset style="border:1px solid                                    8796
        var(--line);border-radius:6px;padding:10px                           8797
        14px;margin:0 0 10px;">                                              8798
          <legend style="padding:0 6px;font-weight:600;">{{                  8799
        name }}</legend>                                                     8800
```

```
        <div class="actions">                                                    8801
          <label class="checkbox">                                               8802
            <input type="radio" name="{{ field }}" value="1"                     8803
                   {% if val is none or val %}checked{% endif                    8804
    %}>                                                                          8805
            <span>通过</span>                                                    8806
          </label>                                                               8807
          <label class="checkbox">                                               8808
            <input type="radio" name="{{ field }}" value="0"                     8809
                   {% if val is not none and not val                             8810
    %}checked{% endif %}>                                                        8811
            <span>不通过</span>                                                  8812
          </label>                                                               8813
        </div>                                                                   8814
      </fieldset>                                                                8815
      {% endfor %}                                                               8816
      <p class="actions" style="margin-top:16px;">                               8817
        <button type="submit" class="btn">保存判定</button>                      8818
        <a class="btn btn--ghost" href="/constraint/">返回</a>                   8819
      </p>                                                                       8820
    </form>                                                                      8821
    {% if record %}                                                              8822
    <div style="margin-top:18px;padding:14px;border-radius:6px                   8823
      ;border-left:4px solid;                                                    8824
                  {% if record.passed %}background:var(--ok-soft                 8825
      );border-left-color:var(--ok);                                             8826
                  {% else %}background:var(--danger-soft);border                 8827
      -left-color:var(--danger);{% endif %}">                                    8828
      <p style="margin:0 0 4px;font-weight:600;">                                8829
        判定结论：                                                               8830
        {% if record.passed %}                                                   8831
        <span class="ok">通过，可进入定量评价</span>                             8832
        {% else %}                                                               8833
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>               8834
        {% endif %}                                                              8835
      </p>                                                                       8836
      <p class="muted" style="margin:0;">{{ record.reason                        8837
      }}</p>                                                                     8838
    </div>                                                                       8839
    {% endif %}                                                                  8840
</div>                                                                           8841
{% endblock %}                                                                   8842
# ==== app/templates/ahp/index.html ====                                         8843
{% extends "base.html" %}                                                        8844
{% block title %}AHP 权重计算{% endblock %}                                      8845
{% block content %}                                                              8846
<div class="card" style="max-width:720px;">                                      8847
    <h2>AHP 权重计算 · 选择指标层级</h2>                                        8848
    <form method="post">                                                         8849
      {{ form.hidden_tag() }}                                                    8850
```

```
      <div class="field">                                                    8851
        <label class="field__label" for="{{ form.unit_type.id                8852
      }}">微地质单元类型</label>                                              8853
        {{ form.unit_type(id=form.unit_type.id) }}                           8854
      </div>                                                                 8855
      <div class="field">                                                    8856
        <label class="field__label" for="{{ form.level.id                    8857
      }}">指标层级</label>                                                    8858
        {{ form.level(id=form.level.id) }}                                   8859
      </div>                                                                 8860
      <div class="field">                                                    8861
        <label class="field__label" for="{{ form.l1.id }}">                  8862
          一级指标（层级选「二级指标」时需选择）                                 8863
        </label>                                                             8864
        {{ form.l1(id=form.l1.id) }}                                         8865
      </div>                                                                 8866
      <p class="actions" style="margin-top:16px;">                           8867
        {{ form.submit(class_="btn") }}                                      8868
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维              8869
      护</a>                                                                 8870
        <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a            8871
      >                                                                      8872
      </p>                                                                   8873
    </form>                                                                  8874
</div>                                                                       8875
<div class="card">                                                           8876
    <h2>各微地质单元当前生效权重</h2>                                          8877
    <table>                                                                  8878
      <thead>                                                                8879
        <tr>                                                                 8880
          <th scope="col">微地质单元</th>                                     8881
          <th scope="col">当前权重来源</th>                                   8882
          <th scope="col">操作</th>                                          8883
        </tr>                                                                8884
      </thead>                                                               8885
      <tbody>                                                                8886
        {% for r in weight_rows %}                                           8887
        <tr>                                                                 8888
          <td style="text-align:left;">{{ r.unit_type }}</td>                8889
          <td style="text-align:left;">                                      8890
            {% if r.custom %}                                                8891
            <span class="tag tag--warn">{{ r.label }}</span>                 8892
            {% else %}                                                       8893
            <span class="tag tag--muted">{{ r.label }}</span>                8894
            {% endif %}                                                      8895
          </td>                                                              8896
          <td class="nowrap">                                                8897
            <span class="row-actions">                                       8898
              <a class="link-btn" href="/ahp/matrix?unit_type=               8899
      {{ r.unit_type }}&level=1">一级指标权重</a>                             8900
```

```
                <a class="link-btn" href="/ahp/matrix?unit_type=
    {{ r.unit_type }}&level=1&calc=1">查看一致性</a>
                {% if r.custom %}
                <form method="post" action="/ahp/weights/reset"
                      onsubmit="return confirm('确定把「{{
    r.unit_type }}」恢复为论文表 3 的默认权重吗？');">
                  <input type="hidden" name="csrf_token"
    value="{{ csrf_token() }}">
                  <input type="hidden" name="unit_type"
    value="{{ r.unit_type }}">
                  <button type="submit" class="link-btn
    link-btn--danger">恢复默认</button>
                </form>
                {% endif %}
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
    {% endblock %}
    # ==== app/templates/ahp/matrix.html ====
    {% extends "base.html" %}
    {% block title %}判断矩阵录入{% endblock %}
    {% block content %}
    <div class="card">
      <h2>
        判断矩阵录入
        · {{ unit_type }}
        · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%
        endif %}
      </h2>
      <p class="muted">
        采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
        2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
        矩阵已按论文表 3 的权重预填。
      </p>
      <form method="post">
        {{ form.hidden_tag() }}
        <table style="width:auto;">
          <thead>
            <tr>
              <th></th>
              {% for name in names %}<th scope="col">{{ name
    }}</th>{% endfor %}
            </tr>
          </thead>
          <tbody>
            {% for i in range(names|length) %}
```

```
        <tr>                                                                8951
          <th scope="row">{{ names[i] }}</th>                               8952
          {% for j in range(names|length) %}                                8953
            {% if i == j %}                                                 8954
            <td class="muted num">1</td>                                    8955
            {% elif i < j %}                                                8956
            <td>                                                            8957
              <label class="muted" style="position:absolute;                8958
  left:-9999px;"                                                            8959
                     for="m-{{ i }}-{{ j }}">{{ names[i] }}                 8960
  相对 {{ names[j] }} 的重要性</label>                                      8961
              <input id="m-{{ i }}-{{ j }}" type="number"                   8962
  step="0.01" min="0.01"                                                    8963
                     name="m_{{ i }}_{{ j }}" value="{{                     8964
  default[i][j] }}">                                                        8965
            </td>                                                           8966
            {% else %}                                                      8967
            <td class="muted num">{{ default[i][j] }}</td>                  8968
            {% endif %}                                                     8969
          {% endfor %}                                                      8970
        </tr>                                                               8971
        {% endfor %}                                                        8972
      </tbody>                                                              8973
  </table>                                                                  8974
  <p class="actions">                                                       8975
    <button type="submit" class="btn" name="action"                         8976
  value="calc">计算权重</button>                                            8977
    {% if result and result.consistent %}                                   8978
    <button type="submit" class="btn" name="action"                         8979
  value="save">保存并生效</button>                                          8980
    {% endif %}                                                             8981
    <a class="btn btn--ghost" href="/ahp/">重新选择</a>                     8982
  </p>                                                                      8983
</form>                                                                     8984
<p class="muted" style="margin-top:10px;">                                  8985
  当前「{{ unit_type }}」生效权重：<strong>{{ source_label                  8986
  }}</strong>。                                                             8987
  点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；      8988
  如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。           8989
</p>                                                                        8990
{% if result %}                                                             8991
<hr style="margin:22px 0;border:none;border-top:1px solid                   8992
  var(--line);">                                                            8993
<h3>计算结果</h3>                                                           8994
<table style="width:auto;">                                                 8995
  <thead>                                                                   8996
    <tr>                                                                    8997
      <th scope="col">指标</th>                                             8998
      {% for name in names %}<th scope="col">{{ name                        8999
  }}</th>{% endfor %}                                                       9000
```

```
        </tr>                                                                  9001
      </thead>                                                                 9002
      <tbody>                                                                  9003
        <tr>                                                                   9004
          <th scope="row">权重</th>                                            9005
          {% for w in result.weights %}<td class="mono">{{ w                   9006
      }}</td>{% endfor %}                                                      9007
        </tr>                                                                  9008
      </tbody>                                                                 9009
    </table>                                                                   9010
    <div class="stat-row">                                                     9011
      <div>                                                                    9012
        <div class="stat__label">最大特征值 λmax</div>                         9013
        <div class="stat__value mono">{{ result.lambda_max                     9014
      }}</div>                                                                 9015
      </div>                                                                   9016
      <div>                                                                    9017
        <div class="stat__label">一致性指标 CI</div>                           9018
        <div class="stat__value mono">{{                                       9019
      "%.4f"|format(result.ci) }}</div>                                        9020
      </div>                                                                   9021
      <div>                                                                    9022
        <div class="stat__label">一致性比例 CR</div>                           9023
        <div class="stat__value mono">{{                                       9024
      "%.4f"|format(result.cr) }}</div>                                        9025
      </div>                                                                   9026
      <div>                                                                    9027
        <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>          9028
        <div class="stat__value">                                              9029
          {% if result.consistent %}<span class="ok">通过</span>               9030
          {% else %}<span class="warn">不通过</span>{% endif %}                9031
        </div>                                                                 9032
      </div>                                                                   9033
    </div>                                                                     9034
    {% if not result.consistent and result.most_inconsistent                   9035
      %}                                                                       9036
    <p style="margin-top:14px;padding:12px;border-radius:6px;b                 9037
      ackground:var(--danger-soft);                                            9038
                border-left:4px solid var(--danger);">                         9039
      <strong class="warn">建议优先调整：</strong>                             9040
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                        9041
      result.most_inconsistent.j + 1 }} 列的比较                               9042
      （当前 <span class="mono num">{{ result.most_inconsistent.v              9043
      alue|round(3) }}</span>，                                                9044
      理想约 <span class="mono num">{{ result.most_inconsistent.e              9045
      xpected|round(3) }}</span>）                                             9046
    </p>                                                                       9047
    {% endif %}                                                                9048
  {% endif %}                                                                  9049
</div>                                                                         9050
```

```
{% endblock %}                                                                  9051
# ==== app/templates/ahp/indicators.html ====                                   9052
{% extends "base.html" %}                                                       9053
{% block title %}指标定义维护{% endblock %}                                      9054
{% block content %}                                                             9055
<div class="card">                                                              9056
  <h2>指标体系维护</h2>                                                         9057
  <p class="actions" style="margin-bottom:14px;">                               9058
    <a class="btn btn--sm btn--ghost"                                           9059
    href="/ahp/">返回权重计算</a>                                               9060
    <a class="btn btn--sm btn--ghost"                                           9061
    href="/ahp/experts">多专家矩阵</a>                                          9062
    <form method="post" action="/ahp/indicators/reset"                          9063
    style="display:inline;"                                                     9064
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 9065
    确定继续？');">                                                             9066
      {{ form.hidden_tag() }}                                                   9067
      <button type="submit" class="btn btn--sm                                  9068
    btn--danger">按论文取值重置</button>                                        9069
    </form>                                                                     9070
  </p>                                                                          9071
  {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,                 9072
    "三级指标")] %}                                                             9073
  {% set level_items = all_ind|selectattr("level",                              9074
    "equalto", level)|list %}                                                   9075
  {% if level_items %}                                                          9076
  <h3>{{ title }}（{{ level_items|length }} 项）</h3>                           9077
  <table>                                                                       9078
    <thead>                                                                     9079
      <tr>                                                                      9080
        <th scope="col">指标</th>                                               9081
        <th scope="col">上级指标</th>                                           9082
        <th scope="col">单位</th>                                               9083
        <th scope="col">标准化类型</th>                                         9084
        <th scope="col">理想取值</th>                                           9085
        <th scope="col">启用单元</th>                                           9086
        <th scope="col">限制性</th>                                             9087
        <th scope="col">操作</th>                                               9088
      </tr>                                                                     9089
    </thead>                                                                    9090
    <tbody>                                                                     9091
      {% for i in level_items %}                                                9092
      <tr>                                                                      9093
        <td style="text-align:left;">                                           9094
          <span class="mono muted">{{ i.code                                    9095
    }}</span><br>{{ i.name }}                                                   9096
        </td>                                                                   9097
        <td>{{ (all_ind|selectattr("code", "equalto",                          9098
    i.parent_code)|list|first).name if i.parent_code else                       9099
    "—" }}</td>                                                                 9100
```

```
            <td>{{ i.unit or "—" }}</td>                                    9101
            <td>{{ type_label(i.ideal_type) }}</td>                         9102
            <td class="mono">{{ i.ideal_value or "—" }}</td>                9103
            <td style="text-align:left;">{{ i.enabled_units or              9104
      "—" }}</td>                                                           9105
            <td>                                                            9106
              {% if i.is_constraint %}<span class="tag                      9107
      tag--warn">是</span>                                                  9108
              {% else %}<span class="tag tag--muted">否</span>{%            9109
      endif %}                                                              9110
            </td>                                                           9111
            <td class="nowrap">                                             9112
              <span class="row-actions">                                    9113
                <a class="link-btn" href="/ahp/indicators/{{                9114
      i.code }}/edit">编辑</a>                                              9115
              </span>                                                       9116
            </td>                                                           9117
          </tr>                                                             9118
          {% endfor %}                                                      9119
        </tbody>                                                            9120
      </table>                                                              9121
      {% endif %}                                                           9122
      {% endfor %}                                                          9123
</div>                                                                      9124
{% endblock %}                                                              9125
# ==== app/templates/ahp/indicator_edit.html ====                           9126
{% extends "base.html" %}                                                   9127
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}                    9128
{% block content %}                                                         9129
<div class="card" style="max-width:680px;">                                 9130
  <h2>编辑指标 · {{ ind.name }}</h2>                                        9131
  <p class="muted">                                                         9132
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{              9133
    ind.level }} 级，                                                       9134
    由论文指标体系确定，不可修改。                                          9135
  </p>                                                                      9136
  <form method="post">                                                      9137
    {{ form.hidden_tag() }}                                                 9138
    <div class="field">                                                     9139
      <label class="field__label" for="name">指标名称</label>               9140
      <input id="name" type="text" name="name"                              9141
    style="width:100%;" required                                            9142
             value="{{ ind.name }}">                                        9143
    </div>                                                                  9144
    <div class="field">                                                     9145
      <label class="field__label" for="unit">计量单位</label>               9146
      <input id="unit" type="text" name="unit"                              9147
    style="width:180px;"                                                    9148
             value="{{ ind.unit or '' }}">                                  9149
    </div>                                                                  9150
```

```
<div class="field">
  <label class="field__label" for="ideal-type">标准化类型</label>
  <select id="ideal-type" name="ideal_type">
    {% for code, label in [("benefit","越大越优"),("cost","越小越优"),("mid","适中型"),("discrete","离散赋值")] %}
    <option value="{{ code }}" {% if ind.ideal_type == code %}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="ideal-value">理想取值</label>
  <input id="ideal-value" type="text" name="ideal_value" style="width:100%;"
         value="{{ ind.ideal_value or '' }}">
  <p class="muted" style="margin:4px 0 0;">
    如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
  </p>
</div>
<fieldset style="border:1px solid var(--line);border-radius:6px;padding:10px 14px;margin:0 0 12px;">
  <legend style="padding:0 6px;font-weight:600;">启用该指标的微地质单元</legend>
  {% for t in unit_types %}
  <label class="checkbox" style="margin-right:18px;">
    <input type="checkbox" name="enabled_units" value="{{ t }}"
           {% if t in ind.units_list() %}checked{% endif %}>
    <span>{{ t }}</span>
  </label>
  {% endfor %}
  <p class="muted" style="margin:8px 0 0;">
    论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。
  </p>
</fieldset>
<div class="field">
  <label class="checkbox" for="is-constraint">
    <input id="is-constraint" type="checkbox" name="is_constraint" value="1"
           {% if ind.is_constraint %}checked{% endif %}>
    <span>同时作为限制性指标（参与一票否决判定）</span>
  </label>
</div>
<div class="field">
  <label class="field__label" for="description">取值依据说明</label>
```

```
        <textarea id="description" name="description"                    9201
      style="min-height:88px;">{{ ind.description or ''                  9202
      }}</textarea>                                                      9203
      </div>                                                             9204
      <p class="actions" style="margin-top:16px;">                       9205
        <button type="submit" class="btn">保存</button>                  9206
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列       9207
      表</a>                                                             9208
      </p>                                                               9209
    </form>                                                              9210
</div>                                                                   9211
{% endblock %}                                                           9212
# ==== app/templates/ahp/experts.html ====                               9213
{% extends "base.html" %}                                                9214
{% block title %}多专家判断矩阵{% endblock %}                             9215
{% block content %}                                                      9216
<div class="card">                                                       9217
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                    9218
  <form method="get" class="actions" style="margin-bottom:14             9219
    px;">                                                                9220
    <div>                                                                9221
      <label class="field__label" for="unit-type">微地质单元类型</l      9222
    abel>                                                                9223
      <select id="unit-type" name="unit_type"                            9224
    onchange="this.form.submit()">                                       9225
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                   9226
        <option value="{{ t }}" {% if unit_type == t                     9227
    %}selected{% endif %}>{{ t }}</option>                               9228
        {% endfor %}                                                     9229
      </select>                                                          9230
    </div>                                                               9231
    <div>                                                                9232
      <label class="field__label" for="level">指标层级</label>           9233
      <select id="level" name="level"                                    9234
    onchange="this.form.submit()">                                       9235
        <option value="1" {% if level == 1 %}selected{%                  9236
    endif %}>一级指标</option>                                           9237
        <option value="2" {% if level == 2 %}selected{%                  9238
    endif %}>二级指标</option>                                           9239
      </select>                                                          9240
    </div>                                                               9241
  </form>                                                                9242
  <p class="muted">                                                      9243
    当前范围：<strong>{{ unit_type }}</strong> ·                         9244
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif    9245
    %}。                                                                 9246
    若选二级指标，请先到                                                 9247
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。                9248
  </p>                                                                   9249
  {% if rows %}                                                          9250
```

```
<table>
  <caption class="muted">已录入 {{ rows|length }} 位专家的矩阵</caption>
  <thead>
    <tr>
      <th scope="col">专家</th>
      <th scope="col">矩阵规模</th>
      <th scope="col">该矩阵 CR</th>
      <th scope="col">一致性</th>
      <th scope="col">录入时间</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for r in rows %}
    <tr>
      <td>{{ r.expert_name }}</td>
      <td class="num">{{ n }} × {{ n }}</td>
      <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
      <td>
        {% if r.cr is not none and r.cr < 0.1 %}<span class="tag tag--ok">通过</span>
        {% else %}<span class="tag tag--warn">不通过</span>{% endif %}
      </td>
      <td class="mono nowrap">{{ r.created_at | localdt }}</td>
      <td>
        <form method="post" action="/ahp/experts/{{ r.id }}/delete"
              onsubmit="return confirm('确定删除专家「{{ r.expert_name }}」的矩阵吗？');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn btn--sm btn--danger">删除</button>
        </form>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="actions" style="margin-bottom:10px;">
  <a class="btn" href="/ahp/experts/aggregate?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">
    汇总计算（几何平均）
  </a>
</p>
{% else %}
```

```
<p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
{% endif %}
<h3>录入一位专家的判断矩阵</h3>
<form method="post">
  {{ form.hidden_tag() }}
  <input type="hidden" name="action" value="add">
  <input type="hidden" name="unit_type" value="{{ unit_type }}">
  <input type="hidden" name="level" value="{{ level }}">
  {% if l1 %}<input type="hidden" name="l1" value="{{ l1 }}">{% endif %}
  <div class="field">
    <label class="field__label" for="expert-name">专家姓名</label>
    <input id="expert-name" type="text" name="expert_name"
           style="width:220px;" placeholder="如：张工">
  </div>
  <table style="width:auto;">
    <thead>
      <tr>
        <th></th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      {% for i in range(n) %}
      <tr>
        <th scope="row">{{ names[i] }}</th>
        {% for j in range(n) %}
          {% if i == j %}
          <td class="muted num">1</td>
          {% elif i < j %}
          <td>
            <label class="muted" style="position:absolute;left:-9999px;"
                   for="e-{{ i }}-{{ j }}">{{ names[i] }} 相对 {{ names[j] }} 的重要性</label>
            <input id="e-{{ i }}-{{ j }}" type="number" step="0.01" min="0.01"
                   name="m_{{ i }}_{{ j }}" value="{{ default[i][j] }}">
          </td>
          {% else %}
          <td class="muted num">{{ default[i][j] }}</td>
          {% endif %}
        {% endfor %}
      </tr>
      {% endfor %}
    </tbody>
```

```
      </table>                                                          9351
      <p class="actions" style="margin-top:12px;">                      9352
        <button type="submit" class="btn">录入该专家矩阵</button>       9353
        <a class="btn btn--ghost" href="/ahp/indicators">指标维护<      9354
      /a>                                                               9355
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>         9356
      </p>                                                              9357
    </form>                                                             9358
</div>                                                                  9359
{% endblock %}                                                          9360
# ==== app/templates/ahp/experts_aggregate.html ====                    9361
{% extends "base.html" %}                                               9362
{% block title %}多专家矩阵汇总{% endblock %}                           9363
{% block content %}                                                     9364
<div class="card">                                                      9365
    <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%  9366
      else %}二级指标（{{ l1 }}）{% endif %}</h2>                       9367
    {% if not result %}                                                 9368
    <p class="muted">                                                   9369
      该范围内尚未录入专家矩阵。                                        9370
      <a href="/ahp/experts?unit_type={{ unit_type                      9371
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif               9372
      %}">去录入</a>                                                    9373
    </p>                                                                9374
    {% else %}                                                          9375
    <div class="stat-row">                                              9376
      <div>                                                             9377
        <div class="stat__label">参与汇总的专家</div>                   9378
        <div class="stat__value">{{ result.count }} 位</div>            9379
      </div>                                                            9380
      <div>                                                             9381
        <div class="stat__label">λmax</div>                             9382
        <div class="stat__value mono">{{ result.lambda_max              9383
      }}</div>                                                          9384
      </div>                                                            9385
      <div>                                                             9386
        <div class="stat__label">CI</div>                               9387
        <div class="stat__value mono">{{                               9388
      "%.4f"|format(result.ci) }}</div>                                 9389
      </div>                                                            9390
      <div>                                                             9391
        <div class="stat__label">CR（要求 &lt; 0.1）</div>              9392
        <div class="stat__value mono">{{                               9393
      "%.4f"|format(result.cr) }}</div>                                 9394
      </div>                                                            9395
      <div>                                                             9396
        <div class="stat__label">一致性检验</div>                       9397
        <div class="stat__value">                                       9398
          {% if result.consistent %}<span class="ok">通过</span>        9399
          {% else %}<span class="warn">不通过</span>{% endif %}         9400
```

```
      </div>                                                                    9401
    </div>                                                                      9402
  </div>                                                                        9403
  <h3>汇总后权重</h3>                                                           9404
  <table style="width:auto;">                                                   9405
    <thead>                                                                     9406
      <tr>                                                                      9407
        <th scope="col">指标</th>                                               9408
        {% for name in names %}<th scope="col">{{ name                          9409
    }}</th>{% endfor %}                                                         9410
      </tr>                                                                     9411
    </thead>                                                                    9412
    <tbody>                                                                     9413
      <tr>                                                                      9414
        <th scope="row">权重</th>                                               9415
        {% for w in result.weights %}<td class="mono">{{ w                      9416
    }}</td>{% endfor %}                                                         9417
      </tr>                                                                     9418
    </tbody>                                                                    9419
  </table>                                                                      9420
  <h3>各专家矩阵的一致性</h3>                                                   9421
  <table style="width:auto;">                                                   9422
    <thead>                                                                     9423
      <tr><th scope="col">专家</th><th scope="col">该矩阵                       9424
    CR</th><th scope="col">判定</th></tr>                                       9425
    </thead>                                                                    9426
    <tbody>                                                                     9427
      {% for r in rows %}                                                       9428
      <tr>                                                                      9429
        <td>{{ r.expert_name }}</td>                                           9430
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is                      9431
    not none else "—" }}</td>                                                   9432
        <td>                                                                    9433
          {% if r.cr is not none and r.cr < 0.1 %}<span                         9434
    class="ok">通过</span>                                                      9435
          {% else %}<span class="warn">不通过</span>{% endif %}                 9436
        </td>                                                                   9437
      </tr>                                                                     9438
      {% endfor %}                                                              9439
    </tbody>                                                                    9440
  </table>                                                                      9441
  {% if not result.consistent and result.most_inconsistent                      9442
    %}                                                                          9443
  <p style="padding:12px;border-radius:6px;background:var(--                    9444
    danger-soft);                                                               9445
              border-left:4px solid var(--danger);">                            9446
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>          9447
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                           9448
    result.most_inconsistent.j + 1 }} 列的比较                                  9449
    （当前 <span class="mono">{{ result.most_inconsistent.value                 9450
```

```
      |round(3) }}</span>,                                                    9451
      理想约 <span class="mono">{{ result.most_inconsistent.expec             9452
      ted|round(3) }}</span>)。                                              9453
      该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。                    9454
    </p>                                                                     9455
    {% endif %}                                                              9456
    {% endif %}                                                              9457
    <p class="actions" style="margin-top:16px;">                             9458
      <form method="post"                                                    9459
            action="/ahp/experts?unit_type={{ unit_type                      9460
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"                9461
            style="display:inline;"                                          9462
            onsubmit="return confirm('保存后，「{{ unit_type                 9463
      }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">                    9464
        {{ form.hidden_tag() }}                                              9465
        <input type="hidden" name="action" value="save_agg">                 9466
        <button type="submit" class="btn" {% if not                          9467
      result.consistent %}disabled{% endif %}>                               9468
          保存为生效权重                                                       9469
        </button>                                                            9470
      </form>                                                                9471
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={               9472
      { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1                   9473
      }}{% endif %}">返回录入</a>                                            9474
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>                  9475
    </p>                                                                     9476
</div>                                                                       9477
{% endblock %}                                                               9478
# ==== app/templates/indicator/index.html ====                               9479
{% extends "base.html" %}                                                    9480
{% from "_macros.html" import patch_guide %}                                 9481
{% block title %}指标数据采集{% endblock %}                                   9482
{% block content %}                                                          9483
<div class="card">                                                           9484
  <h2>指标数据采集与管理</h2>                                                 9485
  <p class="actions" style="margin-bottom:12px;">                            9486
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV            9487
    / Excel）</a>                                                            9488
  </p>                                                                       9489
  {{ patch_guide(projects_without_patch) }}                                  9490
  {% if not patches %}                                                       9491
  <p class="muted">                                                          9492
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，                          9493
    或运行 <span class="mono">python seed_demo.py</span>                     9494
    载入论文案例。                                                           9495
  </p>                                                                       9496
  {% else %}                                                                 9497
  {% for p in patches %}                                                     9498
  <div class="patch-panel">                                                  9499
    <div class="patch-panel__head">                                          9500
```

```
    <h3 class="patch-panel__name">{{ p.name }}</h3>
    <span class="patch-panel__meta">
      所属项目：{{ p.project.name if p.project else "—" }}
    </span>
    <span class="patch-panel__meta">
      微地质单元：{{ p.units.count() }} 个
    </span>
    {% if p.units.count() %}
    <span class="patch-panel__meta patch-panel__tail">
      <a class="link-btn" href="/indicator/import">批量导入本图斑数据</a>
    </span>
    {% endif %}
  </div>
  {% if not p.units.count() %}
  <div class="notice">
    <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
    <p style="margin:0;">
      指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
      请先完成划分，再回到本页录入。
    </p>
    <p style="margin:10px 0 0;">
      <a class="btn btn--sm" href="/unit/{{ p.id }}">去划分微地质单元</a>
    </p>
  </div>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">数据状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in p.units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{ "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td>
          {% if u.indicator_data %}
          <span class="tag tag--ok">已录入</span>
          {% else %}
          <span class="tag tag--muted">未录入</span>
          {% endif %}
```

```
            </td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/indicator/{{ u.id
    }}">录入 / 编辑</a>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
    </div>
    {% endfor %}
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/indicator/edit.html ====
{% extends "base.html" %}
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}
{% block content %}
<div class="card" style="max-width:820px;">
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>
  <p class="muted">
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for group_name, fields, has_hm in groups %}
    <h3>{{ group_name }}</h3>
    {% for fname, cname, unit_ in fields %}
    <div class="field">
      <label class="field__label" for="f-{{ fname }}">
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}
      </label>
      {% if fname == "aspect" %}
      <select id="f-{{ fname }}" name="aspect">
        <option value="">— 不适用 —</option>
        {% for a in aspect_types %}
        <option value="{{ a }}" {% if current.get("aspect")
    == a %}selected{% endif %}>{{ a }}</option>
        {% endfor %}
      </select>
      {% else %}
      <input id="f-{{ fname }}" type="number" step="any"
    name="{{ fname }}"
             value="{{ current.get(fname) if
    current.get(fname) is not none else '' }}">
      {% endif %}
```

```
      </div>                                                                    9601
      {% endfor %}                                                              9602
      {% endfor %}                                                              9603
      {% for group_name, fields, has_hm in groups %}                            9604
        {% if has_hm %}                                                         9605
        <h3>重金属含量（mg/kg）</h3>                                             9606
        <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>  9607
        <table style="width:auto;">                                             9608
          <thead>                                                               9609
            <tr>                                                                9610
              {% for m in heavy_metals %}<th scope="col">{{ m                   9611
      }}</th>{% endfor %}                                                       9612
            </tr>                                                               9613
          </thead>                                                              9614
          <tbody>                                                               9615
            <tr>                                                                9616
              {% for m in heavy_metals %}                                       9617
              <td>                                                              9618
                <label class="muted" style="position:absolute;                  9619
      left:-9999px;"                                                            9620
                       for="hm-{{ m }}">{{ m }} 含量</label>                    9621
                <input id="hm-{{ m }}" type="number"                            9622
      step="any" name="hm_{{ m }}"                                              9623
                       value="{{ current.get('heavy',                           9624
      {}).get(m) if current.get('heavy', {}).get(m) is not                      9625
      none else '' }}">                                                         9626
              </td>                                                             9627
              {% endfor %}                                                      9628
            </tr>                                                               9629
          </tbody>                                                              9630
        </table>                                                                9631
        {% endif %}                                                             9632
      {% endfor %}                                                              9633
      <p class="actions" style="margin-top:18px;">                              9634
        <button type="submit" class="btn">保存</button>                         9635
        <a class="btn btn--ghost" href="/indicator/">返回</a>                   9636
      </p>                                                                      9637
    </form>                                                                     9638
</div>                                                                          9639
{% endblock %}                                                                  9640
# ==== app/templates/indicator/import.html ====                                 9641
{% extends "base.html" %}                                                       9642
{% block title %}批量导入指标数据{% endblock %}                                  9643
{% block content %}                                                             9644
<div class="card" style="max-width:860px;">                                     9645
  <h2>批量导入指标数据</h2>                                                     9646
  <p class="muted">                                                             9647
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，               9648
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。                       9649
    先下载模板按格式填写，可减少出错。                                         9650
```

```
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/indicator/import/template">下载导入模板（CSV）</a>
    <a class="btn btn--sm btn--ghost"
    href="/indicator/">返回指标数据</a>
  </p>
  <form method="post" enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="patch-id">目标矿山图斑</lab
    el>
      <select id="patch-id" name="patch_id" required>
        <option value="">— 请选择 —</option>
        {% for p in patches %}
        <option value="{{ p.id }}">{{ p.name }} ({{
    p.units.count() }} 个单元）</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="datafile">数据文件</label
    >
      <input id="datafile" type="file" name="datafile"
             accept=".csv,.xlsx,.xls" required>
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">开始导入</button>
    </p>
  </form>
  <h3>模板列说明</h3>
  <p class="muted">
    共 {{ headers|length }} 列。留空表示该指标「不适用」。
    坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
    （按论文注释视为未受污染）。
  </p>
  <p class="mono muted" style="line-height:1.9;">{{
    headers|join(" , ") }}</p>
  <h3>导入规则</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
    <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
    <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
    <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
    <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
    <span class="mono">10cm</span>）。</li>
  </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
```

```
  <h2>导入结果</h2>                                                    9701
  <div class="stat-row">                                               9702
    <div>                                                              9703
      <div class="stat__label">目标图斑</div>                          9704
      <div style="font-size:15px;">{{ report.patch }}</div>            9705
    </div>                                                             9706
    <div>                                                              9707
      <div class="stat__label">成功导入</div>                          9708
      <div class="stat__value ok">{{ report.saved }}                   9709
    个单元</div>                                                       9710
    </div>                                                             9711
    <div>                                                              9712
      <div class="stat__label">问题行</div>                            9713
      <div class="stat__value {% if report.errors %}warn{%             9714
    endif %}">{{ report.errors|length }}</div>                         9715
    </div>                                                             9716
  </div>                                                               9717
  {% if report.errors %}                                               9718
  <h3>未导入的行</h3>                                                  9719
  <ul class="muted" style="padding-left:20px;line-height:1.9           9720
    ;">                                                                9721
    {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}           9722
  </ul>                                                                9723
  {% else %}                                                           9724
  <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>  9725
  {% endif %}                                                          9726
  <p class="actions" style="margin-top:16px;">                         9727
    <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>      9728
  </p>                                                                 9729
</div>                                                                 9730
{% endif %}                                                            9731
{% endblock %}                                                         9732
# ==== app/templates/evaluate/index.html ====                          9733
{% extends "base.html" %}                                              9734
{% from "_macros.html" import patch_guide %}                           9735
{% block title %}适宜性定量评价{% endblock %}                          9736
{% block content %}                                                    9737
<div class="card">                                                     9738
  <h2>适宜性定量评价</h2>                                              9739
  {{ patch_guide(projects_without_patch) }}                            9740
  {% if not patches %}                                                 9741
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>  9742
  {% else %}                                                           9743
  <table>                                                              9744
    <thead>                                                            9745
      <tr>                                                             9746
        <th scope="col">所属项目</th>                                  9747
        <th scope="col">矿山图斑</th>                                  9748
        <th scope="col">前置条件</th>                                  9749
        <th scope="col">最近评价结果</th>                              9750
```

```
            <th scope="col">操作</th>                                                    9751
          </tr>                                                                          9752
        </thead>                                                                         9753
        <tbody>                                                                          9754
          {% for p in patches %}                                                         9755
          {% set pr = progress.get(p.id) %}                                              9756
          <tr>                                                                           9757
            <td>{{ p.project.name if p.project else "—" }}</td>                          9758
            <td style="text-align:left;">{{ p.name }}</td>                               9759
            <td style="text-align:left;">                                                9760
              {% if pr and pr.ready %}                                                   9761
              <span class="tag tag--ok">已具备</span>                                    9762
              {% elif pr %}                                                              9763
              <span class="tag tag--warn">缺：{{                                         9764
        pr.blocking[0].label }}</span>                                                   9765
              <span class="muted">{{ pr.blocking[0].reason                               9766
        }}</span>                                                                        9767
              {% endif %}                                                                9768
            </td>                                                                        9769
            <td>                                                                         9770
              {% if latest.get(p.id) %}                                                  9771
              F = <span class="mono num">{{                                              9772
        latest[p.id].final_score }}</span>,                                              9773
              <span class="tag tag--ok">{{ latest[p.id].grade                            9774
        }}</span>                                                                        9775
              {% else %}                                                                 9776
              <span class="tag tag--muted">未评价</span>                                 9777
              {% endif %}                                                                9778
            </td>                                                                        9779
            <td class="nowrap">                                                          9780
              <span class="row-actions">                                                 9781
                <a class="link-btn" href="/evaluate/{{ p.id                              9782
        }}">执行评价</a>                                                                 9783
                {% if latest.get(p.id) %}                                                9784
                <a class="link-btn" href="/evaluate/result/{{                            9785
        latest[p.id].id }}">查看结果</a>                                                 9786
                {% endif %}                                                              9787
              </span>                                                                    9788
            </td>                                                                        9789
          </tr>                                                                          9790
          {% endfor %}                                                                   9791
        </tbody>                                                                         9792
      </table>                                                                           9793
      {% endif %}                                                                        9794
</div>                                                                                   9795
{% endblock %}                                                                           9796
# ==== app/templates/evaluate/run.html ====                                              9797
{% extends "base.html" %}                                                                9798
{% import "_macros.html" as ui %}                                                        9799
{% block title %}执行评价 · {{ patch.name }}{% endblock %}                               9800
```

```
{% block content %}
<div class="card" style="max-width:880px;">
  <h2>执行适宜性评价 · {{ patch.name }}</h2>
  {{ ui.workflow_steps(progress) }}
  {{ ui.workflow_blocking(progress) }}
  {% if progress.ready %}
  <form method="post">
    {{ form.hidden_tag() }}
    <p class="actions">
      <button type="submit" class="btn">开始评价</button>
      <a class="btn btn--ghost" href="/evaluate/">返回</a>
    </p>
  </form>
  {% else %}
  <p class="actions">
    <button type="button" class="btn" disabled>开始评价</button>
    <a class="btn btn--ghost" href="/evaluate/">返回</a>
  </p>
  {% endif %}
</div>
<div class="card" style="max-width:880px;">
  <h2>当前流程状态</h2>
  <table>
    <thead>
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th
    scope="col">说明</th></tr>
    </thead>
    <tbody>
      <tr>
        <td>限制性条件判定</td>
        <td>
          {% if progress.constraint.passed %}<span
    class="tag tag--ok">已通过</span>
          {% elif progress.constraint.done %}<span
    class="tag tag--warn">未通过</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.constraint.detail }}</td>
      </tr>
      <tr>
        <td>划分微地质单元</td>
        <td>
          {% if progress.units.done %}<span class="tag
    tag--ok">已完成</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
```

```
      progress.units.detail }}</td>                                    9851
        </tr>                                                          9852
        <tr>                                                           9853
          <td>录入指标数据</td>                                        9854
          <td>                                                         9855
            {% if progress.data.ready %}<span class="tag               9856
      tag--ok">已完成</span>                                           9857
            {% else %}<span class="tag tag--muted">待完成</span>{      9858
      % endif %}                                                       9859
          </td>                                                        9860
          <td style="text-align:left;">{{                             9861
      progress.data.detail }}</td>                                     9862
        </tr>                                                          9863
      </tbody>                                                         9864
    </table>                                                           9865
</div>                                                                 9866
{% if weight_sources %}                                                9867
<div class="card" style="max-width:880px;">                            9868
    <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">    9869
      <thead>                                                          9870
        <tr><th scope="col">微地质单元</th><th                         9871
      scope="col">权重来源</th></tr>                                   9872
      </thead>                                                         9873
      <tbody>                                                          9874
        {% for ut, label in weight_sources.items() %}                  9875
        <tr>                                                           9876
          <td style="text-align:left;">{{ ut }}</td>                   9877
          <td style="text-align:left;">                                9878
            {% if "自定义" in label %}                                 9879
            <span class="tag tag--warn">{{ label }}</span>             9880
            {% else %}                                                 9881
            <span class="tag tag--muted">{{ label }}</span>            9882
            {% endif %}                                                9883
          </td>                                                        9884
        </tr>                                                          9885
        {% endfor %}                                                   9886
      </tbody>                                                         9887
    </table>                                                           9888
</div>                                                                 9889
{% endif %}                                                            9890
{% endblock %}                                                         9891
# ==== app/templates/evaluate/result.html ====                         9892
{% extends "base.html" %}                                              9893
{% block title %}评价结果{% endblock %}                                9894
{% block content %}                                                    9895
<div class="card">                                                     9896
    <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>        9897
    <div class="stat-row">                                             9898
      <div>                                                            9899
        <div class="stat__label">最终评价因子 F</div>                  9900
```

```
        <div class="stat__value mono">{{ r.final_score
    }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ r.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
      <div style="font-size:15px;">{{ r.measures }}</div>
    </div>
  </div>
  {% if unit_scores %}
  <h3>各微地质单元得分</h3>
  <table style="max-width:860px;">
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">单元评价因子 F′</th>
        <th scope="col">加权贡献</th>
        <th scope="col">权重来源</th>
      </tr>
    </thead>
    <tbody>
      {% for u in unit_scores %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
        <td class="mono">{{ u.unit_score }}</td>
        <td class="mono">{{ "%.4f"|format(u.ratio *
    u.unit_score) }}</td>
        <td class="muted">{{ u.weight_source or "论文表 3
    默认权重" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  <p class="actions" style="margin-top:18px;">
    <a class="btn btn--ghost" href="/report/{{ r.patch_id
    }}">查看得分明细</a>
    <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/report/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
```

```
{% block title %}报告输出{% endblock %}
{% block content %}
<div class="card">
  <h2>报告与结果输出</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">可生成报告</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set pr = progress.get(p.id) %}
      <tr>
        <td>{{ p.project.name if p.project else "—" }}</td>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if pr and pr.ready %}
          <span class="tag tag--ok">可以</span>
          {% elif pr %}
          <span class="tag tag--warn">暂不可</span>
          <span class="muted">{{ pr.blocking[0].reason
    }}</span>
          {% endif %}
        </td>
        <td class="nowrap">
          {% if pr and pr.ready %}
          <span class="row-actions">
            <a class="link-btn" href="/report/{{ p.id
    }}">查看得分明细</a>
            <a class="link-btn" href="/report/{{ p.id
    }}/pdf">导出 PDF</a>
            <a class="link-btn" href="/report/{{ p.id
    }}/word">导出 Word</a>
          </span>
          {% else %}
          <span class="muted">—</span>
          {% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
```

```
  {% endif %}                                                    10001
</div>                                                           10002
{% endblock %}                                                   10003
# ==== app/templates/report/detail.html ====                     10004
{% extends "base.html" %}                                        10005
{% block title %}得分明细 · {{ patch.name }}{% endblock %}       10006
{% block content %}                                              10007
<div class="card">                                               10008
  <h2>得分明细 · {{ patch.name }}</h2>                           10009
  <div class="stat-row">                                         10010
    <div>                                                        10011
      <div class="stat__label">最终评价因子 F</div>              10012
      <div class="stat__value mono">{{                          10013
    "%.4f"|format(result.final_score) }}</div>                   10014
    </div>                                                       10015
    <div>                                                        10016
      <div class="stat__label">适宜性等级</div>                  10017
      <div class="stat__value ok">{{ result.grade }}</div>       10018
    </div>                                                       10019
    <div style="flex:1;min-width:240px;">                        10020
      <div class="stat__label">推荐措施</div>                    10021
      <div style="font-size:15px;">{{                           10022
    result.grade_info.measure }}</div>                           10023
    </div>                                                       10024
  </div>                                                         10025
  <p class="muted" style="margin-top:12px;">                     10026
    本页与导出报告均取自                                         10027
    <strong>{% if record %}{{ record.created_at |               10028
    localdt("%Y-%m-%d %H:%M:%S") }}{% endif %}                   10029
    记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else  10030
    "—" }}），                                                   10031
    不会因事后修改指标数据而变化。                               10032
  </p>                                                           10033
  <p class="actions" style="margin-top:16px;">                   10034
    <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF    10035
    报告</a>                                                     10036
    <a class="btn btn--ghost" href="/report/{{ patch.id          10037
    }}/word">导出 Word 报告</a>                                  10038
    <a class="btn btn--ghost" href="/report/">返回</a>           10039
  </p>                                                           10040
</div>                                                           10041
<div class="card">                                               10042
  <h2>结果图表</h2>                                              10043
  <h3>各微地质单元得分对比</h3>                                  10044
  <div>{{ chart_units|safe }}</div>                              10045
  <h3>各一级指标平均得分</h3>                                    10046
  <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>  10047
  <div>{{ chart_weight|safe }}</div>                             10048
</div>                                                           10049
{% for u in result.unit_results %}                               10050
```

```
<div class="card">
  <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio * 100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>
  {% if u.weight_source %}
  <p class="muted" style="margin-top:-6px;">权重来源：{{ u.weight_source }}</p>
  {% endif %}
  <table>
    <thead>
      <tr>
        <th scope="col">一级指标</th>
        <th scope="col">二级指标</th>
        <th scope="col">实测值</th>
        <th scope="col">标准化值</th>
        <th scope="col">权重</th>
        <th scope="col">贡献</th>
      </tr>
    </thead>
    <tbody>
      {% for d in u.details %}
      <tr>
        <td>{{ d.l1 }}</td>
        <td style="text-align:left;">{{ d.l2 }}</td>
        <td class="mono" style="text-align:left;">{{ "—" if d.raw is none else d.raw }}</td>
        <td class="mono">{{ "—" if d.std_value is none else "%.4f"|format(d.std_value) }}</td>
        <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
        <td class="mono">{{ "—" if d.contribution is none else "%.4f"|format(d.contribution) }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="muted" style="margin:0;">
    一级指标得分：
    {% for k, v in u.l1_scores.items() %}{{ k }} = <span class="num">{{ "%.4f"|format(v) }}</span>{% if not loop.last %}；{% endif %}{% endfor %}
  </p>
</div>
{% endfor %}
{% endblock %}
# ==== app/templates/report/print.html ====
<!DOCTYPE html>
<html lang="zh-CN">
<head>
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
```

```
  @page { size: A4; margin: 18mm 16mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #23262b; background: #fff;
    font-family: "Microsoft YaHei", "SimSun", sans-serif;
    font-size: 10.5pt; line-height: 1.65;
  }
  h1 { font-size: 17pt; text-align: center; margin: 0 0
    6px; }
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
       border-bottom: 1px solid #dfe3e8; padding-bottom:
    4px; }
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0
    6px; }
  .sub { text-align: center; color: #6b7280; font-size:
    9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
          font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
    text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace;
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０ ·
    生成时间 {{ generated_at | localdt }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</td><td>{{ patch.name }}</td></tr>
    <tr><th>所在行政区</td><td>{{ patch.admin_region or "—"
    }}</td></tr>
    <tr><th>图斑面积（公顷）</td><td>{{ patch.area_ha if
```

```
    patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"
    }}</td></tr>
    <tr><th>开采方式</th><td>{{ patch.mining_method or "—"
    }}</td></tr>
    <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entit
    y_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</th>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
</table>
<h2>二、评价结论</h2>
<table class="kv">
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{
    "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</th><td><strong>{{ result.grade
    }}</strong></td></tr>
    <tr><th>判定依据</th><td>{{ result.grade_info.meaning
    }}</td></tr>
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure
    }}</td></tr>
</table>
<h2>三、各微地质单元得分</h2>
<table>
    <thead>
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
    F'</th><th>加权贡献</th></tr>
    </thead>
    <tbody>
      {% for u in result.unit_results %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.unit_score)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.weighted)
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>五、得分明细</h2>
{% for u in result.unit_results %}
```

```
  <h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *      10201
    100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>    10202
  <table>                                                         10203
    <thead>                                                       10204
      <tr>                                                        10205
        <th>一级指标</th><th>二级指标</th><th>实测值</th>          10206
        <th>标准化值</th><th>权重</th><th>贡献</th>               10207
      </tr>                                                       10208
    </thead>                                                      10209
    <tbody>                                                       10210
      {% for d in u.details %}                                    10211
      <tr>                                                        10212
        <td>{{ d.l1 }}</td>                                       10213
        <td class="left">{{ d.l2 }}</td>                          10214
        <td class="mono left">{{ "—" if d.raw is none else        10215
    d.raw }}</td>                                                 10216
        <td class="mono">{{ "—" if d.std_value is none else       10217
    "%.4f"|format(d.std_value) }}</td>                            10218
        <td class="mono">{{ "%.4f"|format(d.weight) }}</td>       10219
        <td class="mono">{{ "—" if d.contribution is none         10220
    else "%.4f"|format(d.contribution) }}</td>                    10221
      </tr>                                                       10222
      {% endfor %}                                                10223
    </tbody>                                                      10224
  </table>                                                        10225
  <p class="note">                                                10226
    一级指标得分：                                                10227
    {% for k, v in u.l1_scores.items() %}{{ k }} = {{             10228
    "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{%      10229
    endfor %}                                                     10230
  </p>                                                            10231
  {% endfor %}                                                    10232
  <h2>六、评价依据说明</h2>                                       10233
  <ul class="note">                                               10234
    <li>指标标准化按论文式（８） 执行，按指标性质分为越大越优、越小越优、  10235
        适中型与离散赋值四类。</li>                               10236
    <li>单元评价因子按论文式（９） 由二级指标权重加权求得；      10237
        整矿评价因子由各微地质单元按面积占比加权合成。</li>      10238
    <li>适宜性等级按论文表 4 划分：F&lt;０．３ 不适宜、０．３≤F≤０．７ 基本适宜、F≥０．７  10239
    适宜。</li>                                                   10240
    <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>  10241
  </ul>                                                           10242
  <p class="sig">                                                 10243
    评价人（签字）：＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿　日期：＿＿＿＿＿＿＿＿＿＿＿＿  10244
  </p>                                                            10245
</body>                                                           10246
</html>                                                           10247
# ==== app/templates/search.html ====                             10248
{% extends "base.html" %}                                         10249
{% from "_macros.html" import icon %}                             10250
```

```
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="card">
  <h2>搜索结果</h2>
  {% if not q %}
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区 / 矿种、
    标准名称与文号。</p>
  {% else %}
  <p class="muted">
    关键词「<strong>{{ q }}</strong>」共找到
    <span class="num">{{ total }}</span> 条结果。
  </p>
  {% if not total %}
  <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
  {% endif %}
  {% endif %}
</div>
{% if projects %}
<div class="card">
  <h2>评价项目（{{ projects|length }}）</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">项目名称</th>
        <th scope="col">编号</th>
        <th scope="col">状态</th>
        <th scope="col">图斑数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in projects %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td class="mono">{{ p.code or "—" }}</td>
        <td><span class="tag tag--muted">{{ p.status
    }}</span></td>
        <td class="num">{{ p.patches.count() }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/project/{{ p.id
    }}/patches">矿山图斑</a>
            <a class="link-btn" href="/project/{{ p.id
    }}/edit">编辑</a>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
```

```
</div>                                                                                                  10301
{% endif %}                                                                                             10302
{% if patches %}                                                                                        10303
<div class="card">                                                                                      10304
  <h2>矿山图斑（{{ patches|length }}）</h2>                                                             10305
  <table>                                                                                               10306
    <thead>                                                                                             10307
      <tr>                                                                                              10308
        <th scope="col">所属项目</th>                                                                   10309
        <th scope="col">图斑名称</th>                                                                   10310
        <th scope="col">行政区</th>                                                                     10311
        <th scope="col">矿种</th>                                                                       10312
        <th scope="col">操作</th>                                                                       10313
      </tr>                                                                                             10314
    </thead>                                                                                            10315
    <tbody>                                                                                             10316
      {% for m in patches %}                                                                            10317
      <tr>                                                                                              10318
        <td>{{ m.project.name if m.project else "—" }}</td>                                             10319
        <td style="text-align:left;">{{ m.name }}</td>                                                  10320
        <td>{{ m.admin_region or "—" }}</td>                                                            10321
        <td>{{ m.mineral_type or "—" }}</td>                                                            10322
        <td class="nowrap">                                                                             10323
          <span class="row-actions">                                                                    10324
            <a class="link-btn" href="/constraint/{{ m.id                                               10325
  }}">限制性判定</a>                                                                                    10326
            <a class="link-btn" href="/unit/{{ m.id                                                     10327
  }}">微地质单元</a>                                                                                    10328
            <a class="link-btn" href="/evaluate/{{ m.id                                                 10329
  }}">执行评价</a>                                                                                      10330
          </span>                                                                                       10331
        </td>                                                                                           10332
      </tr>                                                                                             10333
      {% endfor %}                                                                                      10334
    </tbody>                                                                                            10335
  </table>                                                                                              10336
</div>                                                                                                  10337
{% endif %}                                                                                             10338
{% if standards %}                                                                                      10339
<div class="card">                                                                                      10340
  <h2>标准与政策文件（{{ standards|length }}）</h2>                                                     10341
  <table>                                                                                               10342
    <thead>                                                                                             10343
      <tr>                                                                                              10344
        <th scope="col">类别</th>                                                                       10345
        <th scope="col">省份</th>                                                                       10346
        <th scope="col">名称</th>                                                                       10347
        <th scope="col">文号</th>                                                                       10348
        <th scope="col">操作</th>                                                                       10349
      </tr>                                                                                             10350
```

```
      </thead>                                                                  10351
      <tbody>                                                                   10352
        {% for s in standards %}                                                10353
        <tr>                                                                    10354
          <td>{{ s.category }}</td>                                             10355
          <td>{{ s.province }}</td>                                             10356
          <td style="text-align:left;">{{ s.name }}</td>                        10357
          <td class="mono">{{ s.doc_number or "—" }}</td>                       10358
          <td class="nowrap">                                                   10359
            <span class="row-actions">                                          10360
              <a class="link-btn" href="/standard/{{ s.id                       10361
      }}/rules">规则</a>                                                        10362
            </span>                                                             10363
          </td>                                                                 10364
        </tr>                                                                   10365
        {% endfor %}                                                            10366
      </tbody>                                                                  10367
    </table>                                                                    10368
</div>                                                                          10369
{% endif %}                                                                     10370
{% endblock %}                                                                  10371
#  ==== seed_demo.py ====                                                       10372
#  -*- coding: utf-8 -*-                                                        10373
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。                                10374
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。                           10375
运行：python seed_demo.py                  幂等，只补写缺失的演示数据，          10376
                                               不会删除用户自建的项目与图斑      10377
      python seed_demo.py --reset          连同标准库与指标定义一并重建          10378
"""                                                                             10379
import json                                                                     10380
import sys                                                                      10381
sys.stdout.reconfigure(encoding="utf-8")                                        10382
from app import create_app                                                      10383
from app.models import (db, User, Project, MinePatch,                           10384
    ReferenceEcosystem,                                                         10385
                           MicroUnit, IndicatorData, Standard,                  10386
    StandardRule,                                                               10387
                           ConstraintCheck, EvaluationResult,                   10388
    ScoreDetail)                                                                10389
app = create_app()                                                              10390
#  内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」                    10391
DEMO_CODE = "DEMO-001"                                                          10392
#  论文表 1：11 项地方标准与政策文件                                             10393
STANDARDS = [                                                                   10394
    dict(category="地方标准", province="辽宁",                                   10395
    name="矿山废弃地植被自然恢复技术规程",                                        10396
         doc_number="DB21/T 3702—2023",                                         10397
    publish_date="2023-03-30",                                                  10398
         implement_date="2023-04-30",                                           10399
         summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",      10400
```

```
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
dict(category="地方标准", province="湖北",
name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
dict(category="地方标准", province="内蒙古",
name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
dict(category="地方标准", province="江西",
name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
publish_date="2026-02-02",
        implement_date="2026-08-01",
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),
dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",
        doc_number=None, publish_date="2026-06-18",
implement_date=None,
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
dict(category="政策文件", province="安徽",
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",
        doc_number="皖自然资修函[2020]34号",
publish_date="2020-03-03", implement_date=None,
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",
        rules=[("责任主体", "==", "政府", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
dict(category="政策文件", province="山东",
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",
        doc_number="鲁自然资字[2022]104号",
publish_date="2022-08-02", implement_date=None,
        summary="条件与安徽省相同，另增加自然恢复认定标准",
        rules=[("无地质灾害", "==", "是", "必要条件", None),
```

```
                    ("无水土污染", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="湖南",
             name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",
             doc_number="湘自资办发[２０２３]５８号",
        publish_date="２０２３-０７-２０", implement_date=None,
             summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
        >７５%、连续３年正向增长",
             rules=[("植被覆盖度", ">", "７５", "必要条件", "%"),
                    ("植被覆盖率趋势", "==", "连续３年正向增长", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None),
                    ("符合国土空间规划", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="新疆",
             name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
             doc_number="新自然资源办函〔２０２４〕２１９号",
        publish_date="２０２４-１２-２０", implement_date=None,
             summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
             rules=[("植被覆盖度", ">=", "周边３０%", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None),
                    ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="广西",
             name="广西壮族自治区历史遗留矿山自然恢复技术指南",
             doc_number="桂自然资办〔２０２４〕２５０号",
        publish_date="２０２４-１２-３０", implement_date=None,
             summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
             rules=[("无地质灾害", "==", "是", "必要条件", None),
                    ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                    ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="西藏",
             name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
             doc_number="藏自然资发〔２０２５〕１９号",
        publish_date="２０２５-０９-２４", implement_date=None,
             summary="含修复成本高、生态修复效益不显著等适用条件",
             rules=[("责任主体", "==", "政府", "必要条件", None),
                    ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
```

```
        "ph": 9.24,                                                    10501
                        "organic_matter": 4.89,                        10502
                        "heavy_metals": {"Hg": 0.05, "As":             10503
    5.17, "Ni": 12.60,                                                 10504
                                                "Cu": 11.70, "Cd":     10505
    0.04, "Pb": 6.50},                                                 10506
                        "coverage": 87.92, "shannon": 0.60,            10507
                        "slope": 28.75, "aspect": "阳坡"},             10508
            },                                                         10509
            {                                                          10510
                "unit_type": "平台", "ratio": 0.2319,                  10511
                "data": {"soil_ratio": 0.56, "thickness": 30,          10512
    "ph": 8.72,                                                        10513
                        "organic_matter": 10.92,                       10514
                        "heavy_metals": {"Hg": 0.02, "As":             10515
    3.13, "Ni": 12.33,                                                 10516
                                                "Cu": 11.14, "Cd":     10517
    0.09, "Pb": 9.98},                                                 10518
                        "coverage": 58.42, "shannon": 0.72},           10519
            },                                                         10520
        ],                                                             10521
}                                                                      10522
def clean_case(reset=False):                                           10523
    """清理数据。                                                      10524
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰      10525
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，  10526
    以便把演示环境恢复到初始状态。                                    10527
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目   10528
    会在每次启动时被删除。                                            10529
    """                                                                10530
    from app.models import EvaluationResult, ScoreDetail,              10531
    ConstraintCheck                                                    10532
    demo_projects = Project.query.filter_by(code=DEMO_CODE)            10533
    all()                                                              10534
    demo_ids = [p.id for p in demo_projects]                           10535
    patch_ids = []                                                     10536
    if demo_ids:                                                       10537
        patch_ids = [p.id for p in MinePatch.query.filter(             10538
            MinePatch.project_id.in_(demo_ids)).all()]                 10539
    if patch_ids:                                                      10540
        unit_ids = [u.id for u in MicroUnit.query.filter(              10541
            MicroUnit.patch_id.in_(patch_ids)).all()]                  10542
        if unit_ids:                                                   10543
            IndicatorData.query.filter(                                10544
                IndicatorData.unit_id.in_(unit_ids)).delete(           10545
                synchronize_session=False)                             10546
        result_ids = [r.id for r in EvaluationResult.query.f           10547
    ilter(                                                             10548
            EvaluationResult.patch_id.in_(patch_ids)).all()]           10549
        if result_ids:                                                 10550
```

```
            ScoreDetail.query.filter(                                   10551
                ScoreDetail.result_id.in_(result_ids)).delet10552
    e(                                                                  10553
                synchronize_session=False)                              10554
        EvaluationResult.query.filter(                                  10555
            EvaluationResult.patch_id.in_(patch_ids)).delete10556
    (                                                                   10557
            synchronize_session=False)                                  10558
        ConstraintCheck.query.filter(                                   10559
            ConstraintCheck.patch_id.in_(patch_ids)).delete(10560
            synchronize_session=False)                                  10561
        ReferenceEcosystem.query.filter(                                10562
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele10563
    te(                                                                 10564
            synchronize_session=False)                                  10565
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_10566
    ids)).delete(                                                       10567
            synchronize_session=False)                                  10568
        MinePatch.query.filter(MinePatch.id.in_(patch_ids))10569
    delete(                                                             10570
            synchronize_session=False)                                  10571
    if demo_ids:                                                        10572
        Project.query.filter(Project.id.in_(demo_ids)).delet10573
    e(                                                                  10574
            synchronize_session=False)                                  10575
    if reset:                                                           10576
        StandardRule.query.delete()                                     10577
        Standard.query.delete()                                         10578
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，                  10579
    #  避免长期运行后 score_details 表堆积无效数据。                   10580
    _purge_orphan_details()                                             10581
    db.session.commit()                                                 10582
    return len(demo_ids)                                                10583
def demo_patch():                                                       10584
    """取内置演示案例的图斑。                                           10585
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），               10586
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。       10587
    没有演示案例时才退回第一个图斑。                                   10588
    """                                                                 10589
    proj = Project.query.filter_by(code=DEMO_CODE).first()              10590
    if proj is not None:                                                10591
        m = MinePatch.query.filter_by(project_id=proj.id).fi10592
    rst()                                                               10593
        if m is not None:                                               10594
            return m                                                    10595
    return MinePatch.query.first()                                      10596
def _purge_orphan_details():                                            10597
    """清理指向已不存在评价记录的孤儿得分明细。                        10598
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。    10599
    """                                                                 10600
```

```
    live_ids = [r.id for r in EvaluationResult.query.all()]
    if live_ids:
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(
live_ids)).delete(
            synchronize_session=False)
    else:
        ScoreDetail.query.delete()
    db.session.commit()
def seed(reset=False):
    with app.app_context():
        db.create_all()        # 全新环境下先建表，保证本脚本可独立运行
        _purge_orphan_details()
        admin = User.query.filter_by(username="admin").first
()
        if not admin:
            from werkzeug.security import
generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_has
h("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        # 幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
        existing = Project.query.filter_by(code=DEMO_CODE).f
irst()
        if existing and not reset:
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")
            _report()
            _seed_reference_data(reset=False)
            return
        clean_case(reset=reset)
        project = Project(name=CASE["name"], code=DEMO_CODE,
                          description="论文第 5 节典型案例",
created_by=admin.id)
        db.session.add(project)
        db.session.flush()
        patch = MinePatch(project_id=project.id,
name=CASE["patch_name"],
                          admin_region="河北省唐山市玉田县",
                          area_ha=CASE["area_ha"],
                          mineral_type=CASE["mineral_type"],
                          mining_method="露天开采",
stop_years=None,
                          responsible_entity_missing=True)
        db.session.add(patch)
        db.session.flush()
        ref = ReferenceEcosystem(patch_id=patch.id,
                                 coverage_benchmark=75.0,
shannon_benchmark=0.7)
```

```
        db.session.add(ref)
        for u in CASE["units"]:
            unit = MicroUnit(patch_id=patch.id,
    unit_type=u["unit_type"],
                             ratio=u["ratio"], area=None)
            db.session.add(unit)
            db.session.flush()
            d = u["data"]
            ind = IndicatorData(
                unit_id=unit.id,
                soil_ratio=d.get("soil_ratio"),
                thickness=d.get("thickness"),
                ph=d.get("ph"),
                organic_matter=d.get("organic_matter"),
                heavy_metals=json.dumps(d["heavy_metals"],
    ensure_ascii=False)
                                if d.get("heavy_metals") else
    None,
                coverage=d.get("coverage"),
                shannon=d.get("shannon"),
                slope=d.get("slope"),
                aspect=d.get("aspect"),
            )
            db.session.add(ind)
        #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。
        db.session.add(ConstraintCheck(
            patch_id=patch.id,
            policy_pass=True, location_pass=True,
    pollution_pass=True,
            geo_pass=True, farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        print("已写入案例：", CASE["name"])
        print("   图斑：", CASE["patch_name"])
        print("   微地质单元：岩质边坡 １３．８６％ / 碎石边坡 ６２．９５％ / 平台
    ２３．１９％")
        print("   管理员账号：admin / admin123")
    _seed_reference_data(reset=reset)
    _report()
def _seed_reference_data(reset=False):
    """写入标准库与指标定义。
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护
    的标准条目与指标定义。
    """
    if reset or Standard.query.count() == 0:
        seed_standards()
    from app.services.indicator_service import
    seed_indicators
    n = seed_indicators()
```

```
    print("指标定义：%d 项" % n)                                                    10701
def _report():                                                                    10702
    total_p = Project.query.count()                                               10703
    total_patch = MinePatch.query.count()                                         10704
    print()                                                                       10705
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))              10706
    if total_p > total_patch:                                                     10707
        print("  提示：项目是归档单位，评价针对的是图斑。")                        10708
        print("        新项目需先建立图斑，才会出现在后续各模块中。")              10709
def seed_standards():                                                             10710
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""                     10711
    StandardRule.query.delete()                                                   10712
    Standard.query.delete()                                                       10713
    db.session.commit()                                                           10714
    for s in STANDARDS:                                                           10715
        std = Standard(category=s["category"],                                    10716
    province=s["province"],                                                       10717
                       name=s["name"],                                            10718
    doc_number=s["doc_number"],                                                   10719
                       publish_date=s["publish_date"],                            10720
                       implement_date=s["implement_date"],                        10721
                       summary=s["summary"])                                      10722
        db.session.add(std)                                                       10723
        db.session.flush()                                                        10724
        for item, op, thr, ctype, note in s["rules"]:                             10725
            db.session.add(StandardRule(standard_id=std.id,                       10726
    item=item,                                                                    10727
                                        operator=op,                              10728
    threshold=thr,                                                                10729
                                                                                  10730
    condition_type=ctype, note=note))                                             10731
    db.session.commit()                                                           10732
    print("已写入标准/政策文件：%d 项，规则 %d 条"                               10733
          % (len(STANDARDS), sum(len(s["rules"]) for s in                         10734
    STANDARDS)))                                                                  10735
if __name__ == "__main__":                                                        10736
    #  --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。         10737
    #  无论是否 --reset，都不会删除用户自建的项目与图斑。                        10738
    seed(reset="--reset" in sys.argv[1:])                                         10739
#  ==== 测试/test_ahp.py ====                                                    10740
#  -*- coding: utf-8 -*-                                                         10741
"""地基自检：验证 AHP 引擎与应用可启动。"""                                      10742
import sys                                                                        10743
sys.stdout.reconfigure(encoding="utf-8")                                          10744
import numpy as np                                                                10745
import os                                                                         10746
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                      10747
#  使 `from app import ...` 与 app/app.db 的定位不受影响。                       10748
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil                      10749
    e__)))                                                                        10750
```

```
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import ahp_service as ahp
LINE = "=" * 66
# ------------------------------------------------------------------
# 1） 完全一致矩阵 → 权重应精确还原，CR=0
# ------------------------------------------------------------------
print(LINE)
print("测试 1　完全一致判断矩阵的权重还原与一致性")
print(LINE)
# 由权重构造完全一致矩阵：a_ij = w_i / w_j
w_true = np.array([0.44, 0.31, 0.25])
A = np.outer(w_true, 1.0 / w_true)
r = ahp.weights_and_consistency(A)
print("　判断矩阵 A：")
for row in A:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print()
print("　计算权重 :", "  ".join(f"{v:.4f}" for v in
    r["weights"]))
print("　真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))
print("　λmax = %.6f　CI = %.3e　CR = %.3e　一致: %s"
      % (r["lambda_max"], r["ci"], r["cr"],
    r["consistent"]))
assert np.allclose(r["weights"], w_true, atol=1e-6),
    "权重未精确还原"
assert r["consistent"], "完全一致矩阵应通过检验"
# ------------------------------------------------------------------
# 2） 上三角输入 → 完整矩阵 → 一致性定位
# ------------------------------------------------------------------
print()
print(LINE)
print("测试 2　上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("　由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("　权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "　CR = %.3f　一致: %s" % (r2["cr"], r2["consistent"]))
print("　最不一致比较项：第(%d,%d) 项，当前 %.3f，理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
    bad["expected"]))
```

```
# ------------------------------------------------------------------
    ------------------
# 3） 多专家聚合（几何平均）
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合（几何平均）")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")
print("  聚合后上三角： a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ------------------------------------------------------------------
    ------------------
# 4） 论文表 3 权重可还原性
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 （0.44， 0.31， 0.25）")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f  CR = %.3e  一致： %s" %
    (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------------
    ------------------
# 5） 应用可启动 + 建表
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表：" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
```

```
print(LINE)
#  ==== 测试/test_evaluate.py ====
#  -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/  子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
#  ------------------------------------------------------------------
#  论文第 5 节案例数据（表 5，逐字抄录）
#  ------------------------------------------------------------------
#  三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
#  注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
```

```
                "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
                "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                                 "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
                "coverage": 58.42, "shannon": 0.72,
            },
    },
]
# ----------------------------------------------------------------
    ----------------
# 逐单元输出标准化与得分
# ----------------------------------------------------------------
    ----------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
```

```
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score 中集中定义、可配置")
print(LINE)
# ==== 测试/test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
```

```
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」:", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）:", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
```

```
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
# 工作台：页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含账户 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
```

```
print("   项目列表状态码：", r.status_code)
print("   含案例项目：", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
#  新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("   新建项目后含「测试项目A」：", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
#  清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("   已清理测试项目")
print()
print(LINE)
print("测试 4   微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
    irst()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("   单元列表含三类单元：", all(t in body for t in ("岩质边坡",
    "碎石边坡", "平台")))
print("   含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5   标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("   含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南",
    "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("   口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("   含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6   限制性指标判定（一票否决）")
print(LINE)
```

```
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件:", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
```

```
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1　指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("　状态码：", r.status_code)
print("　含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("　含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2　指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("　岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("　岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("　碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("　碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3　执行评价")
print(LINE)
```

```
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fi
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
```

```
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("        逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
```

```
print("  状态码：", r.status_code)                                    11351
print("  Content-Type:", r.headers.get("Content-Type",                11352
    "")[:60])                                                         11353
data = r.data                                                         11354
print("  文件大小：", len(data), "字节")                               11355
print("  是否为有效 docx（PK 头）:", data[:2] == b"PK")                11356
assert r.status_code == 200 and data[:2] == b"PK"                     11357
# 用 python-docx 打开验证内容                                         11358
import docx                                                           11359
d = docx.Document(io.BytesIO(data))                                   11360
text = "\n".join(p.text for p in d.paragraphs)                        11361
print("  含报告标题：", "自然恢复适宜性评价报告" in text)              11362
print("  含评价结论章节：", "评价结论" in text)                        11363
print("  含得分明细章节：", "得分明细" in text)                        11364
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))     11365
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3        11366
print()                                                               11367
print(LINE)                                                           11368
print("报告输出全部测试通过 √")                                      11369
print(LINE)                                                           11370
# ==== 测试/test_ui_fixes.py ====                                     11371
# -*- coding: utf-8 -*-                                               11372
"""界面修复自检：「记住我」与各模块删除功能。"""                      11373
import os                                                             11374
import re                                                             11375
import sys                                                            11376
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，           11377
# 使 `from app import ...` 与 app/app.db 的定位不受影响。             11378
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil          11379
    e__)))                                                            11380
if ROOT not in sys.path:                                              11381
    sys.path.insert(0, ROOT)                                          11382
from app import create_app                                            11383
sys.stdout.reconfigure(encoding="utf-8")                              11384
LINE = "=" * 70                                                       11385
class TestConfig:                                                     11386
    SECRET_KEY = "test"                                               11387
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(            11388
        ROOT, "app", "app.db")                                        11389
    SQLALCHEMY_TRACK_MODIFICATIONS = False                            11390
    WTF_CSRF_ENABLED = False                                          11391
    TESTING = True                                                    11392
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede          11393
    lta(days=30)                                                      11394
app = create_app(TestConfig)                                          11395
client = app.test_client()                                            11396
# ------------------------------------------------------------------  11397
    ----------------                                                  11398
print(LINE)                                                           11399
print("一、「记住我」")                                               11400
```

```
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
             follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
    r3.status_code))
```

```
assert r3.status_code == 200
# ------------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
# 图斑列表：地址用的是项目主键，不是图斑主键
with app.app_context():
    from app.models import MinePatch, Project
    import seed_demo
    patch = seed_demo.demo_patch()
    patch_id = patch.id
    project_id = patch.project_id
html = client.get("/project/%d/patches" %
    project_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "/project/patch/" in html and "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
```

```
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
```

```
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------
    ----------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
```

```
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
       data={"role": "operator", "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "admin",
    "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
```

```
r = c.post("/project/patch/%d/image" % patch_id,
                data={"image": (io.BytesIO(b"x"), "bad.txt")},
                content_type="multipart/form-data",
      follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
      follow_redirects=True)
with app.app_context():
      ok(db.session.get(MinePatch, patch_id).image_path is
      None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------------
      ------------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
      n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
      项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
     "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
          data={"name": "植被覆盖度", "unit": "%", "ideal_type":
      "benefit",
                "ideal_value": "75", "enabled_units": ["岩质边坡"],
                "description": "自检修改"})
with app.app_context():
      ok(Indicator.query.filter_by(code="coverage").first().de
      scription == "自检修改",
          "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
      ok("自检修改" not in (Indicator.query.filter_by(code="covera
      ge").first().description or ""),
          "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
      d = dict(base); d.update({"action": "add", "unit_type":
      "碎石边坡",
                                "level": "1", "expert_name":
      name})
      c.post("/ahp/experts", data=d)
with app.app_context():
      ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
      level=1).count() == 3,
```

```
        "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
       "导入数据已写入数据库")
```

```
r = c.post("/indicator/import",                                          11751
            data={"patch_id": str(patch_id),                             11752
                  "datafile": (io.BytesIO(b"x"), "d.pdf")},              11753
            content_type="multipart/form-data",                          11754
      follow_redirects=True)                                             11755
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")                11756
# ------------------------------------------------------------------11757
      ------------------                                                 11758
print()                                                                  11759
print(LINE)                                                              11760
print("图表与 PDF 导出")                                                    11761
print(LINE)                                                              11762
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,          11763
      "0.8376")])                                                        11764
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")                11765
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,        11766
      "0.4000")])                                                        11767
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")                    11768
# 报告是评价结论的呈现，需先真实执行一次评价才能打开                                11769
with app.app_context():                                                  11770
      from app.views.evaluate import load_patch_input,                   11771
      save_evaluation                                                    11772
      save_evaluation(patch_id, load_patch_input(patch_id))              11773
r = c.get("/report/%d" % patch_id)                                       11774
body = r.get_data(as_text=True)                                          11775
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %              11776
      body.count("<svg"))                                                11777
exe = browser.browser_path()                                             11778
if exe is None:                                                          11779
      print("  [跳过] 未找到浏览器，PDF 导出无法验证")                          11780
else:                                                                    11781
      r = c.get("/report/%d/pdf" % patch_id)                             11782
      data = r.data                                                      11783
      ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %             11784
      len(data))                                                         11785
      d = fitz.open(stream=data, filetype="pdf")                         11786
      txt = "".join(d[i].get_text() for i in                             11787
      range(d.page_count))                                               11788
      ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")                    11789
      ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")       11790
      ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)      11791
      d.close()                                                          11792
print()                                                                  11793
print(LINE)                                                              11794
print("补齐模块自检全部通过 √")                                              11795
print(LINE)                                                              11796
# ==== 测试/test_workflow.py ====                                        11797
# -*- coding: utf-8 -*-                                                  11798
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。                              11799
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。                 11800
```

```
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于  测试/  子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
```

```
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
        if pr:
            db.session.delete(pr)
        db.session.commit()
```

```
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ------------------------------------------------------------
    ----------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
```

```
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
   "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.3, thickness=10,
                                 ph=8.0,
    organic_matter=5.0, coverage=80.0,
                                 shannon=0.6, slope=20.0,
    aspect="阳坡"))
    db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
```

```
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("  重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")   # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
    (r.headers.get("Location") or ""),
   "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
# 报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
# 也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
   "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
   "直接访问 /report/<id> 被退回，且不产生任何评价结果",
   "HTTP %d" % r.status_code)
# 真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
```

```
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
   "报告数值取自评价记录，而非重新计算",
   "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
# 回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
# 导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
# 演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).fi
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
```

```
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
            follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
            follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
       follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first()
    ).id
r = c.post("/unit/edit/%d" % uid,
            data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
# ==== 测试/test_weights.py ====
# -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
```

```
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
```

```
        db.session.add(patch); db.session.flush()                    12201
        db.session.add(MicroUnit(patch_id=patch.id,                  12202
    unit_type=UNIT_TYPE,                                             12203
                                     area=1.0, ratio=1.0))           12204
        db.session.flush()                                           12205
        u = MicroUnit.query.filter_by(patch_id=patch.id).fir         12206
    st()                                                             12207
        db.session.add(IndicatorData(                                12208
            unit_id=u.id, soil_ratio=0.34, thickness=10,             12209
    ph=9.24,                                                         12210
            organic_matter=4.89, coverage=87.92,                     12211
    shannon=0.60,                                                    12212
            slope=28.75, aspect="阳坡"))                              12213
        db.session.add(ConstraintCheck(                              12214
            patch_id=patch.id, policy_pass=True,                     12215
    location_pass=True,                                              12216
            pollution_pass=True, geo_pass=True,                      12217
    farmland_pass=True,                                              12218
            passed=True, reason="全部通过"))                          12219
        db.session.commit()                                          12220
        return patch.id                                              12221
def cleanup(pid):                                                    12222
    with app.app_context():                                          12223
        from app.models import ScoreDetail                           12224
        rids = [r.id for r in EvaluationResult.query.filter_         12225
    by(patch_id=pid)]                                                12226
        if rids:                                                     12227
            ScoreDetail.query.filter(ScoreDetail.result_id.i         12228
    n_(rids)).delete(                                                12229
                synchronize_session=False)                           12230
        EvaluationResult.query.filter_by(patch_id=pid).delet         12231
    e()                                                              12232
        ConstraintCheck.query.filter_by(patch_id=pid).delete         12233
    ()                                                               12234
        uids = [u.id for u in MicroUnit.query.filter_by(pat          12235
    h_id=pid)]                                                       12236
        if uids:                                                     12237
            IndicatorData.query.filter(IndicatorData.unit_id         12238
    .in_(uids)).delete(                                              12239
                synchronize_session=False)                           12240
        MicroUnit.query.filter_by(patch_id=pid).delete()             12241
        m = db.session.get(MinePatch, pid)                           12242
        if m:                                                        12243
            db.session.delete(m)                                     12244
        p = Project.query.filter_by(code="WEIGHT-%s" %               12245
    TAG).first()                                                     12246
        if p:                                                        12247
            db.session.delete(p)                                     12248
        db.session.commit()                                          12249
def run_eval(pid):                                                   12250
```

```
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
    label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
   "土壤指标 %.2f" % w["l1"]["土壤指标"])
pid = build_case()
base_score = run_eval(pid)
print("  默认权重下的评价值 F = %.4f" % base_score)
print()
print(LINE)
print("二、保存 AHP 算出的权重后，评价应改用新权重")
print(LINE)
with app.app_context():
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",
    0.20),
                                ("terrain", 0.10)])
    n_items = WeightItem.query.count()
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
    label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留论文取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
```

```
import json                                                                  12301
with app.app_context():                                                      12302
    rec = EvaluationResult.query.filter_by(patch_id=pid) \                   12303
        .order_by(EvaluationResult.id.desc()).first()                        12304
    stored = json.loads(rec.unit_scores)                                     12305
ok(any(u.get("weight_source") for u in stored),                              12306
   "评价记录中标注了权重来源",                                                  12307
   stored[0].get("weight_source") or "")                                     12308
print()                                                                      12309
print(LINE)                                                                  12310
print("四、恢复默认后应回到论文取值")                                              12311
print(LINE)                                                                  12312
with app.app_context():                                                      12313
    ws.reset_weights(UNIT_TYPE)                                              12314
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)                    12315
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",              12316
   label3)                                                                   12317
back_score = run_eval(pid)                                                   12318
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,                  12319
   base_score))                                                              12320
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")               12321
print()                                                                      12322
print(LINE)                                                                  12323
print("五、权重首页概览")                                                      12324
print(LINE)                                                                  12325
with app.app_context():                                                      12326
    rows = ws.summary()                                                      12327
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))      12328
r = c.get("/ahp/")                                                           12329
body = r.get_data(as_text=True)                                              12330
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")        12331
print()                                                                      12332
print(LINE)                                                                  12333
print("六、一致性不通过的矩阵不得保存生效")                                      12334
print(LINE)                                                                  12335
with app.app_context():                                                      12336
    ws.reset_weights(UNIT_TYPE)                                              12337
n = 3                                                                        12338
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限                        12339
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]                          12340
data = {"csrf_token": "", "action": "save"}                                  12341
for i, j, v in mats:                                                         12342
    data["m_%d_%d" % (i, j)] = v                                             12343
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,                   12344
    data=data,                                                               12345
           follow_redirects=True)                                            12346
with app.app_context():                                                      12347
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)                     12348
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)                      12349
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")       12350
```

```
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
#  ==== app/services/__init__.py ====
#  -*- coding: utf-8 -*-
#  ==== app/utils/__init__.py ====
#  -*- coding: utf-8 -*-
#  ==== app/utils/timefmt.py ====
#  -*- coding: utf-8 -*-
"""时间显示：库里一律存 UTC，展示时换成服务器本地时间。
为什么不在写库时直接用本地时间：一个系统里时间基准只能有一个。
`app/models.py` 里所有 `created_at` / `checked_at` 都用
    `datetime.utcnow`，
存的是 UTC；如果写库改成本地时间，已经存进库的几千条记录会新旧混着差 8 小时。
所以统一在「展示」这一层换算，老数据不用动。
用法（Jinja）：
    {{ row.created_at | localdt }}                    →
    2026-09-26 23:33
    {{ row.created_at | localdt("%Y-%m-%d %H:%M:%S") }} →
    2026-09-26 23:33:28
"""
from datetime import datetime, timezone
DEFAULT_FORMAT = "%Y-%m-%d %H:%M"
def to_local(value, fmt=DEFAULT_FORMAT, empty="—"):
    """把 UTC 时间换算成服务器本地时间并格式化。
    传进来的值按「无时区即 UTC」处理；带时区的原样换算，便于以后改用
    带时区的时间戳。空值返回占位符，免得页面上出现 None。
    """
    if value is None:
        return empty
    if value.tzinfo is None:
        value = value.replace(tzinfo=timezone.utc)
    return value.astimezone().strftime(fmt)
def now_utc():
    """取当前 UTC 时间（带时区），用于只作展示、不落库的时间戳。"""
    return datetime.now(timezone.utc)
#  ==== app/views/__init__.py ====
#  -*- coding: utf-8 -*-
```